CHAPTER 5

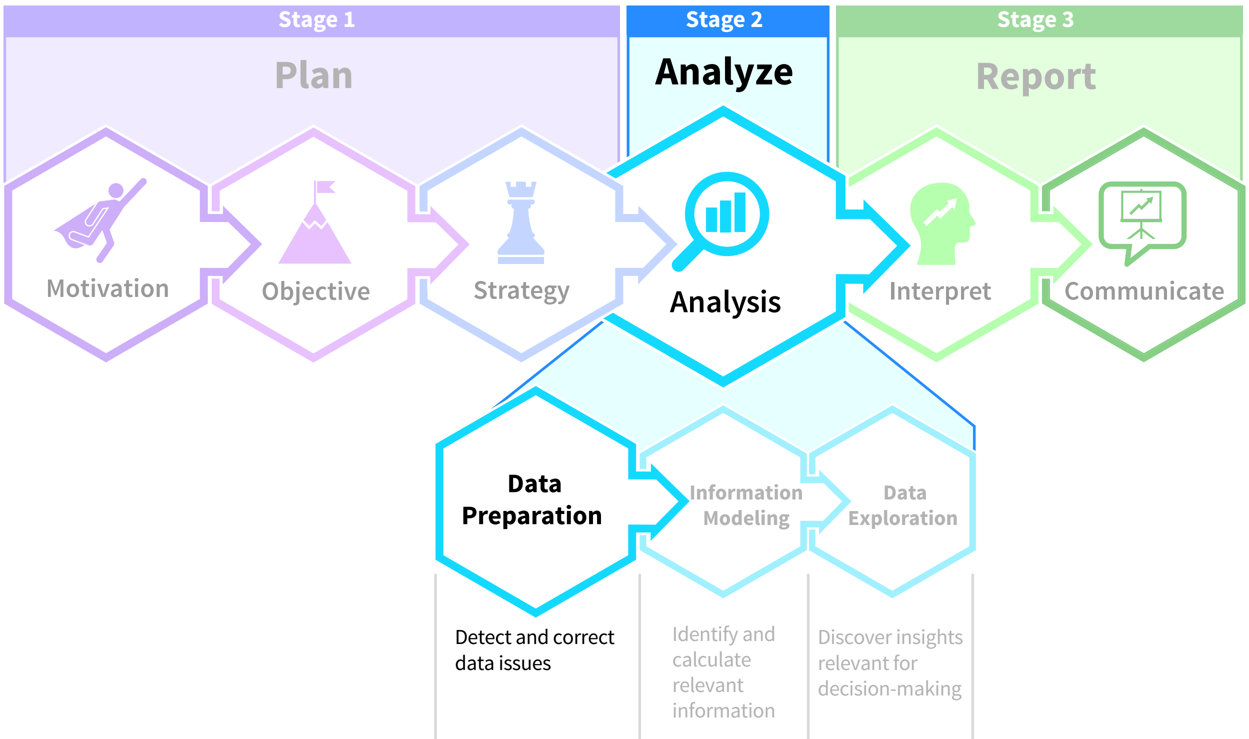
Analysis: Data Preparation

Chapter Preview

So far in this course, you have learned to plan a data analysis project by articulating what is motivating it, identifying its objectives, and designing a strategy to successfully complete it. Now that the plan is in place, it is time to move on to the data analysis stage. There are three tasks in this stage: preparing the data, building the information model, and exploring the data. Here, we will focus on the first task of preparing the data for analysis. Data preparation can be the most time-consuming activity in a data analytics project. You could spend over 75% of the project's total work time on this task alone! But there is a good reason for this, as data preparation involves multiple activities. There are two key factors to be aware of when preparing data for analysis:

1. The quality of the data will affect the quality of the insights and the decisions based on them. In other words, bad data leads to bad decisions.
2. The structure of the data will determine how effectively they can be analyzed.

First, we explore the two key data preparation processes in detail. Next, the chapter presents twenty data extraction, transformation, and loading patterns you can use to prepare data for analysis.



Professional Insight: How Can You Approach the Challenges of Data Preparation?

Bill is a director in the audit data analytics practice at one of the large public accounting firms.

> After graduating with an accounting degree, I gradually moved to our data analytics practice. **I love how accounting and technology blend together in our audit engagements and that there are always new opportunities and challenges to work on.**
>
> A key challenge in working with data at a public accounting firm is that our clients come in all shapes and sizes, with ERP systems ranging from legacy proprietary systems to increasingly modern, cloud-based integrated systems with sophisticated reporting and analytics capabilities. This means that **the cleanliness, quality, granularity, volume, and structure of the data can vary significantly. This increases the importance of ETL (Extract-Transform-Load) solutions to help our professionals efficiently extract data from client systems, transform that data into a common format, and ultimately load the data into our analytics platform.**
>
> To be successful, the data professionals at my firm must be able to understand and evaluate our clients' data to ensure that they are relevant and reliable. This includes knowing how to have the right conversations with our clients, often including representatives from the client's IT department and our internal data extraction specialists, in order to effectively communicate our data requirements. The skills you are building in this chapter are all highly relevant to a successful career as a data analyst in today's public accounting profession.

Chapter Roadmap

LEARNING OBJECTIVES	TOPICS	APPLY IT
LO 5.1 Explain the process of data profiling.	• Investigate Data Quality • Investigate Data Structure • Decide and Inform	**Identify Data Quality Issues**
LO 5.2 Describe the extract-transform-load (ETL) process.	• Extract Data • Transform Data • Load Data	**Combine Tables for Analysis**
LO 5.3 Apply patterns to extract data.	• Two Data Extraction Patterns	**Extract Data with Patterns**
LO 5.4 Apply patterns to transform columns.	• Eight Column Transformation Patterns	**Use Column Transformation Patterns**
LO 5.5 Apply patterns to transform tables.	• Four Table Transformation Patterns	**Transform Tables with Patterns**
LO 5.6 Apply patterns to transform models.	• Three Model Transformation Patterns	**Draw a Star Schema**
LO 5.7 Apply patterns to data loading Issues.	• Three Data Loading Patterns	**Evaluate Relationships Between Tables**

Data The Data tag appears in the chapter when the data for an example, illustration, or application are available on Wiley's online learning platform.

Data analytics software is continuously changing, and there may be more recent versions of the software referenced in this chapter. For more information access the accompanying video on Wiley's online learning platform.

5.1 What is Data Profiling?

LEARNING OBJECTIVE 1

Explain the process of data profiling.

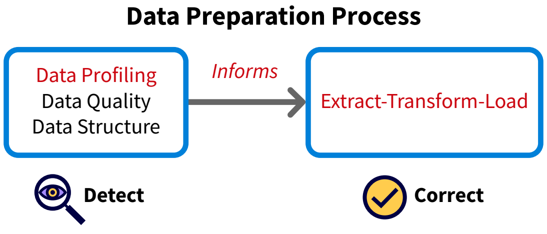
Businesses lose billions of dollars every year due to **dirty data**, or data that provide inaccurate or incomplete descriptions of the economic activities of a business. Using dirty data can result in issues ranging from inaccurate pricing, to sending bills to the wrong customers (or not collecting them at all), to an inability to detect fraud. Data preparation helps avoid these issues and more.

Data preparation is the process of profiling, cleaning, restructuring, and integrating data prior to processing and analysis. It helps ensure high-quality data and, therefore, improved decision-making. **Data profiling** is the process of investigating data quality and structure. It has three parts:

1. **Investigating data quality:** Search for anomalies in the data. That is, are the data dirty?
2. **Investigating data structure:** Find the best way to organize the data and improve analytics.
3. **Deciding and informing:** Make decisions about whether it is possible to address the identified issues, what the cost of doing so would be, and consider the possible consequences if the issues are not addressed.

Decisions made in the final phase will guide the extract, transform, load (ETL) process by determining what must be changed. As **Illustration 5.1**. shows, data preparation is an ongoing collaboration between the data profiling process, which is the subject of this discussion, and the ETL process, which is covered next. For now, keep in mind that data profiling detects data issues and ETL corrects them.

ILLUSTRATION 5.1 The Data Preparation Process



Investigate Data Quality

When discussing the "quality" of the data we work with, we are referring to the suitability of using the data for decision-making. Assessing data quality identifies flawed values in the data set, which reveals if there are data that must be cleaned. There are different methods for doing this:

- The rule-driven method is a top-down approach. A logical relationship, or rule, is defined among data and tested to determine if the data conform to it. The number of rules that can be specified is nearly unlimited. Examples include:
  - Segregation of duties.
  - Quantity on hand cannot be negative.
  - A car cannot be rented to someone who is younger than 16 years old.
  - Sales can only be executed with a valid order.
- The exploration and inference methods are bottom-up approaches. The goal is to find anomalies by examining the data from many different perspectives. Sorting, frequency

distributions, and outlier analysis are examples of powerful techniques for exploration purposes. The second bottom-up approach, inference, is a method that relies on computer algorithms to identify anomalies.

These different methods identify data anomalies, which occur when the data do not meet expectations of correctness, validity, consistency, and completeness.

Correctness

Data describe facts about entities, such as the name of a customer, the price of a product, or the date of a transaction. Data are incorrect when the value assigned to an entity's characteristic is wrong. For instance, a customer might live in New Jersey (NJ), but New York (NY) is recorded instead, or the price of a product is $252, but it is listed as $225.

Incorrect data are common, as these real-world examples of data glitches show. For the businesses involved, data mistakes can be significant:

- Delta Airlines sold a round trip plane ticket from mainland U.S. to Hawaii for less than $7. In a similar example, a United Airlines customer was able to purchase a first-class round-trip ticket from the U.S. to Hong Kong for just seven frequent flyer miles.
- A customer booked a luxury hotel room in Pasadena, California for $10 a night.
- Walmart sold a treadmill for less than $35.

There are several ways to detect such errors. When pricing is incorrect, such as when a normally expensive airfare is available at a deep discount due to an error, then a sudden surge in demand might be an indication of a problem. Profiling techniques, such as exploring outliers, can also be useful. Regardless, always think critically about the data being analyzed and watch for anomalies.

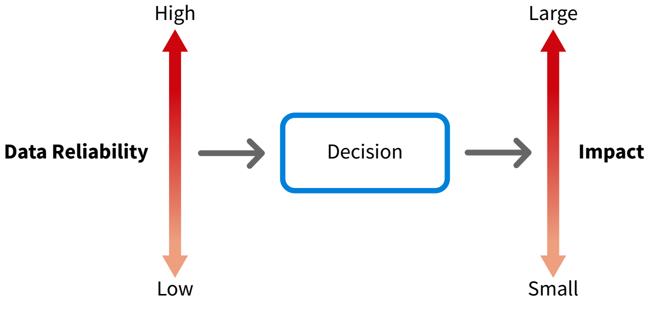
APPLYING CRITICAL THINKING 5.1: Assess Data Quality

Using sound data removes biases and guesswork from decision-making. Using poor data in the process increases risk. So, it is crucial to assess risk associated with less-than-perfect-data. There are two characteristics to consider **(Risks)**:

- Reliability: Data extracted from an enterprise resource planning (ERP) system that have been subjected to hundreds of internal controls are more reliable than sentiment data extracted from Facebook.
- Impact: Similarly, the impact of launching a new product line is much larger than the decision to invest in new business cards.

The higher the impact, the more you should be willing to invest in improving the data's reliability.

Assessing Data Reliability and Impact



Validity

Another integral part of data profiling is designing **validation rules**. These rules define what values are and are not acceptable and they can take many shapes and forms. For example, an order should occur before a shipment, or an email address should contain the symbol @.

Validity and correctness are sometimes confused. Imagine that a business ships their items to midwestern states only: IL, IN, IA, KS, MI, MN, MO, NE, ND, OH, SD, and WI. Here are some scenarios:

- Goods are shipped to Ohio (OH), and OH is entered as the value for the state field in the system. The value is both *valid* and *correct*.
- Someone accidentally records Minnesota (MN) as the destination for a shipment that went to Michigan (MI). The value entered is *valid* (MN is an accepted state) but *incorrect*.
- The business receives an order from a customer in Pennsylvania (PA), and the goods are shipped. PA is entered into the system as the state. While the system will consider the value as *invalid*, it is *correct* (PA is the correct state, but the company cannot sell goods there).
- The same order from a customer in PA is received and the goods will be shipped there. NA–Not Applicable–is entered into the system as the state. In this case, the value entered is both *invalid* and *incorrect*.

An incorrect value means the wrong value is assigned, and an invalid value means an unacceptable value is assigned.

Consistency

In addition to being correct and valid, data should be consistent. Data inconsistency occurs when the same characteristic is represented multiple ways. Using both MI and Michigan to refer to the same state would create data inconsistency. Using mgr, mngr, and manager to describe the same job position would do the same. These inconsistencies create challenges during analysis. For example, it would be difficult to determine total revenues per state if there were separate totals for MI and Michigan.

How can we identify inconsistencies like these? Here are two profiling techniques:

- Create a list with all distinct values, then sort and review. The inconsistent values of mgr, mngr, and manager will likely be noticeable immediately.
- Build a frequency table, or a table that counts how many times a value occurs. Values with a low frequency might indicate inconsistent data.

Completeness

Data that is correct, valid, and consistent are only accurate if they are also complete, as non-complete data might result in distorted insights. Data can be incomplete in two ways.

A missing instance is when a concept occurred but is not recorded, such as when goods were sold, but the sales transaction was not recorded. A missing instance such as a missing sales transaction can be identified with gap analysis. If there is a sequential number for each sales invoice, then a missing number might indicate a sale that occurred but was not recorded.

A missing value occurs when the transaction is recorded, but we do not have information for all characteristics. The result is empty cells. This might happen if a customer is recorded, but the record does not include the customer's email address. The term null typically indicates a missing or unknown value. It is important to assess how the missing information affects decision-making.

Investigate Data Structure

Along with their quality, data are investigated to assess whether they are structured in a way that makes data analytics easy and efficient.

Unambiguous Descriptions

During data exploration and interpretation, column names in tables become variables. We apply mathematical operations, such as sum, to them and use them as part of visualizations, such as pie charts. However, poorly named columns make it complicated to develop information models and to conduct analysis. Column names should also be correct, intuitive, and clear because they are part of the analytical database, which might have many different users. An **analytical database** is the integrated data set used for analysis purposes. Unambiguous descriptions should be a priority when developing an analytical database. Here are some examples of correct and clear column titles:

- An age column should contain numbers (e.g., 32) and not dates (e.g., 6/2/1990).
- CustomerID is more descriptive than ID.
- Birthday is easier to understand than bday.

APPLYING CRITICAL THINKING 5.2: Create A Shared, Comprehensible Vocabulary

Creating a shared vocabulary requires identifying who is going to use the analytical database **(Stakeholders)**:

- If you prepare the data for others, then make sure that you create a vocabulary for column and table names that is easy to understand and work with.
- For example, it is common for data scientists to use table names such as DCustomer and FSales with D and F referring to certain types of tables. However, such terms might be confusing to other stakeholders who are not data scientists!

Table Structures That Make Analysis Easier

Analysis includes aggregating, or grouping, data and then examining them from different viewpoints in many ways. This is known as **slicing**. Here is an example of slicing sales transaction data for a business with sales in multiple regions:

- **Aggregate:** Calculate the total sales amount.
- **Slice:** Break the total down by region to examine the regional sales in more detail.

Some data structures are a good fit for these aggregate and slice processes, while others are not, so learning best practices for structuring data is crucial. Two common best practices are single-valued columns and flat tables.

Single-Valued Columns For analytical purposes, each cell should contain one value describing one characteristic. That is, each column should be **single-valued**. Two or more values in the same cell makes analysis more challenging. Here are two scenarios that violate the single-valued rule.

A **composite column** combines values for two or more characteristics. The Name column in **Illustration 5.2** (A) combines last name, first name, and certification characteristics. Which data set would you prefer if you needed to generate a list with all employees who have a CPA–panel (A) or (B)? While the data are identical for both data sets, panel (B) makes it easier to answer this question.

ILLUSTRATION 5.2 Composite vs. Single-Valued Column

(A) Composite Column

Name
Cole, Lakeisha, NA
Lopez, Alejandro, CPA
Key, Kim, NA
David, Julie, CPA
Malone, Moses, NA
Buslepp, Bill, CPA
Potoms, Keme, NA
Despontin, Marc, CPA
Kaminski, Ivanka, NA

(B) Single-Valued Column

Name	Certification
Cole, Lakeisha	NA
Lopez, Alejandro	CPA
Key, Kim	NA
David, Julie	CPA
Malone, Moses	NA
Buslepp, Bill	CPA
Potoms, Keme	NA
Despontin, Marc	CPA
Kaminski, Ivanka	NA

In a **multi-valued column**, a cell contains multiple values of the same characteristic. The Certifications column in **Illustration 5.3** (A) lists all certifications for an employee. Again, which data set would you prefer if you needed to generate a list with all employees who have a CPA–panel (A) or (B)? While the data are identical for both data sets, it is easier to answer this question with the single-valued column in panel (B).

ILLUSTRATION 5.3 Multi-Valued vs. Single-Valued Column

(A) Multi-Valued Column

Name	Certifications
Cole, Lakeisha	CISA
Lopez, Alejandro	CPA, CISA
Key, Kim	CMA
David, Julie	CPA, CMA, CISA
Malone, Moses	CPA
Buslepp, Bill	CPA, CMA
Potoms, Keme	CMA
Despontin, Marc	CPA
Kaminski, Ivanka	CMA, CISA

(B) Single-Valued Column

Name	Certifications
Cole, Lakeisha	CISA
Lopez, Alejandro	CISA
Lopez, Alejandro	CPA
Key, Kim	CMA
David, Julie	CISA
David, Julie	CMA
David, Julie	CPA
Malone, Moses	CPA
Buslepp, Bill	CMA
Buslepp, Bill	CPA
Potoms, Keme	CMA
Despontin, Marc	CPA
Kaminski, Ivanka	CISA
Kaminski, Ivanka	CMA

Flat Tables Another best practice for structuring data is using **flat tables**, in which column headers do not contain data values useful for analysis purposes. For data analysis, flat table structures are preferred over crosstabulation tables

Illustration 5.4 (A) shows a crosstabulation table representing a relationship between the two variables of car model and salesperson:

- In the first column are the values for the different car models: Focus, Mustang, Escape, and Explorer.
- The column headers in the top row show the values for salespeople.
- A cross-section cell shows how many units a salesperson has sold of a specific model. Salesperson Elodie has sold five units of the Focus model.

Crosstabulation tables have limitations for analysis. The names in the panel (A) column headers are examples of data values useful for data analysis. Column headers cannot be used for filter or group-by operations.

ILLUSTRATION 5.4 Crosstabulation Table Structure vs. Flat Table Structure

(A) Crosstabulation

	Jim	Jane	Elodie	Carlos
Focus	0	1	5	4
Mustang	1	2	3	3
Escape	4	1	0	2
Explorer	2	2	0	1

(B) Flat

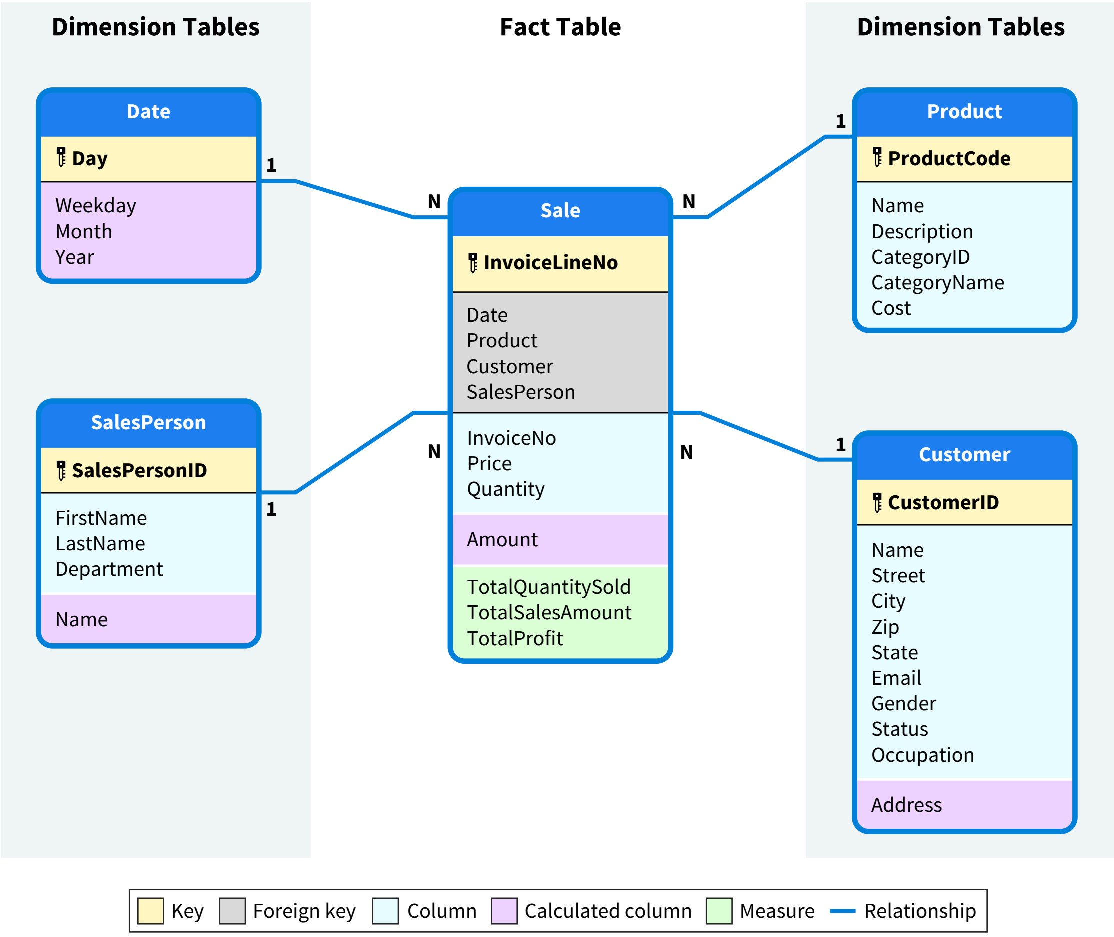
Model	SalesPerson	UnitsSold
Focus	Carlos	4
Mustang	Carlos	3
Escape	Carlos	2
Explorer	Carlos	1
Focus	Elodie	5
Mustang	Elodie	3
Escape	Elodie	0
Explorer	Elodie	0
Focus	Jane	1
Mustang	Jane	2
Escape	Jane	1
Explorer	Jane	2
Focus	Jim	0
Mustang	Jim	1
Escape	Jim	4
Explorer	Jim	2

Illustration 5.4 (B) shows the same data using a flat structure. Carlos, Elodie, Jane, and Jim have become values of the Salesperson column. They can be used for filter and group-by operations. Would you prefer the table in panel (A) or the table in panel (B) to answer a question like "How many Mustangs did Jane and Jim sell?" The flat structure in Illustration 5.4 (B) makes analysis easier because we can filter by salesperson and then sum the values in the UnitsSold column.

Data Models That Make Analysis Easier

A **data model** defines how different tables relate to each other. Analytical data models should be easy to understand by business users such as accountants, and computers should be able to efficiently process them. The recommended structure for analytical data models is the **star schema**, which has both characteristics (**Illustration 5.5**). As the top of Illustration 5.5 shows, a star schema consists of two types of tables, fact tables and dimension tables, and the relationships between them.

ILLUSTRATION 5.5 Star Schema

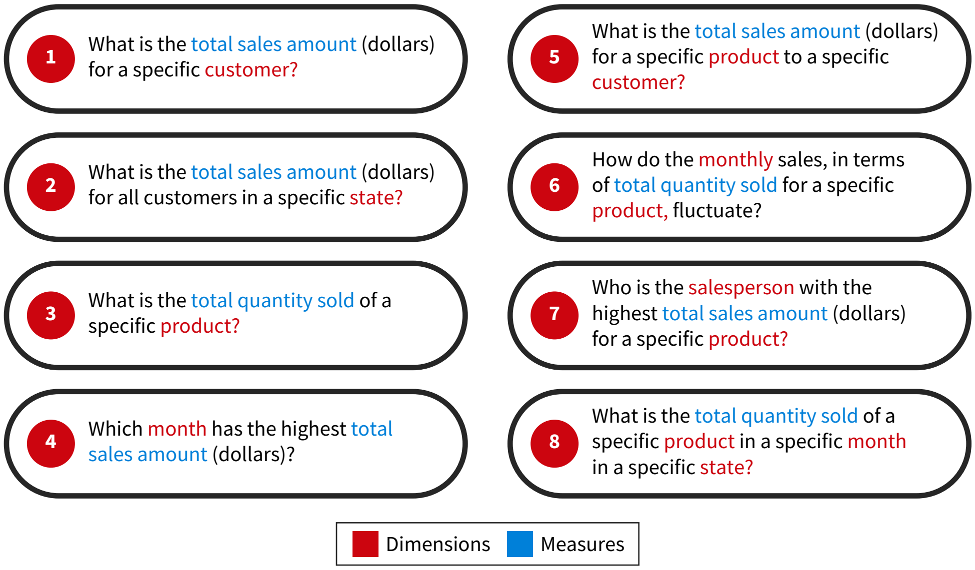


Fact Tables In an accounting context, **facts** correspond to business transactions such as orders, sales, purchases, and payments. The fact table in the center of Illustration 5.5 contains sales data. Data in a fact table are captured at a high granularity level to avoid imposing additional constraints on the analysis. (The level of detail is known as the grain.) In Illustration 5.5 the grain is an invoice line, which is a specific product shipped to a specific customer, on a specific day, and involving a specific salesperson.

While the number of columns in a fact table is generally small, they usually have many rows. Companies often have thousands of sales on the same day, if not more–think of online retail giants such as Amazon. The columns in fact tables are also mostly quantitative, and more specifically, additive. The values in these columns can be easily aggregated, or grouped together, as measures. These measures can then be sliced using the columns in the next type of tables.

Dimension Tables **Dimensions** provide context to analysis and give meaning to facts. The dollar amount for a sale of $100 does not provide much information. Businesses want to know when the sales transaction occurred, who must pay the $100, and for what goods. The star schema in Illustration 5.5 has dimension tables for the date the sale occurred, the product that was sold, the customer, and the salesperson. Columns in dimension tables are variables that can be combined in multiple ways to slice the measures. **Illustration 5.6** demonstrates eight questions that can be answered by slicing a sale measure by one or more of the dimensions shown in Illustration 5.5.

ILLUSTRATION 5.6 Slicing Measures by Dimensions



While the number of rows in a dimension table is usually small, they often have many columns. The Customer dimension table in Illustration 5.5 has ten columns but no measures. In practice, customer dimension tables often have more than 100 columns.

Relationships Relationships, the final element of data models, link tables and are represented by lines in Illustration 5.5. All relationships in a star schema have a one-to-many (1-N) cardinality pattern. A **cardinality** is a constraint that defines how many times an instance of an entity can participate in a relationship. It can take two possible values, N or 1:

- N: An instance can participate many times in the relationship. There is no restriction.
- 1: An instance can participate only one time in the relationship.

A 1-N cardinality pattern for a relationship between a dimension and a fact table can be interpreted as follows:

- 1: For every fact, there is just one corresponding value in each dimension table. For each instance of the Sale table (an invoice line), there is one date, product, customer, and salesperson.
- N: There can be many facts for each dimension. There can be many sales on the same day, and the same product type can be sold many times. Both customers and salespeople can be involved in many sales transactions.

Decide and Inform

After investigating the quality and structure of the data, it is time to take the next step:

- **Do not move forward:** If the data are unfit, such as the data having poor quality, then using the data for decision-making would be too risky. Similarly, if the data structure is poor, then restructuring and processing the data might be too complicated and economically unfeasible.
- **Redesign the source system:** This decision is usually made because errors generated by the source system need to be corrected. Once fixed, the data source should be profiled again.
- **Confirm the data's fitness for the analysis and move forward:** Moving forward with the data requires being risk-aware. That is, we can move forward with the data with an awareness of potential reliability issues and their impact on decisions.

If the decision is made to move forward, the next step in the process would be informed by any issues that must be addressed.

Apply It 5.1
Identify Data Quality Issues

Stufan is a small store in Gumboro, Delaware that buys and sells stuffed animals. Founder and CEO Shanice Parker wants to apply data analytics to better understand her business. To start, she asks your firm to help prepare the data for analysis.

The table is a sample of Stufan's product data. Identify three quality issues with the data.

Code	Description	QOH
BASHL	Bashful Large, Dwarfs	13
BASHS	Bashful Small, Dwarfs	67
BERL	Berlioz, Aristocats	23
CAPH	Captain Haddock, Tintin	14
DOCL	Doc Large, Dwarfs	23
DOCS	Doc Small, Dwarfs	45
DOPL	Dopey Large, Dwarfs	43
DOPS	Dopey Small, Dwarfs	–3
DUCH	Duchess, Aristocrats	44

SOLUTION

1. Description is not a single-valued column. It combines "description" and "category."
2. There is a consistency issue: "Aristocats" and "Aristocrats."
3. There is a validity issue: QOH (Quantity On Hand) cannot be negative.

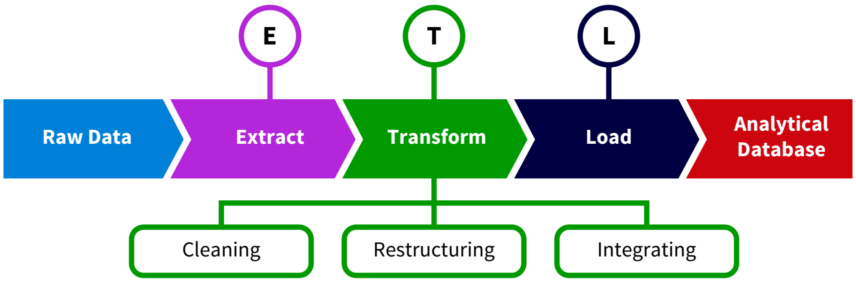
5.2 What Does it Mean to Extract-Transform-Load (ETL) Data?

LEARNING OBJECTIVE 2

Describe the extract-transform-load (ETL) process.

While the data profiling process you just learned about detects issues, the **extract-transform-load (ETL)** process corrects them. As **Illustration 5.7** shows, the ETL process builds an analytical database using one or more sources of raw data.

ILLUSTRATION 5.7 The Extract-Transform-Load (ETL) Process



Extract Data

Data extraction, which involves transferring data, is the first step in the ETL process:

- Source data are moved to the platform where they will be transformed. This platform is normally a **data warehouse**, which is software that stores and analyzes large data sets. Power BI and Tableau are examples of data warehouses.
- The process also includes data validation, or confirming that the data were transferred completely and correctly.

ETL tools make it easy to extract data from databases, spreadsheets, text files, and many other data sources by providing **data connectors** that are intuitive software programs designed to extract data. **Illustration 5.8** (A) shows some data connectors that are available in Excel, while panel (B) does the same for Power BI.

Transform Data

Raw data are rarely ready for analysis after they are extracted. **Data transformation** improves the raw data for analysis through cleaning, restructuring, and integration.

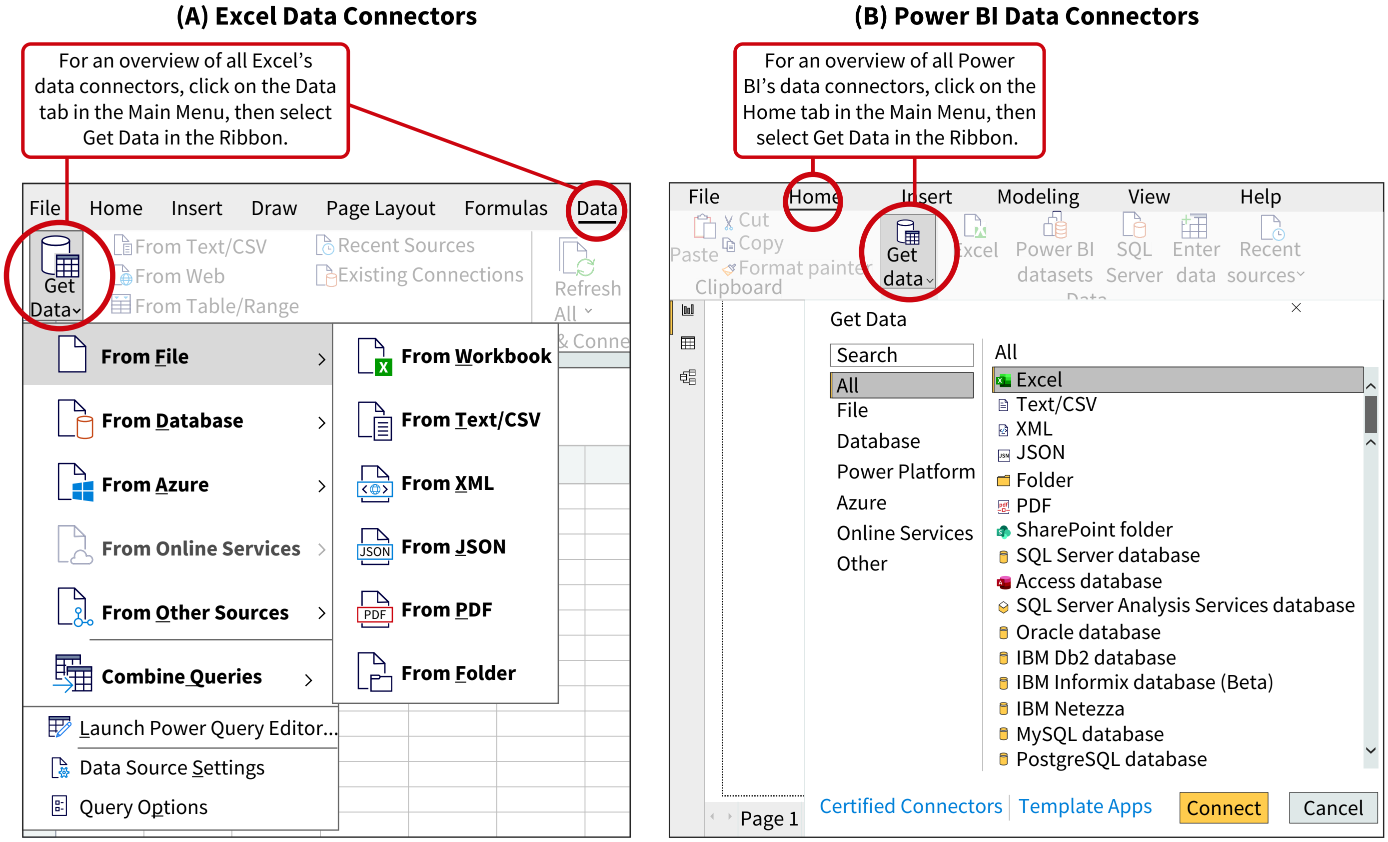
Cleaning Data

Data can be incorrect, invalid, inconsistent or incomplete. Cleaning data, one of the most important and time-consuming aspects of data transformation, is also known as data cleansing or data scrubbing. It involves adding, modifying, and deleting data:

- In the case of incomplete data, such as a missing sales transaction, data might need to be added. A specific strategy for dealing with incomplete data is imputation, which is when estimated values are substituted for missing data.

- Modifying data is necessary when a current value must be replaced with a new value if data are incorrect, invalid, inconsistent, or incomplete. Replacing the value NY with NJ in a column that records a customer's state is an example of this.
- Delete data that are not relevant for analysis. Redundant data, such as a duplicate sales transaction, should be removed.

ILLUSTRATION 5.8 Data Extraction using Data Connectors



ETL tools provide built-in functions for a range of data cleaning tasks. What if you wanted to remove rows from a table? **Illustration 5.9** shows that Power Query, Excel's and Power BI's ETL tool, have commands for removing duplicate rows, blank rows, or rows containing errors. Other tools offer similar operations.

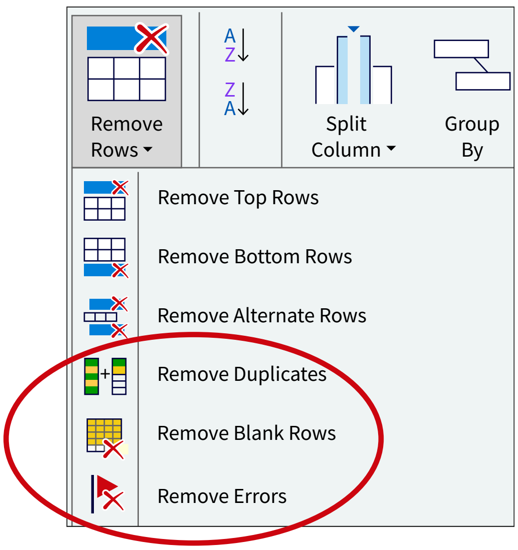
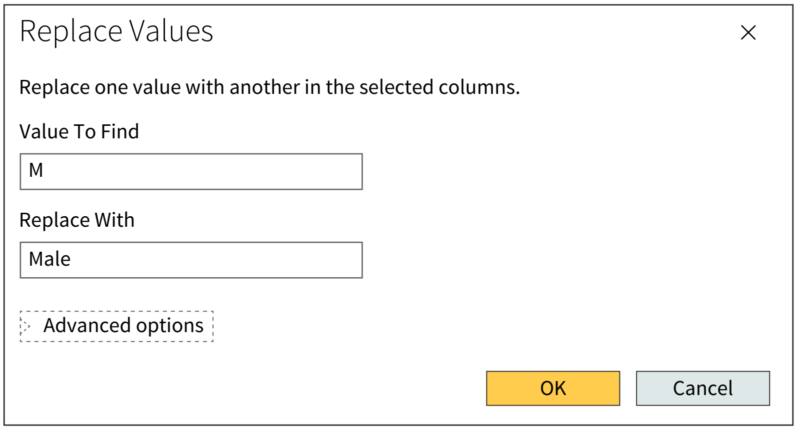


ILLUSTRATION 5.9 Removing Rows with Power Query

What about replacing specific values? **Illustration 5.10** shows how Power Query's Find and Replace option can fix an inconsistency problem.

ILLUSTRATION 5.10 Replacing Values with Power Query



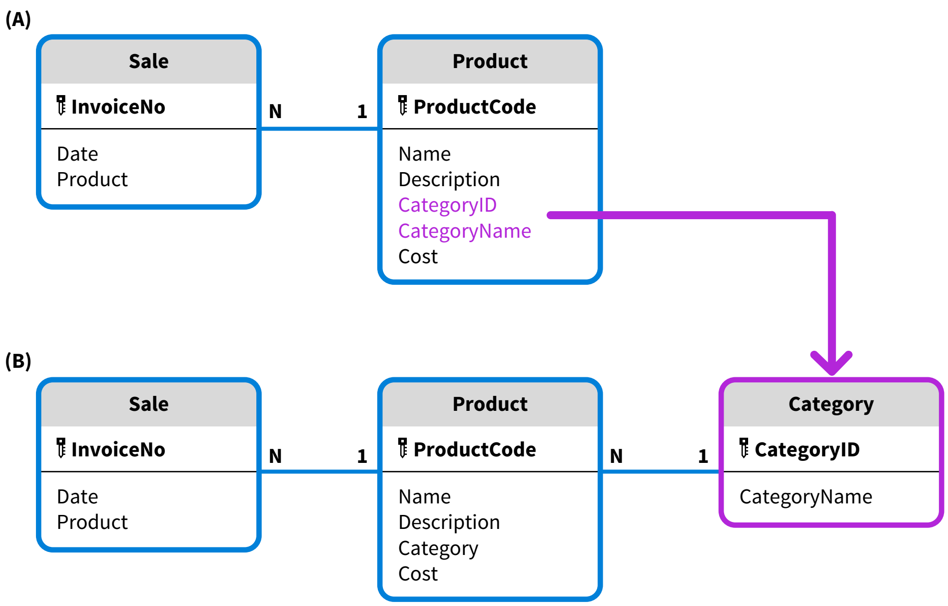
If "M" is sometimes used for "Male," and sometimes the word "Male" is used in the column for gender in the data set, then all M values in the Gender column can be replaced with "Male." Using this tool quickly resolves the inconsistent data by finding the inconsistent values and replacing them.

APPLYING CRITICAL THINKING 5.3: Choose the Best Method

Data preparation is not an exact science. Judgment is an important part of building an analytical database. Always consider alternatives, examine their strengths and weaknesses, and rank them. In the following, panels (A) and (B) represent the same data but organize them differently **(Alternatives)**:

- In panel (B), category becomes its own entity and the name of a category is recorded only once.
- On the other hand, adopting the data model in panel (A) means the category name is repeated for all products in the same category. The data model in panel (B) saves space at the price of complexity, given that there is an extra relationship that needs to be considered during analytics.

Alternative Data Models for the Same Data Set



Restructuring Data

Clean data are not necessarily structured in a way that makes analytics easy and efficient. **Data restructuring**, which is also known as data wrangling or data munging, does not change data values, but it does change how the data are organized.

ETL tools provide a variety of techniques to make restructuring easier:

- Adding and deleting columns.
- Renaming columns and tables.
- Splitting and merging columns.
- Splitting and combining tables.
- Transposing and unpivoting tables.

Integrating Data

Most data analytics projects involve multiple tables, often from different data sources, which must be integrated before performing analysis. **Data integration** is the process of connecting related data. There are two distinctive forms of integration:

- Linking two tables by defining a relationship between them. Relationships are created using primary and foreign keys. Other aspects of relationships that must be specified are cardinalities.
- Combining two or more tables unites information about the same entity. Tables can be combined two ways. A **union** combines different tables with the same data structure. The result is a table with more rows. Recall from the foundational data analysis chapter that a join, or **merge**, combines data elements or columns from different tables. The result is a table with more columns.

A challenge specific to integration is **data matching**, a process that compares data and determines whether they describe the same entity. Consider the two customer data sets shown in **Illustration 5.11**. Ilustration 5.11 (A) contains financial information, and panel (B) contains demographic information about customers. How would we reconcile, or match, the customer names? Specific issues to be addressed include:

- Nicknames: Jen Pollack versus Jenny Pollack.
- Typos: Carlos Panetta versus Carlos Paretta.
- Reversed names: Margarita David versus David Margo.

Panel (C) shows the merged table. Most ETL tools provide advanced support for data matching.

ILLUSTRATION 5.11 Data Matching Issues

(A) Customer Table: Financial Information

Names	Status	Discount
William McCarthy	A	5
Margarita David	C	1
Ann White	B	3
Carlos Panetta	A	5
Jen Pollack	B	3

(B) Customer Table: Demographic Information

Names	Gender	Age
Carlos Paretta	M	44
David Margo	M	22
Jenny Pollack	F	57
Bill McCarthy	M	38
Ann White	F	28

(C) Merged Customer Table

Names	Status	Discount	Gender	Age
Ann White	B	3	F	28
Carlos Panetta	A	5	M	44
Jen Pollack	B	3	F	57
Margarita David	C	1	M	22
William McCarthy	A	5	M	38

Load Data

Once the data are cleaned and transformed, they are loaded into the software for analysis. **Data loading** is the process of making the analytical database available for use. Analytical databases are often posted in the cloud where they can be used simultaneously by many users. Like extraction, part of transferring the data is validating whether all records were transferred and whether they were transferred correctly.

Apply It 5.2
Combine Tables for Analysis

Stufan CEO Shanice suggests organizing product data in three separate tables, with one table for each product category. She asks if there would be issues combining the three tables for analysis. How would you respond?

Aristocats		
Code	**Description**	**QOH**
BERL	Berlioz	23
DUCH	Duchess	44
MAR	Marie	34
THO	Thomas	19
TOU	Toulouse	32

Dwarfs	
Code	**Description**
BASHL	Bashful Large
BASHS	Bashful Small
DOCL	Doc Large
DOCS	Doc Small
DOPL	Dopey Large
DOPS	Dopey Small
GRUML	Grumpy Large
GRUMS	Grumpy Small
HAPL	Happy Large
HAPS	Happy Small
SLB	Sleepy Big
SLS	Sleepy Small
SNEL	Sneezy Large
SNES	Sneezy Small
SNOWL	Snow White Large
SNWS	Snow White Small

Tintin		
Code	**Description**	**QOH**
CAPH	Captain Haddock	14
SNO	Snowy	22
TINL	Tintin Large	23
TINS	Tintin Small	44

SOLUTION

- The three tables cannot be combined using a union since the table structure for the Dwarfs table is different from the table structures of the Aristocats and Tintin tables. The QOH column is missing.
- A product's category is determined by its table name and cannot be used for analysis purposes. A Category column needs to be added to each of the three tables.

5.3 Which Patterns Extract Data?

LEARNING OBJECTIVE 3

Apply patterns to extract data.

Each data analytics project has unique data preparation challenges. While there is not a single, common approach for all projects, a structured set of data preparation patterns can address most challenges. These patterns signal potential problems and provide guidance for finding them within the data set and correcting them. Each pattern identifies a data issue, discusses how to detect it with a profiling method, and explains one or more ETL methods to correct it. Think of the patterns as a menu; you can select those which are most appropriate for your needs.

Data Let's use a case to illustrate how to apply data preparation patterns in a real-world scenario. You can also use the available data to work through each pattern yourself.

Beans is an accounting firm in Okemos, Michigan providing accounting and tax services. Petra, their managing partner, has made improving analysis of their services a priority in the new year. Petra does not mind doing the analytics but struggles to integrate the different data sources. Imagine you are one of the staff accountants helping with the data preparation.

The data set for Beans consists of four worksheets in an Excel file (**Illustration 5.12**).

ILLUSTRATION 5.12 Worksheets in the Beans Data Set

Worksheet	Description
ClData	Information about Beans' clients.
Employee	General information about Beans' employees.
E-Dem	Demographic information about Bean's employees.
Service	Information about the services provided for the period January-July 2025.

The first step is identifying the available data and creating a **data dictionary**, a chart that indicates what data are available and where they can be found. A data dictionary records different pieces of information for each field, including a name, a brief description of content, the data type, whether the field is a primary or foreign key, and whether the field is mandatory or optional.

A data dictionary is often referred to as metadata–it is data about data. It is built gradually throughout the data preparation process, but creating column names and descriptions is a good starting point. **Illustration 5.13** shows the first draft of a data dictionary for the Beans data set. (See Illustration 5.44 at the end of the chapter for an example of the completed data dictionary.)

ILLUSTRATION 5.13 Beans Data Dictionary

ClData

Name	Description
ID	A client's unique ID.
Name	A client's name.
Industry Name	A client's industry.

Employee

Name	Description
ID	An employee's ID.
Name	An employee's name.
Job Title	An employee's job title.
Rate	An employee's hourly rate.
Office	An employee's office.
MS	An employee's marital status.

E-Dem

Name	Description
First	An employee's first name.
Last	An employee's last name.
Age	An employee's age.
CertList	List of an employee's certifications.
University	The university from which the employee received an undergraduate degree.

(Continued)

ILLUSTRATION 5.13 *(Continued)*

Service

Name	Description
ID	A unique ID for a service provided.
Date	The date the service occurred.
Actual Time	The actual time spent on a service provided.
Budgeted Time	The budgeted time for a service provided.
ID	The ID of the task performed.
Area	The task's area.
Task	The name of the task performed.
EmployeeID	The ID of the employee performing the task.

Starting from the data dictionary in Illustration 5.13, we are ready to prepare the data by applying twenty patterns. The first two are extraction patterns to determine if all data have been extracted and if they were transferred correctly.

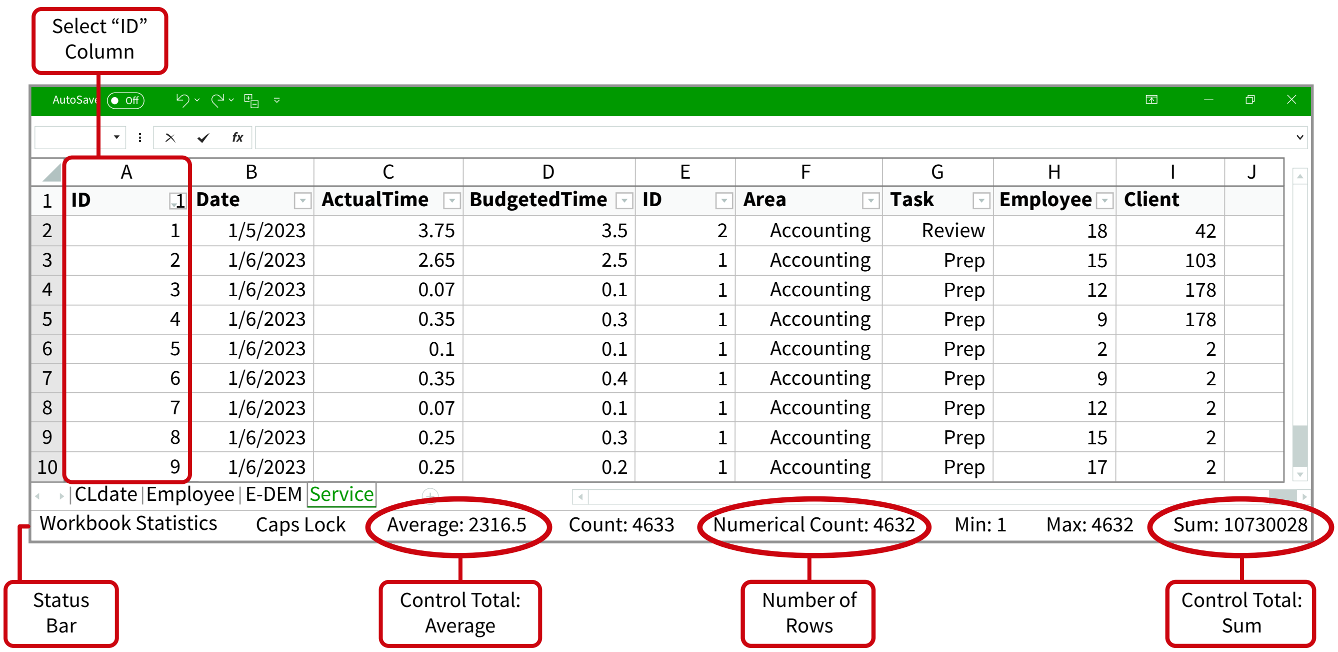
Data Preparation Pattern 1: Incomplete Data Transfer

Extraction transfers data from the source files to the ETL tool for further processing. An incomplete transfer of data and missing data leads to unreliable results.

Compare Row Counts

Comparing row counts is one way to check completeness. Working with the Beans data set, test completeness by comparing the Excel worksheet's row count with the corresponding table's row count in the ETL tool. **Illustration 5.14** shows the Service Excel worksheet. When the ID column is selected, Excel's status bar shows that there are 4,632 rows in the worksheet.

ILLUSTRATION 5.14 Row Count for Excel Worksheet



	A	B	C	D	E	F	G	H	I	J
1	ID	Date	ActualTime	BudgetedTime	ID	Area	Task	Employee	Client	
2	1	1/5/2023	3.75	3.5	2	Accounting	Review	18	42	
3	2	1/6/2023	2.65	2.5	1	Accounting	Prep	15	103	
4	3	1/6/2023	0.07	0.1	1	Accounting	Prep	12	178	
5	4	1/6/2023	0.35	0.3	1	Accounting	Prep	9	178	
6	5	1/6/2023	0.1	0.1	1	Accounting	Prep	2	2	
7	6	1/6/2023	0.35	0.4	1	Accounting	Prep	9	2	
8	7	1/6/2023	0.07	0.1	1	Accounting	Prep	12	2	
9	8	1/6/2023	0.25	0.3	1	Accounting	Prep	15	2	
10	9	1/6/2023	0.25	0.2	1	Accounting	Prep	17	2	

Next, determine the row count for the Service table in the ETL tool used for extraction. **Illustration 5.15** shows that information as part of the ID column profile in Power Query. The matching numbers indicate that all rows have been transferred. If the numbers do not match, we must determine which data were not transferred and what caused the problem. Given the sequential nature of the IDs, gap analysis might be a useful tool to do this. In this case, there were no transfer issues when extracting the data from the Beans data set into Power Query.

ILLUSTRATION 5.15 Row Count with Power Query

Column Statistics	...
Count	4,632
Error	0
Empty	0
Distinct	4,632
Unique	4,632
NaN	0
Zero	0
Min	1
Max	4,632
Average	2,316.5
Standard deviation	1,337.28...
Even	2,316
Odd	2,316

Number of Rows

Control Total: Average

Add Missing Rows

If the row counts don't match, add the missing rows to the source data, the Service worksheet in the Beans data set file, or to the ETL's data set.

Pattern 1 Summary

Problem	All the data did not transfer.
Detect (Data Profiling)	Compare row counts.
Correct (ETL)	Add the missing rows.

Data Preparation Pattern 2: Incorrect Data Transfer

Even when all the rows have been transferred, the data might not have been transferred correctly. This is often caused by differences in data types.

Compare Control Amounts

This issue can be detected using control amounts. In the Beans data set, compare the Average in the Excel worksheet (Illustration 5.14) with the same number for the Service table in the ETL tool (Illustration 5.15). Matching numbers indicate a correct transfer of the values for the ID column in the Service table. Similar tests can be run for the other columns.

Modify Incorrect Values

There were no differing control amounts when pulling the data from the Beans Excel spreadsheet into Power Query. But if the numbers do not match, the next step is to determine which data were transferred incorrectly and what caused the problem. Once identified, the incorrectly transferred values in the ETL's data set can be modified.

Pattern 2 Summary

Problem	Data was not correctly transfered.
Detect (Data Profiling)	Compare control amounts.
Correct (ETL)	Modify incorrect values.

Checking the completeness and correctness of data transfers is a common practice for auditors. Clients provide data through emails, cloud sharing, USB transfers, etc. Comparing data statistics generated at the originator and receiver sides helps detect transfer issues.

Apply It 5.3
Extract Data with Patterns

Data Shanice gives you, an accountant working for Stufan, a text file with Stufan's sales transaction information. She also provides the following information:

- Number of transactions: 28
- Average price charged: 25.32

1. Extract the sales data into your ETL tool.

2. Determine if all transactions are transferred and whether all data transferred correctly.

SOLUTION

1. First, extract the data into Excel. Next, open the Power Query editor and examine the profile for the Price column.

Column Statistics	...
Count	28
Error	0
Empty	0
Distinct	8
Unique	4
NaN	0
Zero	0
Min	10
Max	40
Average	25.3214...

2. As the visual shows, the information provided indicates all transactions were transferred–Count: 28– and that they were transferred correctly–Average: 25.32.

5.4 Which Patterns Transform Columns?

LEARNING OBJECTIVE 4

Apply patterns to transform columns.

Once all data are transferred to the ETL tool, it is time to transform them. Transformation has two purposes–cleaning the data by correcting values and restructuring and integrating the data for analytics.

An analytical database consists of an integrated set of tables, which is a data model, and each table has several columns. This means that transformations can be conducted gradually at the column level, table level, and model level. This section focuses on transformation patterns at the column level. These patterns look for data problems in a single column, such as ambiguous names, data type issues, and incorrect, inconsistent, incomplete, or invalid values.

Data Preparation Pattern 3: Irrelevant and Unreliable Data

Data that are irrelevant for decisions bloat the data model. It is also important to avoid integrating unreliable data into the data model (recall the old adage, "garbage in, garbage out"). Keep in mind that excluding data from an analytical database is not the same as deleting the data. The raw data still exist and can be integrated if necessary.

Scan Columns for Irrelevant and Unreliable Data

Irrelevant columns can be identified primarily by scanning the data visually. The data dictionary can also be a helpful tool. For example, the Employee table contains information regarding offices. How would you use the data in this column for decision making?

Scanning data can also determine whether a column contains unreliable data. In **Illustration 5.16** (A), the Age column in the E-Dem table mixes null values, numbers, dates, and text, which would make it challenging to generate reliable insights. Most ETL tools provide statistics about errors, null values, and more, that can help determine a column's reliability. (**Data** See How to 5.1 at the end of the chapter to learn how to use Power Query to profile data.)

How To

Remove Columns With Irrelevant or Unreliable Data

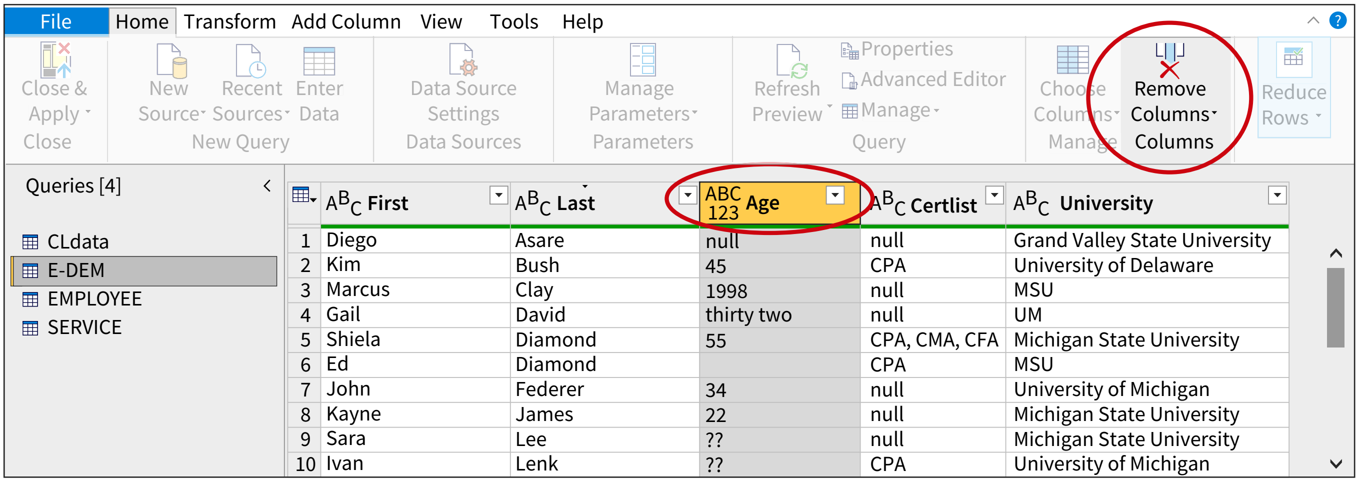
To correct the issue, delete the columns with irrelevant and unreliable data from the analytical database. Illustration 5.16 (B) shows how to do this with Power Query. Select a column, such as Age, and then select Remove Columns. For the Beans data set, the irrelevant column, Office, and the unreliable column, Age, were deleted.

ILLUSTRATION 5.16 Detecting and Correcting Unreliable Data

(A) Column with Unreliable Data

Age
45
1998
thirty two
55
34
22
??
??
51
9/2/1998

(B) Removing a Column with Unreliable Data



	First	Last	Age	Certlist	University
1	Diego	Asare	null	null	Grand Valley State University
2	Kim	Bush	45	CPA	University of Delaware
3	Marcus	Clay	1998	null	MSU
4	Gail	David	thirty two	null	UM
5	Shiela	Diamond	55	CPA, CMA, CFA	Michigan State University
6	Ed	Diamond		CPA	MSU
7	John	Federer	34	null	University of Michigan
8	Kayne	James	22	null	Michigan State University
9	Sara	Lee	??	null	Michigan State University
10	Ivan	Lenk	??	CPA	University of Michigan

Pattern 3 Summary

Problem	Irrelevant data bloat a data set. Unreliable data increases the risk of making bad decisions.
Detect (Data Profiling)	Visually scan columns for irrelevant and unreliable data.
Correct (ETL)	Remove columns with irrelevant or unreliable data.

Data Preparation Pattern 4: Incorrect and Ambiguous Column Names

Column names become variables during data exploration and interpretation. Their names are important because other people might use the analytical database. Essentially, column names become part of the database's vocabulary. Recall that Beans' managing partner wants to perform the necessary analytics but struggles with the data preparation. Column names that are correct, intuitive, and unambiguous make it easier for other people to use a database and explore data.

Scan Columns for Incorrect or Ambiguous Names

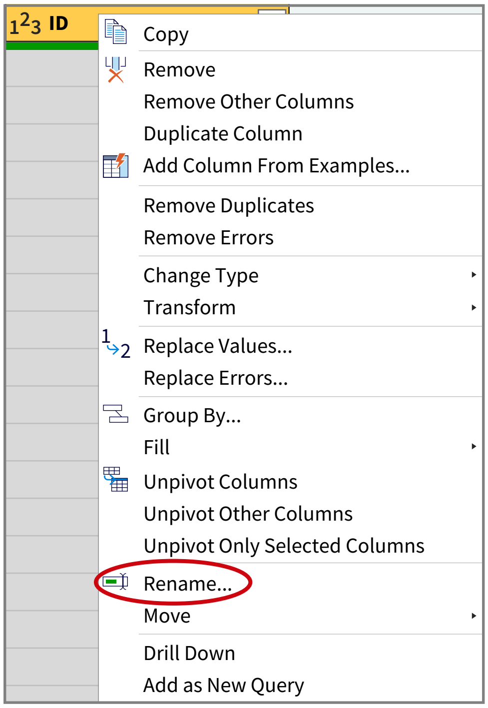
Visually scanning a column's content and its data dictionary definition can reveal whether the column name accurately reflects its content. Here are four rules for naming columns:

1. Names should accurately describe the column's content.
2. They should be intuitive to business people.
3. Use only common abbreviations that are understood by everyone, such as YTD.
4. Eliminate spaces, underscores, or other symbols. For example, use CustomerName instead of Customer_Name.

Rename Columns

ETL tools make it easy to correct this problem by renaming the column. **Illustration 5.17** shows how to rename a column in Power Query. First, right-click on the ID column in the ClData table and then select **Rename** and enter the new name: "ClientID." Note that a column name change propagates automatically to all formulas and reports.

ILLUSTRATION 5.17 Modify Column Names



Not all column names in the Beans data set are intuitive and easy to understand. **Illustration 5.18** shows some potential name changes for columns. Note that, because two columns in the same table cannot have the same name, Power Query automatically changed the second ID column in the Service table to ID-1 when loading the data.

ILLUSTRATION 5.18 Column Name Suggestions for Beans

Table	ColumnName	RevisedColumnName
ClData	ID	ClientID
ClData	Industry Name	IndustryName
Employee	ID	EmployeeID
Employee	Job Title	JobTitle
Employee	MS	MaritalStatus
E-Dem	First	FirstName
E-Dem	Last	LastName
E-Dem	CertList	CertificationList
Service	ID	ServiceID
Service	Actual Time	ActualTime
Service	Budgeted Time	BudgetedTime
Service	ID-1	TaskID
Service	Task	TaskName

Pattern 4 Summary

Problem	Incorrect or ambiguous column names make it harder to understand and work with a data set.
Detect (Data Profiling)	Scan columns for incorrect or ambiguous names.
Correct (ETL)	Rename columns.

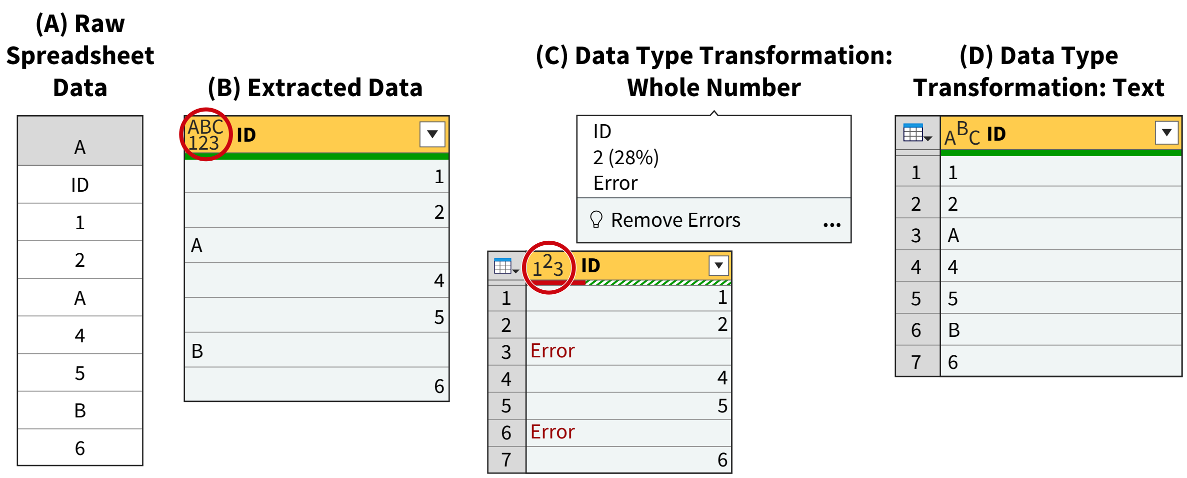
Data Preparation Pattern 5: Incorrect Data Types

Verifying data types is an essential part of data preparation. Data types are an integral part of column definitions because they determine what we can and cannot do with the data in a column. For example, mathematical operators such as SUM and AVERAGE require a numerical field, but time and date functions require a column with a date data type.

Inspect the Data Type

ETL tools automatically assign a data type to each column during extraction, but sometimes either the assignment is incorrect or the ETL tool is unable to determine a data type. Let's illustrate the last scenario with an example. **Illustration 5.19** (A) shows the raw spreadsheet data. Panel (B) shows the same data set after extraction in Power Query. Notice that the data type is $^{\text{ABC}}_{\text{123}}$, also known as the Any data type. It indicates that Power Query cannot identify the data type, which means calculations cannot be performed on the data in the column.

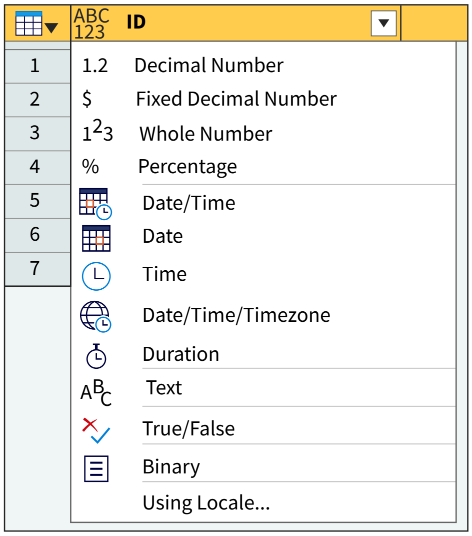
ILLUSTRATION 5.19 Inspecting and Changing Data Types



Change the Data Type

Correct this problem by changing the data type with an ETL tool. **Illustration 5.20** shows the different data types available in Power Query. Illustration 5.19 (C) shows what happens if Whole Number is selected for the data set in panel (B). Notice that some values are converted while other values generate an error. This is helpful because checking for errors should be part of data preparation. The red bar indicates there are errors, and clicking on it reveals the percentage of errors (28%). They can also be filtered out, which is useful for large data sets. Illustration 5.19 (D) shows what happens when the Data Type is converted to Text. The errors disappear, but what can be done with the data is limited.

ILLUSTRATION 5.20 Power Query Data Types



There are no issues with Bean's data types, so no changes must be made. However, make sure to add a column to the data dictionary that specifies the data types.

Pattern 5 Summary

Problem	An incorrect data type limits what can be done with the data in a column.
Detect (Data Profiling)	Inspect the data type.
Correct (ETL)	Change the data type.

Data Preparation Pattern 6: Composite and Multi-Valued Columns

Each cell should contain one value describing one characteristic because two or more values in the same cell makes analysis more challenging. Two specific scenarios that violate the single-valued rule and make analysis more complex are composite columns and multi-valued columns.

Scan for Composite and Multi-Valued Columns

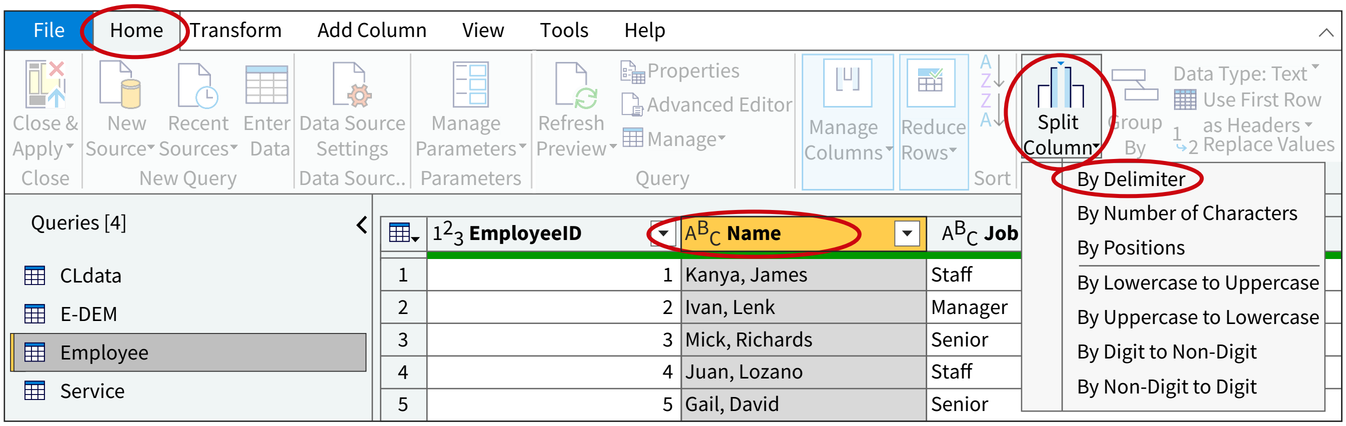
The best method to detect composite or multi-valued columns is visual scanning. Examining the Beans data set this way would reveal:

- The Name column in the Employee table mixes an employee's first and last names. While this might not be an issue from an analytical perspective, it is an issue when matching them with the FirstName and LastName columns in the E-Dem table.
- The CertificationList column in the E-Dem table is multi-valued.

Restructure the Data

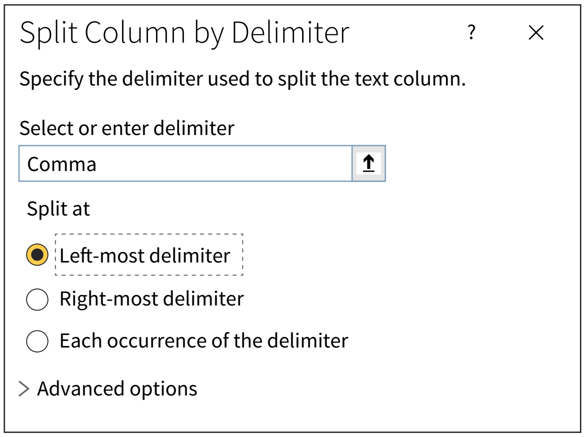
How a column is restructured depends on whether the column is composite or multi-valued. The solution for a composite column is to split it. In Power Query, click on the Name column. Then select the **Home** tab in the Main Menu. Click **Split Column** in the ribbon, and select **By Delimiter** (Illustration 5.21).

ILLUSTRATION 5.21 Power Query | Split Column | Setup



The window shown in Illustration 5.22 will appear. Power Query already selected **Comma** as the delimiter, and since there is only one comma for each name, it does not matter which of the three **Split at** options is selected. In this example the **Left-most delimiter** option was selected. Once split, rename Name.1 as “FirstName” and Name.2 as “LastName.”

ILLUSTRATION 5.22 Power Query | Split Column | Select Delimiter



On the other hand, the CertificationList column’s multi-valued nature in the E-Dem table requires creating a new table. But because the employee data needs additional work, it makes sense to do that later. This example shows the non-sequential nature of applying the patterns–the order in which they are applied will depend on each project. The other columns in the Beans data set are single-valued, and this pattern does not apply to them.

Pattern 6 Summary

Problem	Columns that are not single-valued make analytics difficult.
Detect (Data Profiling)	Examine the values in the columns.
Correct (ETL)	Split combined columns. Create a separate table for multi-valued columns.

Data Preparation Pattern 7: Incorrect Values

At times the wrong value is assigned to one of the entities’ characteristics. The certification for an employee could be recorded as CMA instead of CPA, which are both valid certifications. Incorrect values can have dramatic consequences, including shipping and billing errors.

Detect Incorrect Values with Outliers

Incorrect data are difficult to identify within the content of a single column, so it is helpful to look for outliers, values that stand out in numeric data. If a product with a price of $19 is instead recorded as $91, and most product prices are between $10 and $30, that will stand out. A more formal definition for an **outlier** is a value that falls more than 1.5 times the interquartile range below the first quartile or above the third quartile.

Let's profile the ActualTime column in the Service table. **Illustration 5.23** (A) displays the column profile statistics generated by Power Query. Panel (B) shows the values of the ActualTime in descending order. The top three values in the column (panel (B)) are outliers and most likely incorrect. Also, the actual time for the service with ID "1325", 25 hours, is invalid. It should be 2.5 hours.

(A) ActualTime Column: Statistics

Column Statistics	...
Count	4,632
Error	0
Empty	0
Distinct	190
Unique	48
NaN	0
Zero	0
Min	0.05
Max	25
Average	1.77672...
Standard deviation	2.03039...

(B) ActualTime Column: Values in Descending Order

1²3 ID	1.2 ActualTime
1325	25
3277	22.5
4387	20
2828	11.4
2452	10.75

ILLUSTRATION 5.23 Profiling for Incorrect Data

Modify Incorrect Values

Once a questionable value is identified, there are a few options. The first is identifying the error's root cause and eliminating it. A user could be warned that an outlier is being entered into the system. This would significantly eliminate errors in the raw data. Another option is to correct the value in the source data. Finally, the value could be corrected in the analytical database, but not in the source data.

Illustration 5.24 shows the ActualTime values modified for the Beans example.

ID	IncorrectValue	CorrectValue
1325	25	2.5
3277	22.5	2.25
4387	20	2

ILLUSTRATION 5.24 Correcting the ActualTime Field in the Service Table

Pattern 7 Summary

Problem	Incorrect data might result in poor decision-making.
Detect (Data Profiling)	Detect incorrect values with outliers.
Correct (ETL)	Modify incorrect values.

Data Preparation Pattern 8: Inconsistent Values

The next pattern addresses data inconsistency, which occurs when two or more different representations of the same value are mixed in the same column. For example, determining the total sales amount in dollars for MI customers when both MI and Michigan are used as values could lead to an underestimation, which could in turn result in poor decisions.

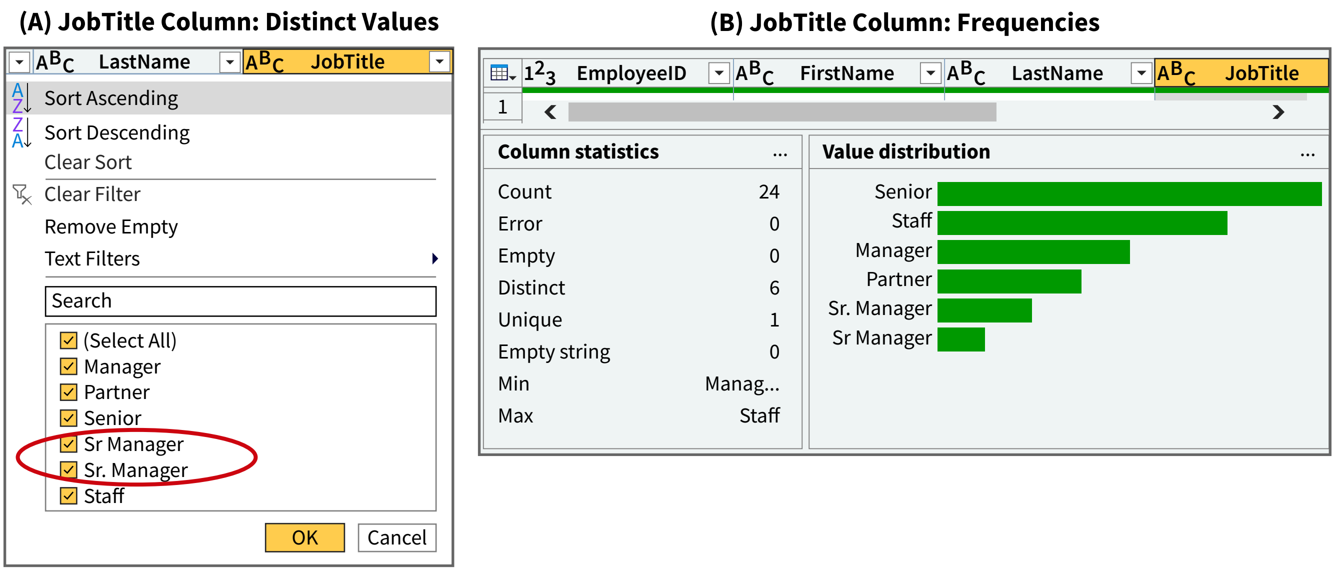
Identify Inconsistent Values

Two profiling techniques are useful for detecting inconsistent values:

- Distinct values: Visually scanning the distinct values of a column is an effective way to identify inconsistent data.
- Frequencies: Values with a low frequency could indicate inconsistent data.

Illustration 5.25 (A) shows the distinct values for the JobTitle column in the Employee table. This information is available for all columns in Power Query. The illustration shows that Sr. Manager is inconsistently represented. Low frequencies might also indicate a misspelling resulting in an inconsistency. As panel (B) shows, the frequency for Sr Manager is 1. The value distribution shown in panel (B) is part of a column's profile in Power Query. Another column with inconsistencies issues in the Beans data set is University.

ILLUSTRATION 5.25 Profiling for Inconsistent Data



Modify Inconsistent Values

Correct the inconsistent data by identifying the root cause and eliminating it or modifying the values in either the source data or the analytical database. When modifying the values, first determine which representation to keep. For the JobTitle column, keep Sr. Manager. The changes shown in **Illustration 5.26** were made to the University column.

ILLUSTRATION 5.26 Changes to Address Inconsistencies in the University Column

Current Value	New Value
MSU	Michigan State University
UM	University of Michigan

Most ETL software, including Power Query, offer find and replace tools to modify values.

Pattern 8 Summary

Problem	Inconsistent data might result in poor decision-making.
Detect (Data Profiling)	Identify inconsistent values.
Correct (ETL)	Modify inconsistent values.

Data Preparation Pattern 9: Incomplete Values

This pattern addresses incompleteness that might make data unusable and unreliable. For example, without customers' addresses, a business cannot send them marketing materials. The different meanings given to null values might also result in unreliable data.

There are a few dimensions of incompleteness worth exploring:

- Should null values be allowed, and if not, are there any?
- If null values are allowed, what percentage of the values are null? If the percentage is high, should the column be loaded?
- How are incomplete values represented: nulls, or a specific code? Are the representations consistent?

Investigate Null Values

ETL tools reveal, on a column-by-column basis, the percentage of null values. Using Power Query, **Illustration 5.27** shows this information for the MaritalStatus column in the Employee table. This information is useful in many ways:

- For each column, decide whether null values are allowed. Primary keys cannot contain null values. Also, the Rate column in the Employee table cannot contain any null values because it is used to determine the fee to be paid by the client. The percentage shows whether this is the case.
- A high or higher than expected percentage might make the column unusable.

It is also important to analyze how incompleteness is represented and how this affects the data's reliability. What does a null value in the MaritalStatus column imply? Is this a different way to state that an employee is single, or do we not know the employee's marital status?

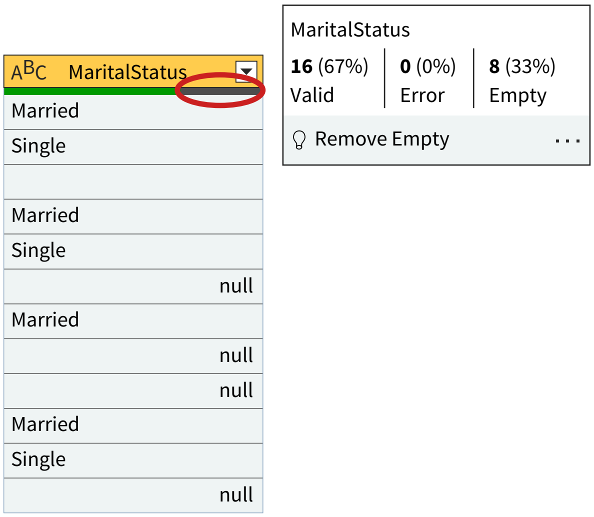


ILLUSTRATION 5.27
Incompleteness Profiling

Remove the Column or Replace the Null Values

The correction scenario depends on the situation. If null values are not allowed but they exist, they should be replaced. If the number of null values is too high to be useful, then remove the column from the analytical database. If there is inconsistency in representing missing values, design a consistent schema and correct the values in terms of that schema. For the MaritalStatus column, values could be Married, Single, and blank. Blank values indicate that the values are unknown.

Pattern 9 Summary

Problem	Incompleteness might make data unusable and unreliable and result in poor decision-making.
Detect (Data Profiling)	Investigate null values.
Correct (ETL)	Remove the column or replace the null values.

Data Preparation Pattern 10: Invalid Values

Domain-specific rules that determine whether data are acceptable can be created for most columns. Data that do not meet these expectations are considered invalid.

Create and Apply Validation Rules

For some validation rules, we can rely on the profiling information automatically generated by the ETL tool. For a mandatory column, which cannot contain null values, the statistics about null values provided by ETL tools can be used for validation. However, in most cases the validation rule must be implemented with a scripting language. **Illustration 5.28** shows a few examples of validation rules that apply to the Beans case. The syntax used in the sample code column is general, as no specific scripting language is used.

ILLUSTRATION 5.28 Design and Implementation of Validation Rules

Description	Sample Code
The actual hours for a service must be positive and can't exceed 14.	ACTUALHOURSVALID = IF SERVICE.ACTUALTIME > 0 AND SERVICEACTUALTIME <= 14, THEN "YES", ELSE "NO"
The minimum employee rate is $150, and the maximum employee rate must be lower than $500.	RATEVALID = IF EMPLOYEE.RATE >= 150 AND EMPLOYEE.RATE < 500, THEN "YES", ELSE "NO"
Valid job titles are: {Manager, Partner, Senior, Sr. Manager, Staff}.	JOBTITLEVALID = IF EMPLOYEE.JOBTITLE IN {"MANAGER", "PARTNER", "SENIOR", "SR. MANAGER", "STAFF"}, THEN "YES", ELSE "NO"

Modify Invalid Values

If a questionable value is identified, eliminate the root cause, change the value in the source, or change the value in the analytical database. In the Beans data set, no invalid data were detected.

Pattern 10 Summary

Problem	Invalid data might result in poor decision-making.
Detect (Data Profiling)	Create and apply validation rules.
Correct (ETL)	Modify invalid values.

Apply It 5.4
Use Column Transformation Patterns

Data One of Shanice's assistants at Stufan prepared four files called Customer, Item, Sale, and Salesperson and gave them to you. These files have several data issues. Identify them using the column transformation patterns in this section.

For each data issue you identify:

1. Describe the issue.
2. Identify the data preparation pattern you applied to detect the issue.
3. Explain how you would correct it.

SOLUTION

Issue 1:

1. The data in the LoyaltyRating column in the Customer table are not useful for analytical purposes.
2. Data Preparation Pattern 3: Irrelevant and Unreliable Data.
3. Remove the LoyaltyRating column from the analytical database.

Issue 2:

1. The Sales table has a column with ambiguous heading-SP.
2. Data Preparation Pattern 4: Incorrect and Ambiguous Column Names.
3. Replace the ambiguous abbreviation SP with SalesPerson.

Issue 3:

1. The State column in the Customer table contains the value DN, which might be incorrect, inconsistent, and/or invalid.
2. Data Preparation Patterns 7, 8, and 10: Incorrect, Inconsistent, and Invalid Values.
3. Replace the DN value with DE.

5.5 Which Patterns Transform Tables?

LEARNING OBJECTIVE 5

Apply patterns to transform tables.

Transformation patterns at the table level look for data issues in a single table, such as ambiguously named tables, missing primary keys, and overlapping columns.

Data Preparation Pattern 11: Non-Intuitive and Ambiguous Table Names

Like column names, table names are part of both the data model and the data set's vocabulary, so they must be correct, intuitive, and clear.

Scan Tables for Incorrect or Ambiguous Names

Examining a table's content and its data dictionary definition can help determine whether the name accurately reflects its content. The rules for naming tables are the same as those for naming columns. They should be intuitive, avoid spaces, underscores, and special coding:

- For example, use CashReceipt instead of Cash Receipt.
- Avoid special coding like DCustomer (the D refers to a dimension table, but not everyone will understand that).

Rename Tables

To correct this issue, rename the table with an ETL tool, as name changes are automatically propagated to all formulas. For the Beans example, make two changes. First, replace ClData with Client and then replace E-Dem with EmployeeDemographics.

Pattern 11 Summary

Problem	Incorrect or ambiguous table names make it harder to understand and work with a data set.
Detect (Data Profiling)	Visually scan tables for incorrect or ambiguous names.
Correct (ETL)	Rename tables.

Data Preparation Pattern 12: Missing Primary Keys

This pattern focuses on primary keys. Tables are descriptions of entities, and each instance of an entity should be uniquely identified. To be a primary key, a column must have a unique value for each instance and no null values. Primary keys are normally already in place when data are extracted from a relational database. However, a primary key will not be in place when data are extracted from a spreadsheet, such as in the Beans case. To establish a primary key, the field must be selected and both rules must be validated.

Identify Tables with a Primary Key Missing

Every table needs at least one column–or combination of columns–that meet the two criteria discussed earlier. Columns that meet both criteria are candidate keys, and ETL tools can help identify them. The column profile provided by Power Query, first shown in Illustration 5.15 and repeated in **Illustration 5.29**, provides the information necessary to identify a candidate key:

- The value for Empty should be zero.
- The values for Count, Distinct, and Unique should be the same. A unique value is a value that occurs only once in the column.

ILLUSTRATION 5.29 Column Profile in Power Query

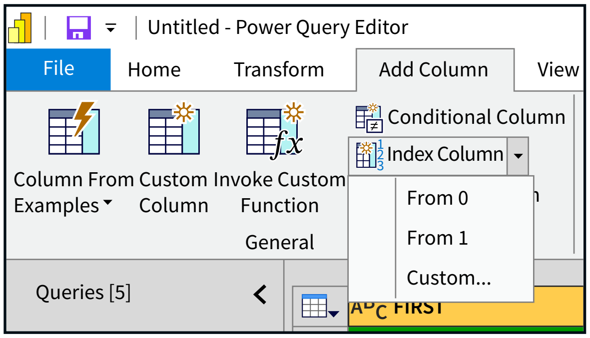
Column Statistics	. . .
Count	4,632
Error	0
Empty	0
Distinct	4,632
Unique	4,632
NaN	0
Zero	0
Min	1
Max	4,632
Average	2,316.5
Standard deviation	1,337.28...
Even	2,316
Odd	2,316

Create a Primary Key

The EmployeeDemographics table does not have a candidate key. We could combine the first and last names into the same field (Name), but names generally should not be used as primary keys because they are rarely unique. For scenarios like this, create an artificial key, such as a number. As **Illustration 5.30** shows, ETL tools can help with that:

- In Power Query, click **Add Column** in the Main Menu.
- Select **Index Column**.
- Click **From 1**.

ILLUSTRATION 5.30 Creating a Primary Key



The situation for the Beans data set is more complicated, so there is an alternative solution, which is presented with Pattern 15.

Pattern 12 Summary

Problem	Some tables do not have a primary key.
Detect (Data Profiling)	Identify tables missing a primary key.
Correct (ETL)	Create a primary key.

Data Preparation Pattern 13: Redundant Content Across Columns

This pattern looks for redundancies that create inconsistencies. Data inconsistencies occur when the same data are recorded more than once and changed in one place but not the other, such as a customer's email address. Here are two scenarios where two or more columns in a table might have the same content:

- When there is overlap such as an address that contains state information and a separate state field.
- When there is dependency, which exists when one column's values are dependent on the values of another column in the same table. Assume both age and date of birth are recorded. However, the values of age change when time passes and the data will become inconsistent. Therefore, instead of transferring age from the data source, it should be calculated as part of the analytical database.

Perform Column-By-Column Comparisons

Performing column-by-column comparisons for overlaps or dependencies would detect this issue. Doing so within the Beans data set would show there is currently no redundant content.

Delete Redundant and Dependent Columns

Columns that contain redundant information can be deleted. When there is dependency, delete the column that contains the dependent value. Instead, use a formula to recreate the column in the analytical database.

Pattern 13 Summary

Problem	Redundant content among columns in a table might result in inconsistencies.
Detect (Data Profiling)	Perform column-by-column comparisons.
Correct (ETL)	Delete redundant and dependent data.

Data Preparation Pattern 14: Find Invalid Values with Intra-Table Rules

Pattern 14 is similar to Pattern 10 in that it also defines acceptable values for a column. However, Pattern 14 determines the validity of a column's values based on the values in one or more other columns in the same table.

Create and Apply Intra-Table Validation Rules

The goal of the validation rule is to identify invalid data. Creating validation rules requires in-depth knowledge of the business, and they are implemented using a scripting language. **Illustration 5.31** shows an intra-table validation rule for the Beans case. The values in the Rate column depend on the values in the JobPosition column. Applying this validation rule to the Bean's data set shows that the rates are too low for five employees–Alex Messi, Thibaut Martens, Paulo Lukaku, Ed Diamond, and Molly McCarthy.

ILLUSTRATION 5.31 Design and Implementation of Intra-Table Validation Rules

DESCRIPTION

The rate of Beans' employees is determined by their job position. The table shows the minimum and maximum rates, in dollars, for each job position.

Position	>=	<
Staff	150	200
Senior	200	250
Manager	250	300
Sr. Manager	300	350
Partner	350	500

SAMPLE CODE

```
RATEVALIDBASEDONJOBTITLE =
IF EMPLOYEE.JOBTITLE = “Staff” AND (EMPLOYEE.RATE >= 150 AND EMPLOYEE.RATE < 200), THEN “YES”, ELSE,
IF EMPLOYEE.JOBTITLE = “Senior” AND (EMPLOYEE.RATE >= 200 AND EMPLOYEE.RATE < 250), “YES”, ELSE,
IF EMPLOYEE.JOBTITLE = “Manager” AND (EMPLOYEE.RATE >= 250 AND EMPLOYEE.RATE < 300), “YES”, ELSE,
IF EMPLOYEE.JOBTITLE = “Sr. Manager” AND (EMPLOYEE.RATE >= EMPLOYEE.RATE < 350), “YES”, ELSE,
IF EMPLOYEE.JOBTITLE = “Partner” AND (EMPLOYEE.RATE >= 350 AND EMPLOYEE.RATE < 500), “YES”, ELSE,
“NO”
)
```

Modify Invalid Values

At Beans, modifying employee rates or the rate policy requires approval from the CEO, so there can be no rate changes until approval is granted.

Pattern 14 Summary

Problem	Invalid data might result in poor decision-making.
Detect (Data Profiling)	Create and apply intra-table validation rules.
Correct (ETL)	Modify the invalid values.

Apply It 5.5
Transform Tables with Patterns

At Stufan, CEO Shanice wonders whether it would be possible to integrate sales and purchases information for analysis. Purchase information is currently recorded in an Excel worksheet titled PTS, a sample of which is shown.

	A	B	C	D	E
1	Invoice	Item	Price	Quantity	Vendor
2	1	DOCL	20	10	DITV
3	1	DOCS	15	10	DITV
4	2	TINL	10	10	BIC
5	2	SNOWL	7	10	BIC
6	3	DUCH	20	10	STWS
7	4	GRUML	20	10	DELT
8	4	SNOWL	25	10	DELT
9	5	TINL	9	10	STIMP

PTS

Use the table transformation patterns to identify two data issues in the PTS file.

1. First, describe the data issue.
2. Next, identify the data exploration pattern you could apply to detect it.
3. Finally, explain how you would correct the problem.

SOLUTION

Issue 1:

1. The current name of the worksheet/table is ambiguous. PTS stands for PurchaseTransactions, but that is not clear from the title.
2. Use Data Exploration Pattern 11: Non-Intuitive and Ambiguous Table Names.
3. Rename the worksheet "Purchases."

Issue 2:

1. The table currently does not have a primary key, and none of the columns contain unique values. Each row in the table represents an invoice line. The first two rows in the table represent two line items for invoice 1, one for the item DOCL and another for the item DOCS.
2. Use Data Exploration Pattern 12: Missing Primary Keys.
3. Create an artificial key consisting of unique sequential numbers.

5.6 Which Patterns Transform Models?

LEARNING OBJECTIVE 6

Apply patterns to transform models.

Transformation patterns at the model level search for data issues across tables, such as data that describe the same entity spread across multiple tables, data models with a structure that is difficult to understand, and data models that do not support efficient processing.

Data Preparation Pattern 15: Data Spread Across Tables

Analysis becomes more challenging when data that describe the same entity are spread across multiple tables. **Illustration 5.32** shows two possible scenarios.

1. In panel (A) both tables, JanuarySales and FebruarySales, have the same structure but different rows. It is the sales table shown in panel (C) split horizontally. Questions about total sales amounts are easier to answer if all data are in one table.
2. In panel (B) the two tables describe different characteristics of the same entity–Product. Some information for the product with ID 1 is in the ProductDescriptions table. Other information for the same product, ID = 1, is in the ProductAccounting table. In this case, it is the product table shown in panel (D) split vertically.

ILLUSTRATION 5.32 Combining Tables

(A) Monthly Sales Data Across Tables

JanuarySales

Number	Date	Amount
189	1/10/2025	$ 17,450
190	1/15/2025	$ 23,890
191	1/24/2025	$ 19,001

FebruarySales

Number	Date	Amount
192	2/9/2025	$ 25,451
193	2/10/2025	$ 34,881
194	2/14/2025	$ 7,282
195	2/23/2025	$ 13,209

(B) Product Information Across Tables

ProductDescriptions

ID	Type	Category
1	Stuffed Animals	Toys
2	Radio	Electronics
3	Soccer Balls	Toys

ProductAccounting

ID	QOH
1	84
2	133
3	354

(C) Monthly Sales Data Union

Sales

Number	Date	Amount
189	1/10/2025	$ 17,450
190	1/15/2025	$ 23,890
191	1/24/2025	$ 19,001
192	2/9/2025	$ 25,451
193	2/10/2025	$ 34,881
194	2/14/2025	$ 7,282
195	2/23/2025	$ 13,209

(D) Product Information Merge

Product

ID	Type	Category	QOH
1	Stuffed Animals	Toys	84
2	Radio	Electronic	133
3	Soccer Balls	Toys	354

Identify Similarly Structured Tables/Tables Describing Different Characteristics of the Same Entity

To identify similiarly structured tables, look for two or more tables with the same structure. These tables would have the same columns and similar data. Another option is to search for tables that describe different characteristics of the same entity. In the Beans case, Employee and EmployeeDemographics are such tables. The goal is to create a single table with all the employee information.

Combine Tables

Recall from earlier in the chapter that combining two tables with a similar structure is called a union. In Illustration 5.32, the table in panel (C) combines the two tables in panel (A). Combining two tables with different characteristics for the same entity is a merge or a join. The table in panel (D) combines the two tables in panel (B). For the Beans case, both the Employee and EmployeeDemographics tables contain employee information, so they should be merged into one table (**Data** See How To 5.2 to learn how to merge these tables with Power Query.).

How To

Illustration 5.33 shows the structure of the combined table, which is named Employee.

ILLUSTRATION 5.33
Combined Employee Table

Employee
EmployeeID
FirstName
LastName
JobTitle
Rate
CertificationList
MaritalStatus
University

Pattern 15 Summary

Problem	An entity's data are spread across multiple tables, which complicates analysis.
Detect (Data Profiling)	Identify tables with a similar structure or tables that describe different characteristics of the same entity.
Correct (ETL)	Union or merge tables.

Data Preparation Pattern 16: Data Models Do Not Comply with Principles of Dimensional Modeling

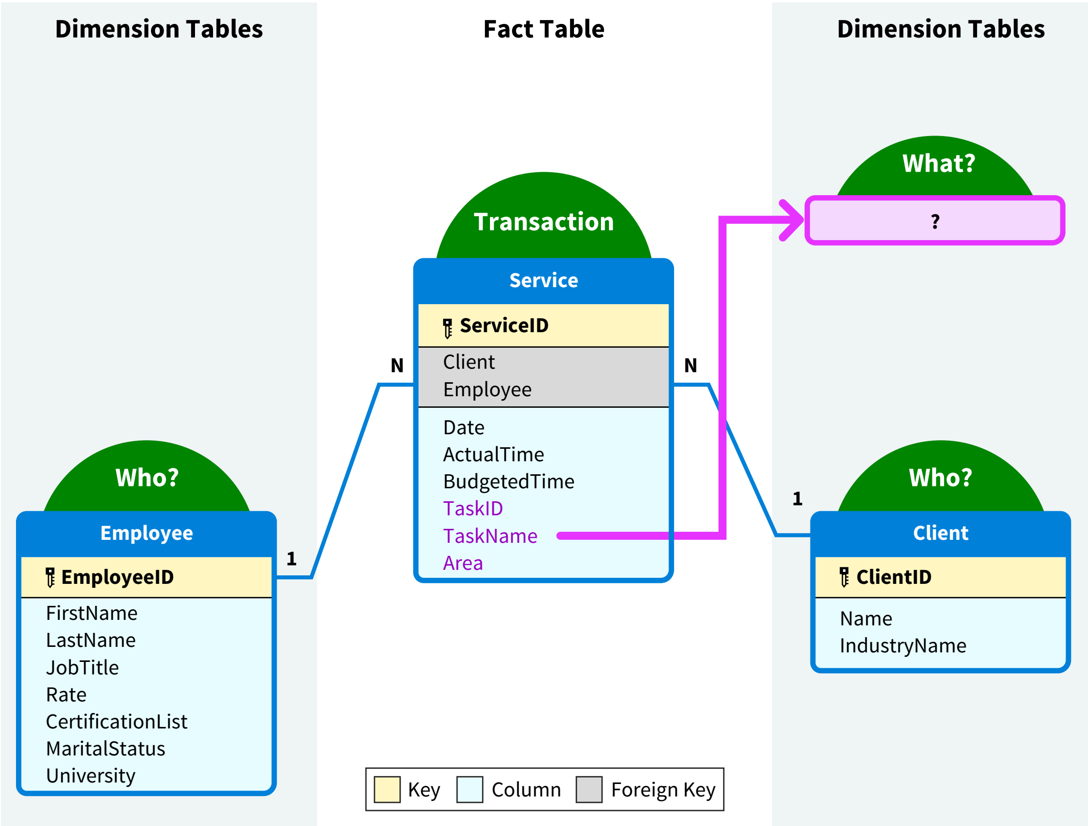
Dimensional modeling is the technique of creating data models with fact tables surrounded by dimension tables. These data models, such as star schemas, are easy to understand and result in efficient data processing.

Analyze a Data Model's Compliance with Dimensional Modeling Principles

Use these dimensional modeling principles by determining the fact and dimension tables and ensuring all fields belong to the correct table. In an accounting context, fact tables correspond to business transactions. Dimension tables, on the other hand, describe *who* participates in the transactions, *when* the transactions occurred, and *what* was given up or acquired. (The chapter on information modeling discusses who, what, and when analysis of accounting transactions in more detail.)

Illustration 5.34 structures the current analytical database as a star schema. The Service table is the fact table, the Employee table is a *who* dimension, and the Client table is also a *who* dimension.

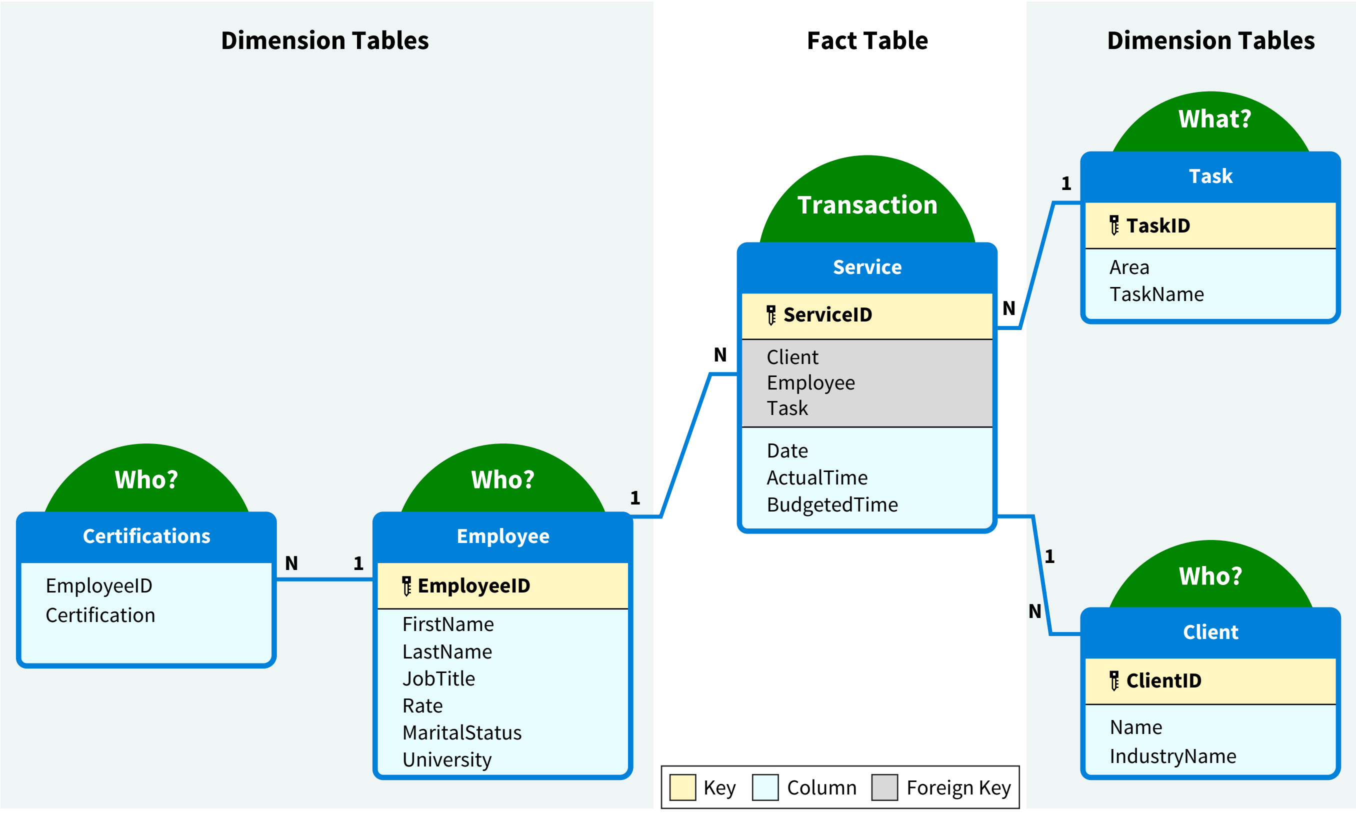
ILLUSTRATION 5.34 Current Beans Star Schema



Note that the *when* and *what* dimensions seem to be missing. There is only the Date field in the Service table, and we opted not to specify a separate *when* dimension. However, TaskId, TaskName, and Area are all *what* descriptions–What does Beans sell to their clients? Therefore, a new dimension table named Task can be created. Further, recall from Pattern 6 that the CertificationList column in the Employee table is a multi-valued column. Creating a new table transforms it into a single-valued column. **Illustration 5.35** shows the data model we would like to create.

Creating a separate table for Certifications, results in a snowflake schema. In a **snowflake schema**, the information for a dimension is spread across multiple tables. The schema in Illustration 5.35 can be created using Power Query as the ETL tool.

ILLUSTRATION 5.35 An Ideal Beans Star/Snowflake Schema

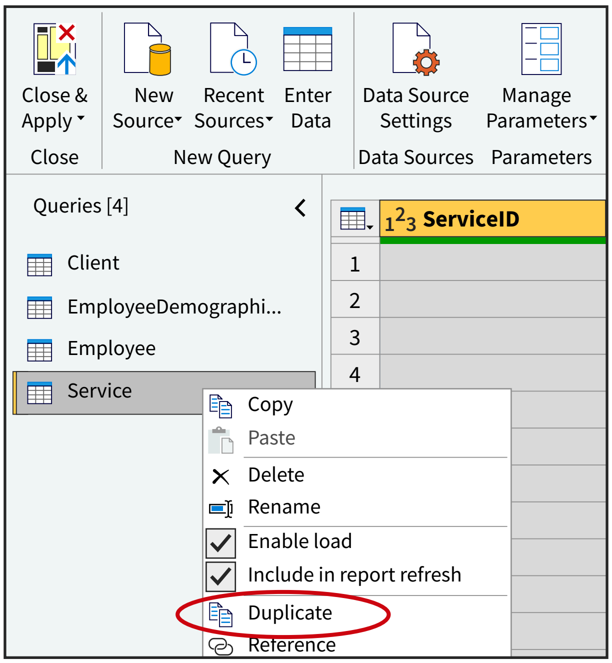


Reconfigure the Data Model as a Star/Snowflake Schema

First, create a Task Dimension table that captures the descriptions for services.

Step 1: Duplicate the Service table and rename the duplication "Task" (**Illustration 5.36**). For both the Service and Task tables, keep only the columns that are shown in Illustration 5.35.

ILLUSTRATION 5.36 Duplicate Service Table



Be sure to retain the TaskID field in the Service table and to rename it "Task." The Service table has now been split into two tables. Service is a fact table, and Task is a dimension table.

Step 2: Delete the duplicate rows in the Task table by selecting the **Home** tab in Power Query's main menu. Select **Remove Rows**, then select **Remove Duplicates**.

The Task table should now mirror the one shown in **Illustration 5.37**.

ILLUSTRATION 5.37 Task Dimension Table

	TaskID	Area	TaskName
1	1	Accounting	Prep
2	2	Accounting	Review
3	3	Accounting	Admin
4	4	Accounting	Other
5	5	Tax	Prep
6	6	Tax	Review
7	7	Tax	Admin
8	8	Tax	Other

Second, address the problem of CertificationList, which is a multi-valued column in the Employee table:

Step 1: Use the same process as before to create the CertificationList table:

- Duplicate the Employee table and name the duplication "Certifications."
- Remove the CertificationList column from the Employee table.
- In the new Certifications table, keep only the EmployeeID and the CertificationsList columns.

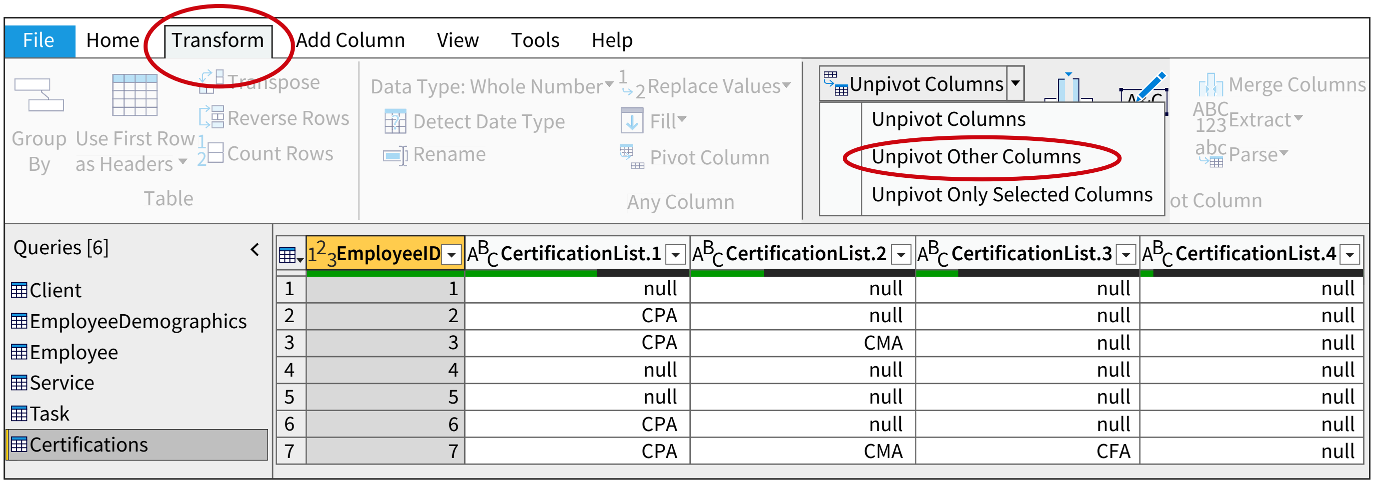
Step 2: Transform the CertificationList from a multi-valued column into a single-valued column:

- In Power Query, split the CertificationList column and select the **Each Occurrence of the delimeter** option.
- Click on the EmployeeID column and select **Transform** in the main menu.
- Finally, select **Unpivot Columns** in the ribbon, and choose **Unpivot other Columns** (**Illustration 5.38**).

Step 3: In the resulting table, delete the Attribute column and change the Value column's name to "Certificate." An employees's certifications are now recorded in a single-valued column.

This completes the transformation of the Bean's data set into a star/snowflake schema.

ILLUSTRATION 5.38 Unpivoting Columns



Pattern 16 Summary

Problem	Data models that are not structured following the principles of dimensional modeling are typically more difficult to analyze.
Detect (Data Profiling)	Analyze a data model's compliance with dimensional modeling principles.
Correct (ETL)	Reconfigure the data model as a star/snowflake schema.

Data Exploration Pattern 17: Find Invalid Values with Inter-Table Rules

Patterns 10, 14, and 17 are similar because they define the acceptable values for a column. However, Pattern 17 determines the validity of a column's values based on the values in one or more other tables. An example of a widely used inter-table validation rule is **referential integrity**, which refers to the fact that all values in a foreign key should also exist as values in the corresponding primary key. (**Data** How To 5.3 at the end of the chapter explains how to implement referential integrity with Microsoft Access.)

How To

Create and Apply Inter-Table Validation Rules

Inter-table validation rules identify invalid data. Their creation requires in-depth knowledge of the business. **Illustration 5.39** shows an example of an inter-table validation rule applied to the Beans case. It is part of the Service table, and it specifies that only managers, sr. managers, and partners can review an engagement. The rule has detected that service 3971 was reviewed by a senior, and a staff employee reviewed service 4193.

ILLUSTRATION 5.39 Design and Implementation of an Inter-Table Validation Rule

Description	Sample Code
Only managers, sr. managers, and partners can review an engagement.	NOAUTHORITY = IF TASK.TASKNAME = “REVIEW” AND EMPLOYEE[JOBTITLE] IN {“MANAGER”, “SR. MANAGER”, “PARTNER”}, THEN “OK”, ELSE “ISSUE”

Modify Invalid Values

If you were working on the Beans case, you would consult with the CEO to find out if a policy violation, incorrect entry of the data, or another reason caused the error. Once identified, the issue can be corrected by putting a control in place to avoid further policy violations, correcting the values in the source data, or correcting the values in the analytical database.

Pattern 17 Summary

Problem	Invalid data might result in poor decision-making.
Detect (Data Profiling)	Create and apply inter-table validation rules.
Correct (ETL)	Modify the invalid rules.

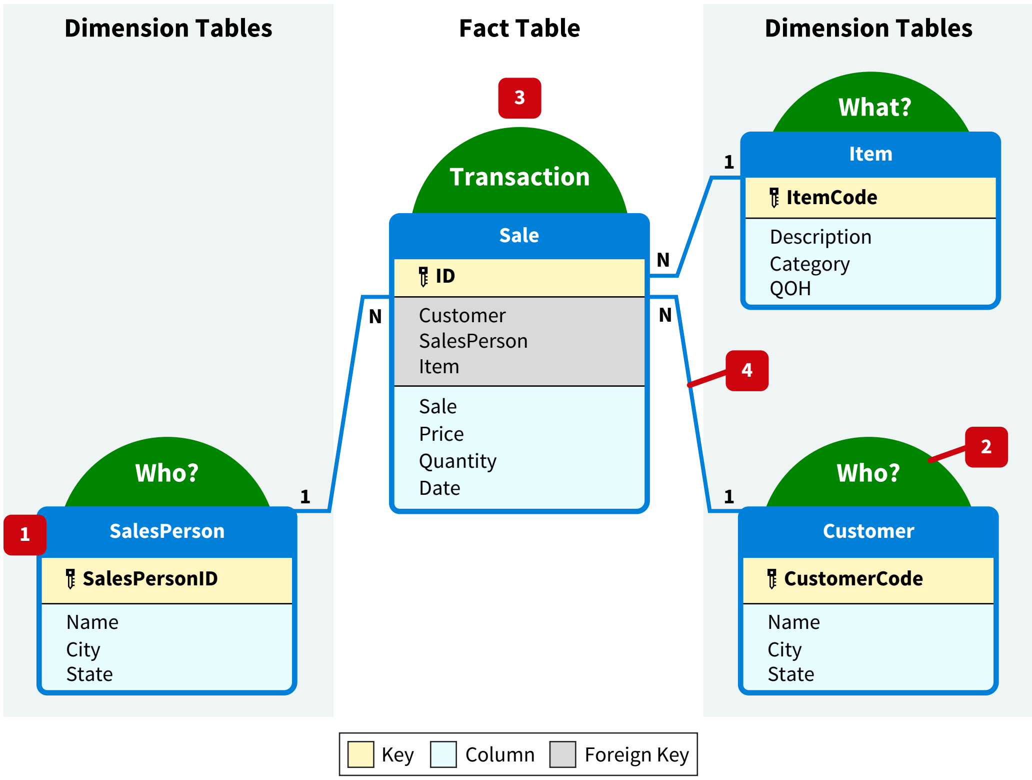
Apply It 5.6

Draw a Star Schema

Data Shanice's assistant at Stufan provides you with revised versions of the Customer, Item, Sale, and Salesperson files. Use the data in these files to draw a star schema.

1. Draw the different tables and their fields.
2. Label each table as fact or dimension and label each dimension table as who, when, or what.
3. Put the fact table in the center.
4. Complete the star schema by connecting the tables and defining their cardinalities.

SOLUTION



5.7 Which Patterns Apply to Data Loading?

LEARNING OBJECTIVE 7

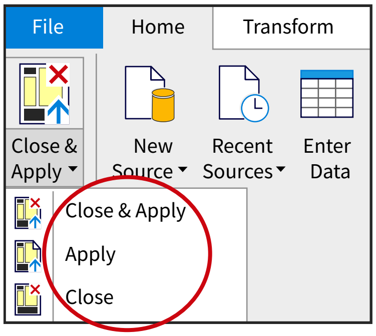
Apply patterns to data loading issues.

Once the data are cleaned and transformed, it is time to load them into the software for analysis. Data loading is the process of making the analytical database available for use. Since both extraction and loading are transfer processes, they have similar issues when it comes to the completeness and correctness of transferred data. It is also important that the data model of the analytical database is validated–that is, that all relationships have been defined.

Data Preparation Pattern 18: Incomplete Data Loading

Loading moves the data from the ETL tool to the analytical database. **Illustration 5.40** shows how to transfer data from the ETL tool, in this case Power Query, to the analytical database.

ILLUSTRATION 5.40 ETL-Analytical Database Transfer



There are three options:

- Close and apply: Close Power Query and apply all transformations to the analytical database.
- Apply: Apply all the transformations to the analytical database, but keep Power Query open.
- Close: Close Power Query without applying any transformations to the analytical database.

If we choose one of the first two options, then it is important to confirm that all data were transferred.

Compare Row Counts

Like Pattern 1, the row count for the analytical database can be compared with the row count of the data set in the ETL tool. The ETL tool will also issue an alert if any errors occurred when the transformations are applied to the analytical database.

Add Missing Rows

If the numbers do not match, determine which rows were not transferred and why. Once identified, add the missing rows to the analytical database.

Pattern 18 Summary

Problem	All the data did not transfer during loading.
Detect (Data Profiling)	Compare row counts.
Correct (ETL)	Add the missing data.

Data Preparation Pattern 19: Incorrect Data Loading

Even when all the rows have been transferred, the data might not have been transferred correctly. Pattern 19 addresses this potential issue.

Compare Control Amounts

Like Pattern 2, an effective way to validate the correct transfer of data is by comparing sums, averages, or any other control amounts.

Modify Incorrect Values

If the numbers do not match, determine which data were transferred incorrectly and what caused the problem. Once identified, modify the incorrectly transferred values in the analytical database.

Pattern 19 Summary

Problem	The correct data did not transfer during loading.
Detect (Data Profiling)	Compare control amounts.
Correct (ETL)	Modify incorrect values.

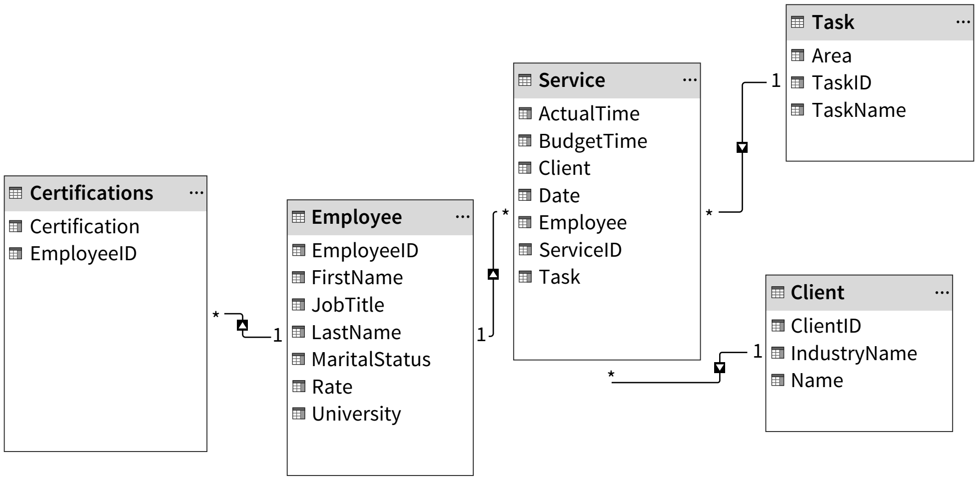
Data Preparation Pattern 20: Missing or Incorrect Data Relationships

Analytics rely heavily on the underlying data model. Missing or incorrectly defined data relationships make analytics challenging or even impossible, so the completeness and accuracy of the data model must be validated after loading.

Investigate the Completeness and Accuracy of the Data Model

A complete and accurate data model is one in which all relationships are correct. **Illustration 5.41** shows the finalized data model for the Beans analytical database created with Power Query. Compare this with your data model, and determine that no relationships are missing, there are no unnecessary relationships, and that all relationships are defined correctly.

ILLUSTRATION 5.41 Beans Data Model



Modify the Data Model

To do this in Power BI, select the **Home** tab in the Main Menu and click **Manage Relationships** in the ribbon. The window shown in Illustration 5.42 will appear. Select the buttons at the bottom of the window to create, edit, or delete a relationship.

ILLUSTRATION 5.42 Manage Relationships

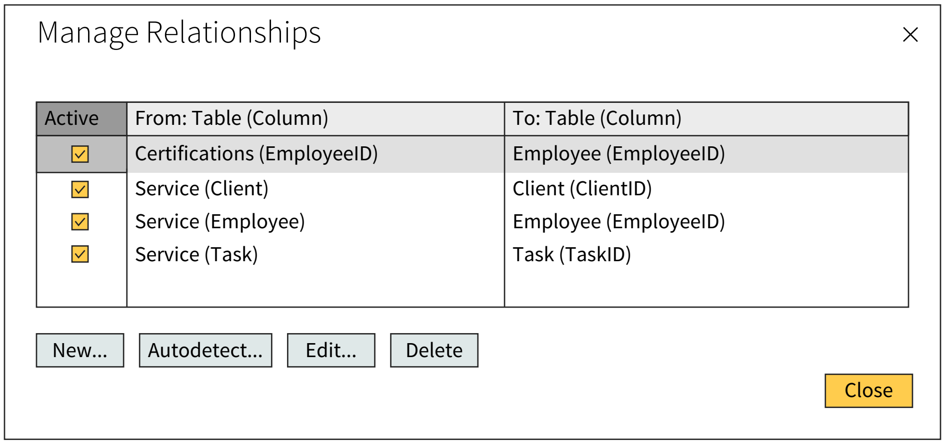
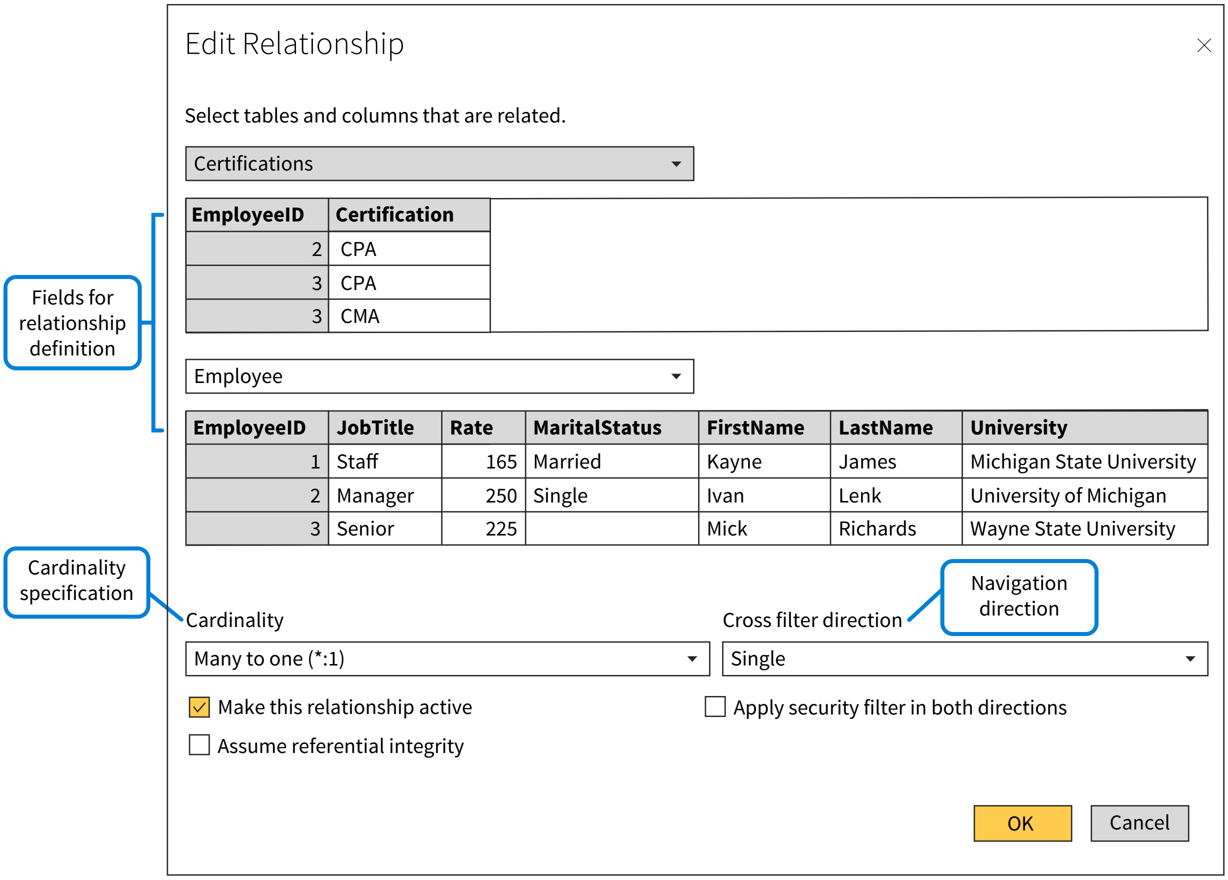


Illustration 5.43 shows some of the aspects of a relationship that can be defined:

- The data fields between which the relationship is specified.
- The cardinalities that apply to the relationship.
- The navigation direction used to aggregate data.

ILLUSTRATION 5.43 Define Relationships



Pattern 20 Summary

Problem	Missing or incorrectly defined data relationships makes analysis challenging or even impossible.
Detect (Data Profiling)	Investigate the completeness and accuracy of the data model.
Correct (ETL)	Modify the data model.

The transformation of the Beans data set and its preparation for analytics is now complete.

Illustration 5.44 shows the revised data dictionary for the analytical database in Illustration 5.41.

ILLUSTRATION 5.44 Beans Revised Data Dictionary

Service

Name	Description	Data Type	Key	Mandatory
Client	A client's unique ID.	Integer	Foreign	Yes
Employee	An employee's unique ID.	Integer	Foreign	Yes
Task	A task's unique ID.	Integer	Foreign	Yes
ServiceID	A service's unique ID.	Integer	Primary	Yes
Date	The date the service occurred.	Date		Yes
ActualTime	The actual time spent on the service provided.	Decimal		Yes
BudgetedTime	The budgeted time for the service provided.	Decimal		Yes

Task

Name	Description	Data Type	Key	Mandatory
TaskID	A task's unique ID.	Integer	Primary	Yes
TaskName	A task's name.	Text		Yes
Area	A task's area.	Text		Yes

Client

Name	Description	Data Type	Key	Mandatory
ClientID	A client's unique ID.	Integer	Primary	Yes
Name	A client's name.	Text		Yes
IndustryName	A client's industry.	Text		Yes

Employee

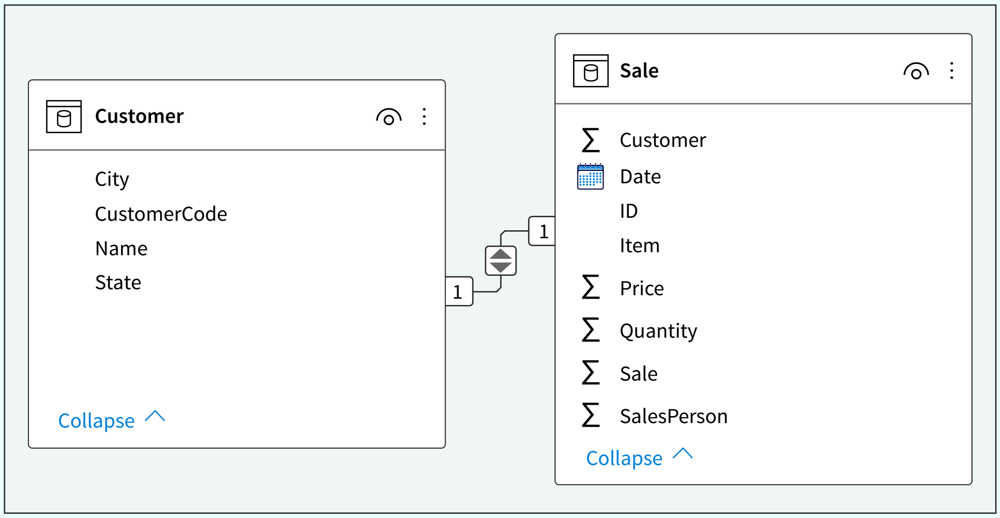
Name	Description	Data Type	Key	Mandatory
EmployeeID	An employee's unique ID.	Integer	Primary	Yes
FirstName	An employee's first name.	Text		Yes
LastName	An employee's last name.	Text		Yes
JobTitle	An employee's job title.	Text		Yes
Rate	The hourly rate charged for the employee.	Integer		Yes
MaritalStatus	An employee's marital status.	Text		No
University	The university from which the employee received an undergraduate degree.	Text		No

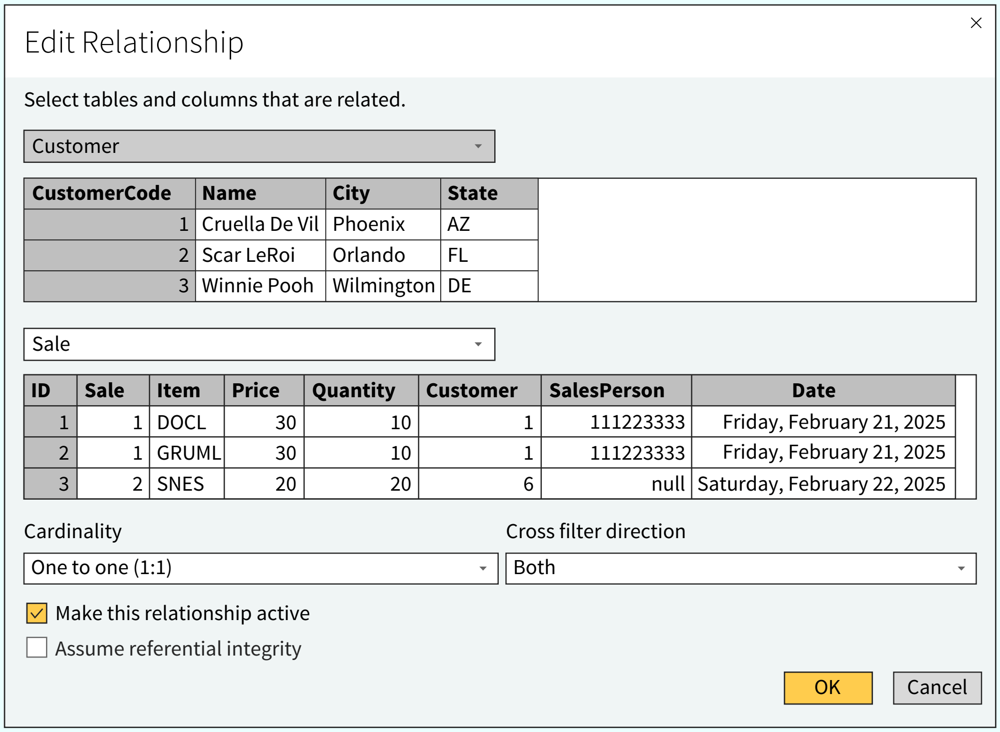
Certifications

Name	Description	Data Type	Key	Mandatory
EmployeeID	An employee's unique ID.	Integer	Foreign	Yes
Certification	A certification earned by the employee.	Text		Yes

Apply It 5.7
Evaluate Relationships Between Tables

The following two illustrations show the current relationship between the Sale and Customer tables for the Stufan data set. There are two issues that would dramatically distort any analytics related to this relationship. Identify them and describe why they are problems.





Edit Relationship

Select tables and columns that are related.

Customer

CustomerCode	Name	City	State
1	Cruella De Vil	Phoenix	AZ
2	Scar LeRoi	Orlando	FL
3	Winnie Pooh	Wilmington	DE

Sale

ID	Sale	Item	Price	Quantity	Customer	SalesPerson	Date
1	1	DOCL	30	10	1	111223333	Friday, February 21, 2025
2	1	GRUML	30	10	1	111223333	Friday, February 21, 2025
3	2	SNES	20	20	6	null	Saturday, February 22, 2025

Cardinality: One to one (1:1)

Cross filter direction: Both

☑ Make this relationship active

☐ Assume referential integrity

OK Cancel

SOLUTION

1. The relationship is defined between the wrong fields: CustomerCode and ID. ID is a sequential number assigned to each row in the Sale table. As a result, the wrong customers are being assigned to the sales. The relationship should have been designated between the CustomerCode field in the Customer table and the Customer field in the Sale table.
2. A 1-N cardinality pattern is expected for a relationship between a dimension table (customer) and a fact table (sale). Customers can make more than one sale, but only one customer is specified for each sale.

Chapter Review and Practice

Learning Objectives Review

1 Explain the process of data profiling.

Data profiling is the process of investigating data quality and structure. It has three parts:

- Investigating data quality: Identify if there are anomalies in the data. Stated differently, are the data dirty?
- Investigating data structure: Improve the organization of the data for the purpose of analysis.
- Deciding and informing: Decide whether to address the issues identified, the cost of doing so, and the consequences of not addressing the issues.

2 Describe the extract-transform-load (ETL) process.

Extract, transform, load (ETL) is a process that corrects data issues:

- Extraction is moving the data to a staging area for transformation purposes.
- Transformation involves improving the data during three subprocesses: cleaning, restructuring, and integration.
- Loading occurs when the data are moved to the area where they will be used for analysis.

3 Apply patterns to extract data.

Data preparation patterns are a robust tool for data preparation projects. They help identify data issues and provide guidance for detecting and correcting them. Data extraction patterns address:

- The incomplete transfer of data.
- The incorrect transfer of data.

4 Apply patterns to transform columns.

Transformation cleans the data by correcting values (quality) and restructuring and integrating the data structures for analytics (structure). It can be gradually conducted at the column, table, and model level.

Column transformation patterns deal with data issues in a single column. Structure-oriented column patterns address:

- Columns that contain irrelevant and unreliable data.
- Incorrect and ambiguous column names.
- Incorrect data types.
- Combined or multi-valued columns.

Quality-oriented column patterns address:

- Columns with incorrect data.
- Inconsistent data in columns.
- Columns with incomplete data.
- Columns with invalid data.

5 Apply patterns to transform tables.

Table transformation patterns deal with data problems in a single table. Structure-oriented table patterns address:

- Non-intuitive and ambiguous table names.
- Tables without a primary key.
- Tables with two or more columns with overlapping content.

Quality-oriented table patterns address:

- Invalid values that can be determined by intra-table rules.

6 Apply patterns to transform models.

Model transformation patterns deal with data issues across tables. Structure-oriented model patterns address:

- Combining tables with a similar structure with a union. Combining tables tables with different characteristics for the same entity with a merge.
- Compliance with dimensional modeling principles.

Quality-oriented table patterns address:

- Invalid values that can be determined by inter-table rules.

7 Apply patterns to data loading issues.

Extracting and loading are both transfer processes and have therefore the same data issues:

- The incomplete transfer of data.
- The incorrect transfer of data.

In addition, once the data have been loaded, it is important to:

- Investigate that the data model is complete and accurate.

Illustration 5.45 provides an overview of the twenty data preparation patterns. The first column shows an identification number for easy reference, and the code in the second column indicates whether a pattern focuses on values (V) or restructuring the data (S). The chart also includes the data issue and how to detect and correct it.

ILLUSTRATION 5.45 Overview of Data Preparation Patterns

Data Preparation Patterns

Extraction				
Id	**Code**	**Issue**	**Detect (Data Profiling)**	**Correct (ETL)**
1	V	All the data did not transfer.	Compare row counts.	Add missing rows.
2	V	Data was not correctly transferred.	Compare control amounts.	Modify incorrect values.
Transformation				
Id	**Code**	**Issue**	**Detect (Data Profiling)**	**Correct (ETL)**
Column				
3	S	Irrelevant or unreliable data.	Visually scan columns for irrelevant and unreliable data.	Remove columns with irrelevant and unreliable data.
4	S	Incorrect or ambiguous data.	Scan column for incorrect or ambiguous names.	Rename column.
5	S	Incorrect data types.	Inspect the data type.	Change data type.
6	S	Columns are not single-valued.	Examine the values in the columns.	Split composite columns. Create a separate table for multi-valued columns.
7	V	Incorrect values.	Detect incorrect values with outliers.	Modify incorrect values.
8	V	Inconsistent values.	Identify inconsistent values.	Modify inconsistent values.
9	V	Incomplete values.	Investigate null values.	Remove the columns or replace the null values.
10	V	Invalid values.	Create and apply validation rules.	Modify invalid values.
Table				
11	S	Incorrect or ambiguous table names.	Visually scan the table for incorrect or ambiguous names.	Rename tables.
12	S	Missing primary keys.	Identify tables missing a primary key.	Create a primary key.
13	S	Redundant column content.	Perform column-by-column comparisons.	Delete redundant and dependent data.
14	V	Invalid values detected with intra-table rules.	Create and apply intra-table validation rules.	Modify the invalid data.
Model				
15	S	Data spread across tables.	Identify tables with a similiar structure or tables that describe different characteristics of the same entity.	Union or merge tables.
16	S	Compliance with dimensional modeling principles.	Analyze a data model's compliance with dimensional modeling principles.	Restructure model as a star/snowflake schema.
17	V	Invalid values detected with inter-table rules.	Create and apply inter-table validation rules.	Modify the invalid rules.
Loading				
Id	**Code**	**Issue**	**Detect (Data Profiling)**	**Correct (ETL)**
18	V	Incomplete data loading.	Compare row counts.	Add the missing data.
19	V	Incorrect data loading.	Compare control amounts.	Modify incorrect values.
20	S	Missing or incorrect data relationships.	Investigate the completeness and accuracy of the data model.	Create, modify, and delete relationships Modify the data model.

Key Terms Review

Analytical database 5-5
Cardinality 5-8
Composite column 5-5
Data connectors 5-10
Data dictionary 5-15
Data extraction 5-10
Data integration 5-13
Data loading 5-14
Data matching 5-13
Data model 5-7
Data preparation 5-2
Data profiling 5-2
Data restructuring 5-13
Data transformation 5-10
Data warehouse 5-10
Dimensions 5-8
Dirty data 5-2
Extract-transform-load (ETL) 5-10
Facts 5-7
Flat tables 5-6
Merge 5-13
Multi-valued column 5-6
Outlier 5-25
Referential integrity 5-39
Single-valued column 5-5
Slicing 5-5
Snowflake schema 5-36
Star schema 5-7
Union 5-13
Validation rules 5-3

How To Walk-Throughs

How To

HOW TO 5.1 Profile Data With Power Query

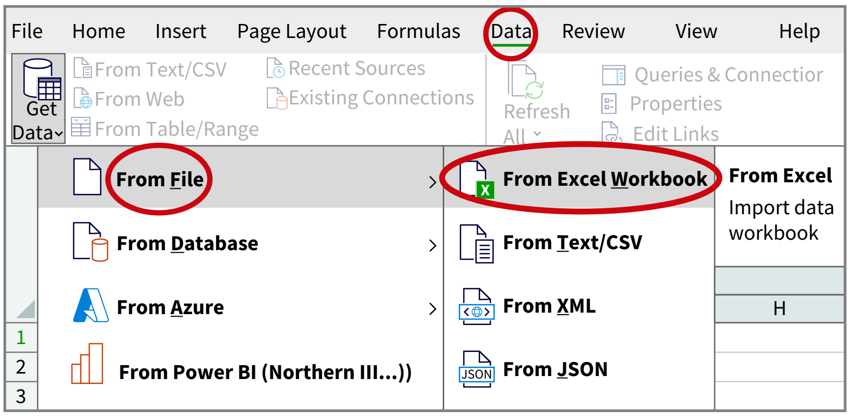
Power Query integrates excellent data profiling tools. Let's examine how it can help with row counts, control amounts, and distribution information.

What You Need: **Data** The How To 5.1 data file.

STEP 1: Extract the data from the data file. There are different steps for extracting data from Excel versus Power BI. If you are using Excel:

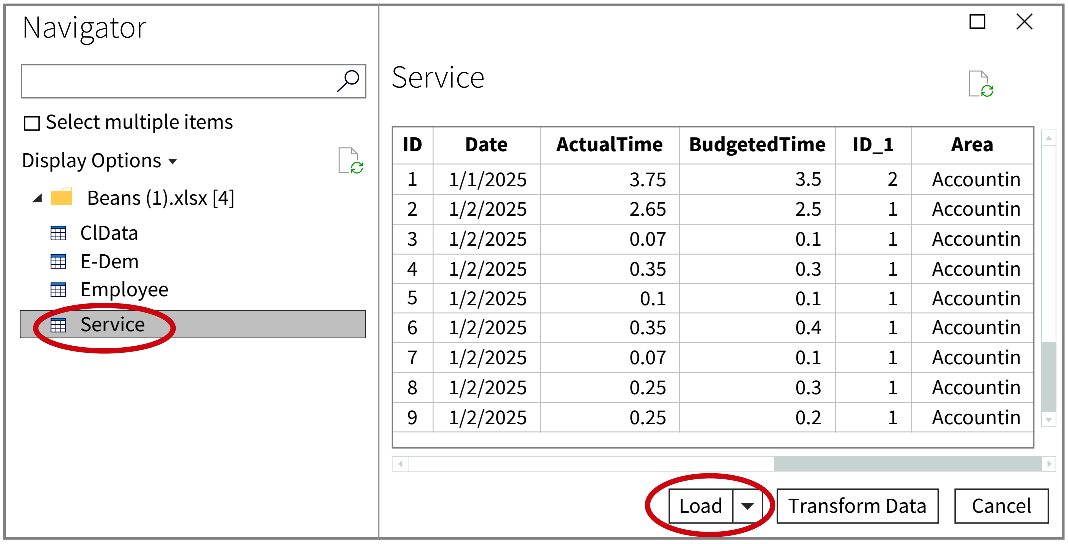
- First, open a Blank workbook and click **Data > From File > From Excel Workbook** (**Illustration 5.46**).

ILLUSTRATION 5.46 Extract Data from an Excel File



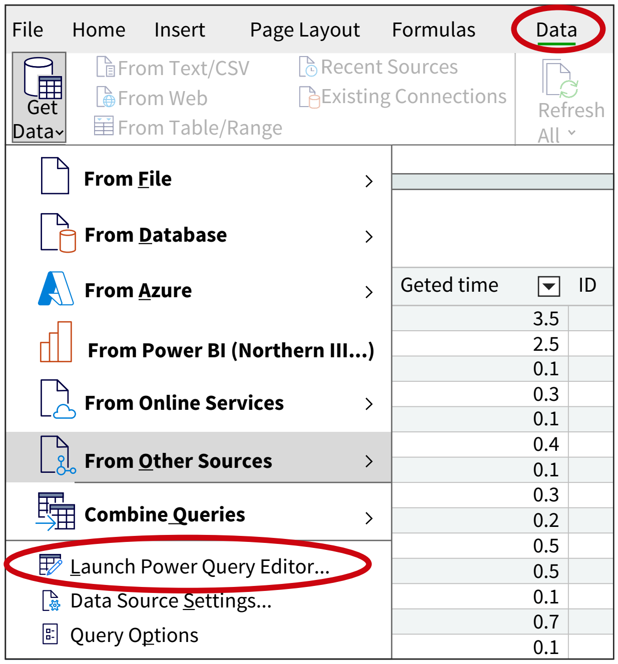
- Next, select the Beans Excel file and click **Import**. A Navigator dialog box will open. Select the **Service** table and click **Load** (**Illustration 5.47**).

ILLUSTRATION 5.47 Select and Load the Service Table



- Finally, click the **Data** tab and then the down arrow in **Get Data**. Next, you will see the option to **Launch Power Query Editor**. Click to open the Query Editor (**Illustration 5.48**).

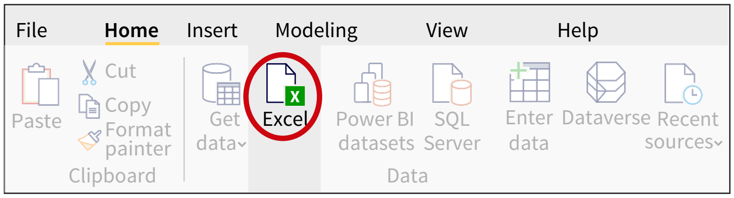
ILLUSTRATION 5.48 Open Power Query



If you are using Power BI:

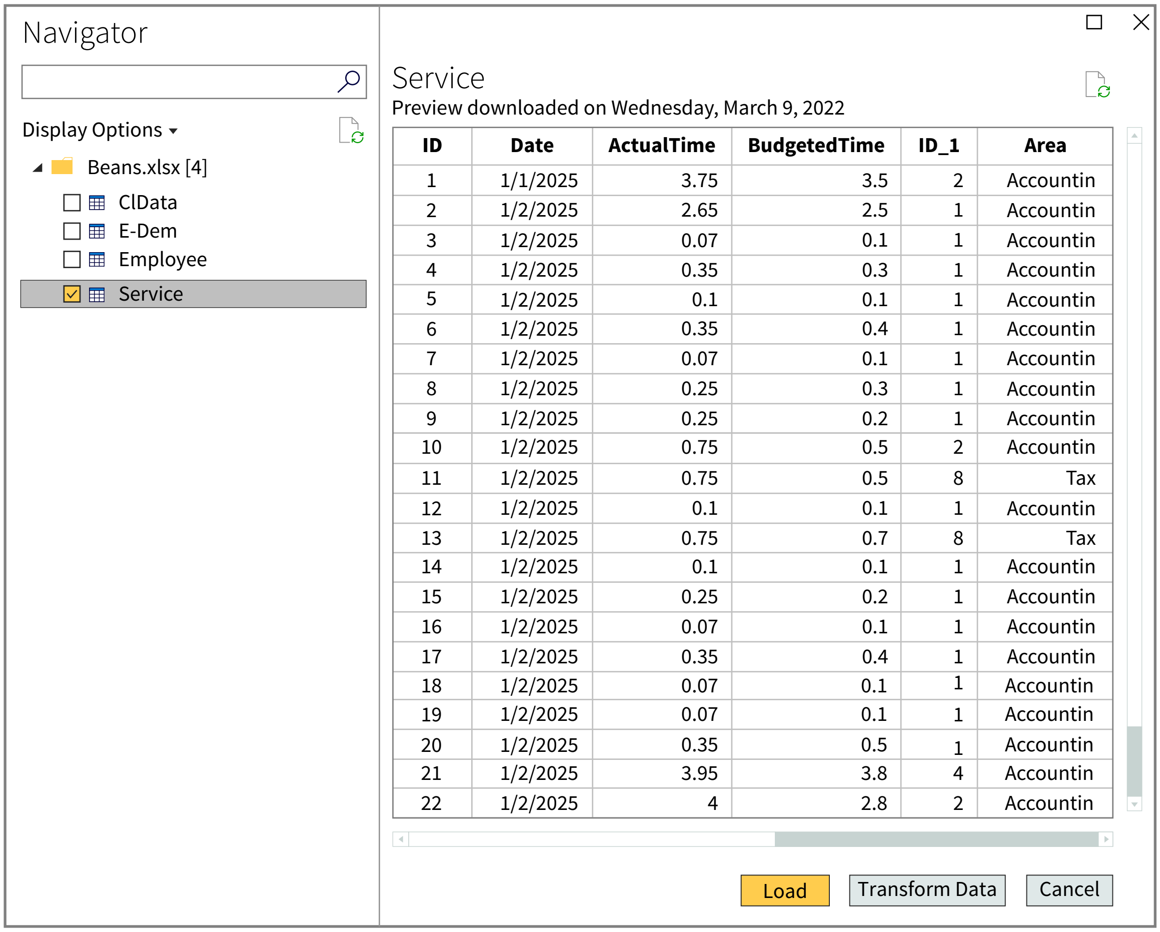
- Open a new Power BI file and select **Excel** in the **Data** group. **Illustration 5.49** shows Power BI's Home tab, which includes the Data group.

ILLUSTRATION 5.49 Extract Data from an Excel File



- Next, select the Beans Excel file and click **Open**. A Navigator dialog box will open. Select the **Service** table and click **Load** (**Illustration 5.50**).

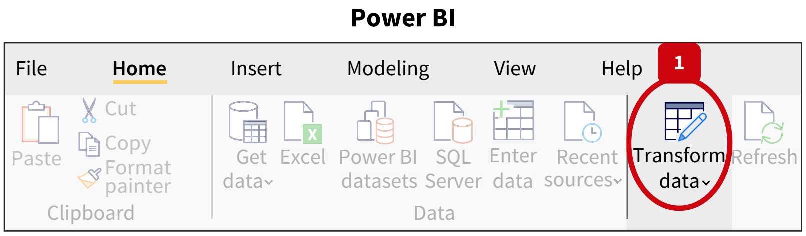
ILLUSTRATION 5.50 Select and Load the Service Table



ID	Date	ActualTime	BudgetedTime	ID_1	Area
1	1/1/2025	3.75	3.5	2	Accountin
2	1/2/2025	2.65	2.5	1	Accountin
3	1/2/2025	0.07	0.1	1	Accountin
4	1/2/2025	0.35	0.3	1	Accountin
5	1/2/2025	0.1	0.1	1	Accountin
6	1/2/2025	0.35	0.4	1	Accountin
7	1/2/2025	0.07	0.1	1	Accountin
8	1/2/2025	0.25	0.3	1	Accountin
9	1/2/2025	0.25	0.2	1	Accountin
10	1/2/2025	0.75	0.5	2	Accountin
11	1/2/2025	0.75	0.5	8	Tax
12	1/2/2025	0.1	0.1	1	Accountin
13	1/2/2025	0.75	0.7	8	Tax
14	1/2/2025	0.1	0.1	1	Accountin
15	1/2/2025	0.25	0.2	1	Accountin
16	1/2/2025	0.07	0.1	1	Accountin
17	1/2/2025	0.35	0.4	1	Accountin
18	1/2/2025	0.07	0.1	1	Accountin
19	1/2/2025	0.07	0.1	1	Accountin
20	1/2/2025	0.35	0.5	1	Accountin
21	1/2/2025	3.95	3.8	4	Accountin
22	1/2/2025	4	2.8	2	Accountin

- Finally, click **Transform data** in the **Queries** group of the **Home** tab to launch Power Query (Illustration 5.51).

ILLUSTRATION 5.51 Open Power Query



The remaining steps are identical for Excel and Power BI.

STEP 2: Click the **View** tab in the main menu (Illustration 5.52). Select **Column quality, Column distribution**, and **Column profile**.

ILLUSTRATION 5.52 Column Profiling Options

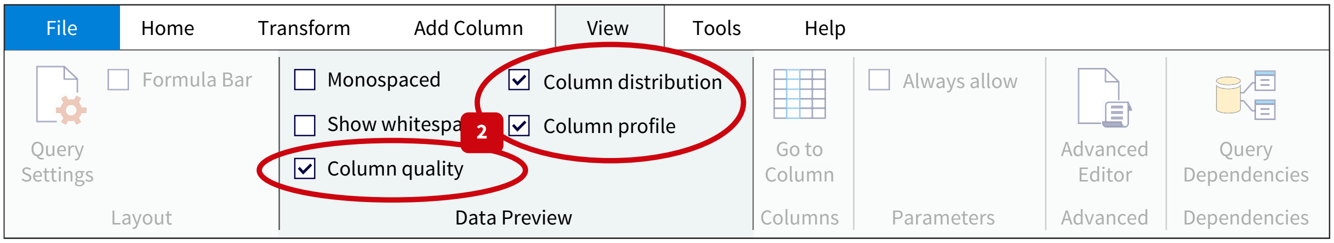
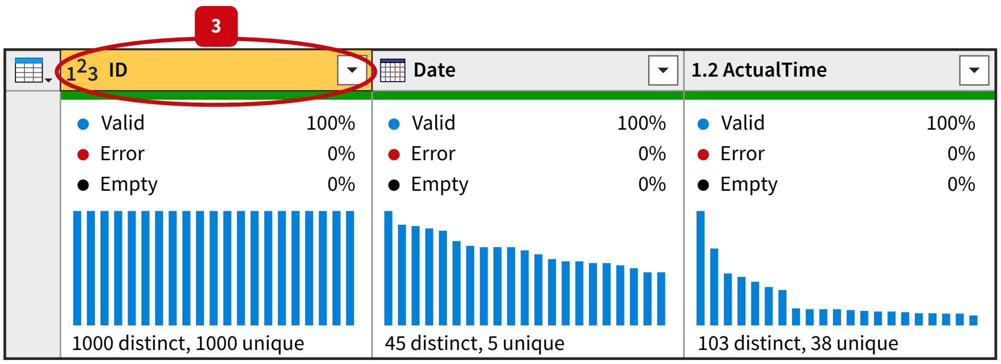


Illustration 5.53 shows that Power Query now displays detailed profiling information for each column. The valid, error, and empty percentages appear when **Column quality** is checked. The frequency distributions appear when you check **Column distribution**.

ILLUSTRATION 5.53 Column Quality and Distribution Information



STEP 3: Select the **ID** column. As the bottom left corner of the Power Query window shows (Illustration 5.54), the column profiling is currently based on the top 1000 row.

ILLUSTRATION 5.54 Column Profiling Based on Sample

10 COLUMNS, 999+ ROWS   Column profiling based on top 1000 rows

Click **Column profiling based on 1000 rows** and then select **Column profiling based on entire data set** (Illustration 5.55).

ILLUSTRATION 5.55 Column Profiling Based on Entire Data Set

- [x] Column profiling based on top 1000 rows
- [ ] Column profiling based on entire data set

The statistics shown in Illustration 5.56 will appear. This results from checking the **Column profile** option. Among others, the information provided includes the row count and the average control amount.

ILLUSTRATION 5.56 Column Profile

Column statistics	...
Count	4,632
Error	0
Empty	0
Distinct	4,632
Unique	4,632
NaN	0
Zero	0
Min	1
Max	4,632
Average	2,316.5
Standard deviation	1,337.28...
Even	2,316
Odd	2,316

How To

HOW TO 5.2 Merge Tables With Power Query

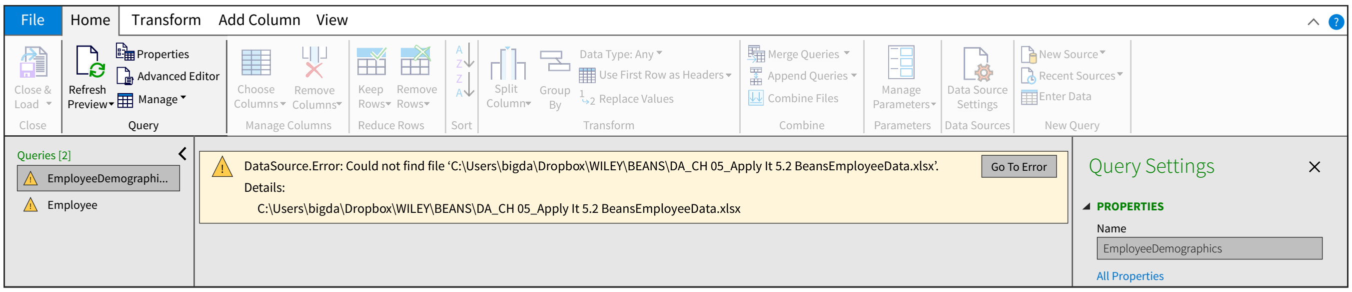
In the Beans case the goal is to join the Employee and EmployeeDemographics tables. If both tables share the same primary key, a join, or merge, is straightforward. But the EmployeeDemographics table does not have a primary key and there are matching issues between the names. Let's explore how these different issues can be addressed with Power Query.

What You Need: **Data** The How To 5.2 data file.

STEP 1: If you are using Excel, open the file and click the **Data** tab in the main menu. Select **Get Data** in the ribbon, then **Launch Power Query Editor**. In Power BI, open the file and click on the **Home** tab in the main menu. Select **Transform data**.

In both cases, Power Query will open. Make sure you are connected to the BeansEmployeeData file. Create this connection by clicking the wheel next to **Source** in the **Applied Steps** pane. Under **File path**, specify where the data set is located on your device. If you don't define this connection correctly, you will receive an error message like the one shown in Illustration 5.57.

ILLUSTRATION 5.57 Incorrect Connection to Data Set



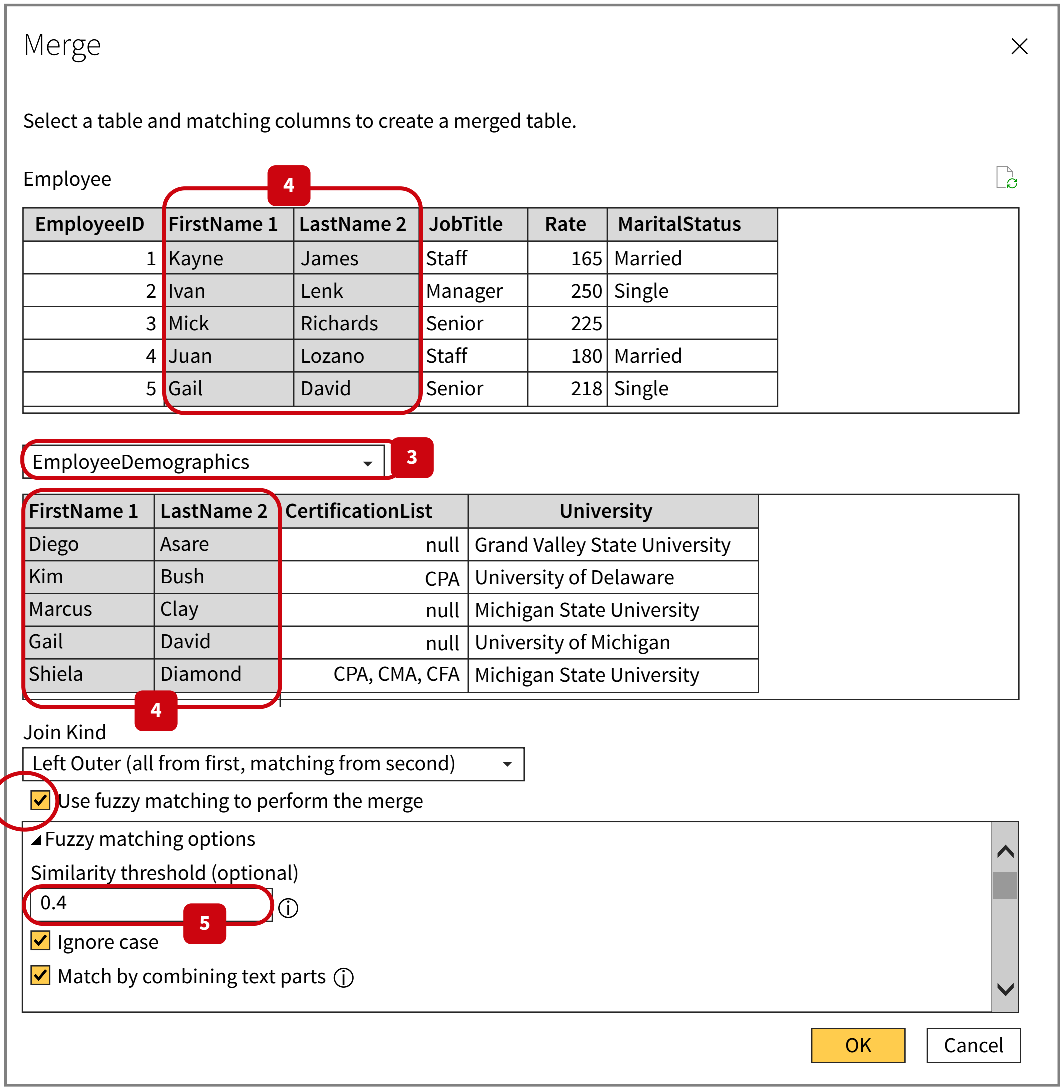
STEP 2: Select the Employee table in the Queries pane. Next, select the **Home** tab in Power Query's main menu and select **Merge Queries** in the **Combine** group.

STEP 3: Select the EmployeeDemographics table to join it with the Employee table.

STEP 4: Choose **FirstName** and **LastName** in both tables. Using Pattern 6 to split the name in the Employee table now pays off!

Illustration 5.58 visually shows how to perform steps 3, 4, and 5.

ILLUSTRATION 5.58 Merge Tables with Fuzzy Matching



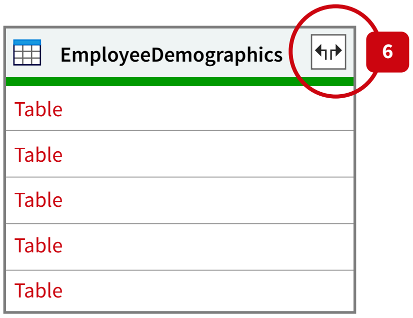
STEP 5: To address data issues resulting from nicknames, typos, and reversed names, click the checkbox **Use fuzzy matching to perform the merge**. Fuzzy matching is an algorithm that measures similarities between two sets of values. The similarity threshold determines how similar the values need to be for them to match. To fully match both sets here, the threshold should be lowered to 0.4. Click **OK** to perform the merge.

STEP 6: As shown in Illustration 5.59, a new column is added to the Employee table as a result of the merge: EmployeeDemographics. All values in this column show **Table** as the value. Next, Click the right upper corner of the column:

- Power Query will let you select any column from the EmployeeDemographics table and add it to the Employee table. Uncheck the **FirstName** and **LastName** columns and keep the **CertificationList** and **University** columns.
- Uncheck **Use original column name as prefix**. This will delete EmployeeDemographics from the CertificationList and University column names.

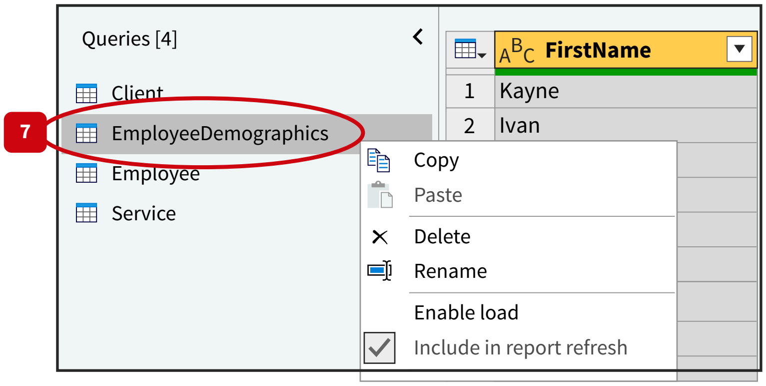
At this point, you have successfully merged the two tables into one.

ILLUSTRATION 5.59 Select Data from the Merged Table



STEP 7: Only the Employee table is needed for analytical purposes. Illustration 5.60 shows how to instruct Power Query not to load a table. Right-click on **EmployeeDemographics** and unclick the checkbox before **Enable Load**. Tables that are not loaded into the analytical database are shown in italics. Note: This option is available in power BI only, not in Excel

ILLUSTRATION 5.60 Select Tables that Should Not be Loaded



How To

HOW TO 5.3 Implement Referential Integrity with Microsoft Access

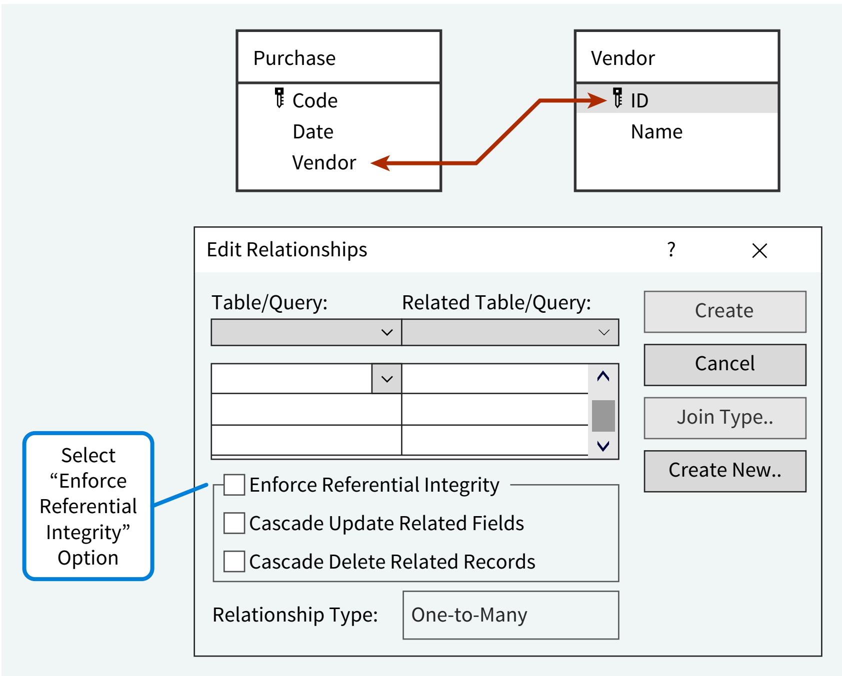
To improve data quality, referential integrity is best implemented at the data entry level. Database software such as Microsoft Access makes this easy.

What You Need: Data The How To 5.3 file.

STEP 1: The Microsoft Access database contains the Vendor and Purchase tables:

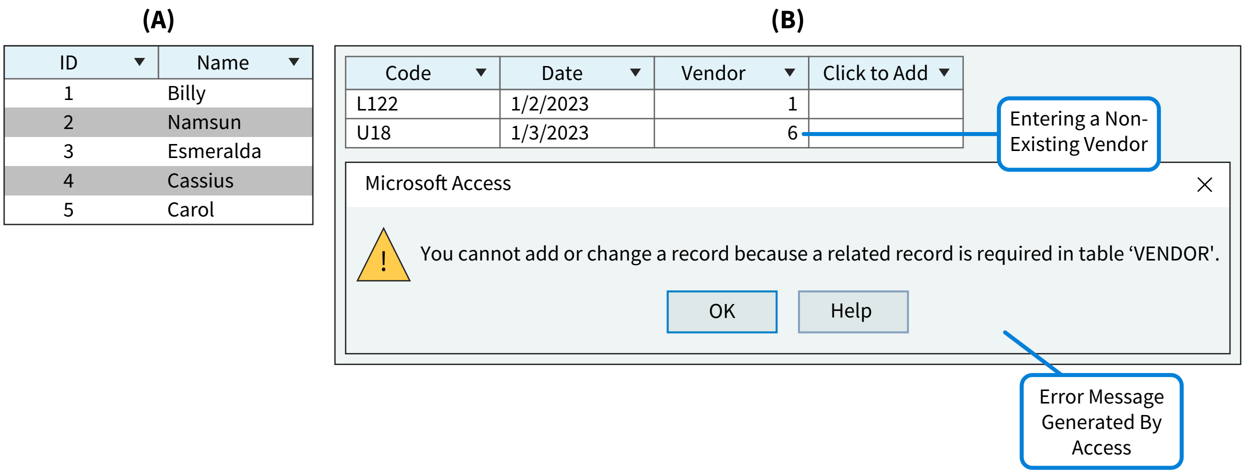
- As shown in Illustration 5.61, the relationship between the two tables is defined between the ID field in the Vendor table and the Vendor field in the Purchase table.
- The bottom of Illustration 5.61 shows that when creating this relationship (by dragging the vendor field into the Purchase table, or the other way around), Microsoft Access provides an **Enforce Referential Integrity** option.

ILLUSTRATION 5.61 Enforcing Referential Integrity With Microsoft Access



STEP 2: Illustration 5.62 demonstrates what happens if you select this option. Panel (A) shows all existing vendors. Panel (B) shows what happens if you try to add a non-existing vendor–the vendor with ID 6–to the Purchase table. Access generates an error preventing you from entering the *invalid* vendor.

ILLUSTRATION 5.62 Entering an Invalid Vendor



Data The Data tag appears when the data to answer a question or complete an exercise are available on Wiley's online learning platform.

Multiple Choice Questions

1. **(LO 1)** Which of the following terms describes the process of profiling, cleaning, restructuring, and integrating data prior to processing and analysis?
   - **a.** Data profiling
   - **b.** Data preparation
   - **c.** Data querying
   - **d.** Data parsing

2. **(LO 1)** Hao, an accounts receivable analyst, is reviewing the customer master file for use in an analysis project. He noticed that in the Address field one customer's state is listed as NM, while another customer's state is listed as New Mexico. Which of the following is true?
   - **a.** Hao has identified an instance of data that is incorrect.
   - **b.** Hao has identified an instance of data that is invalid.
   - **c.** Hao has identified an instance of data that is inconsistent.
   - **d.** Hao has identified an instance of data that is not sizable.

3. **(LO 1)** Spencer, a financial analyst at an athletic apparel store, is preparing a master file of all professional sporting spokespersons for the company. Use the table to identify which of the following statements is true:

Spokesperson	Name	Sport
Bharat Arun, Bowling	Bharat Arun	Bowling
Ramakrishnan Sridhar, Fielding	Ramakrishnan Sridhar	Fielding
Virat Kohli, Batsman	Virat Kohli	Batsman
Rohit Sharma, Batsman	Rohit Sharma	Batsman

   - **a.** The Spokesperson column contains incorrect data, while the Name column contains invalid data.
   - **b.** Spokesperson is a single-valued column, whereas Name is a composite column.
   - **c.** Spokesperson is a composite column, whereas Name is a single-valued column.
   - **d.** Spokesperson represents an aggregate column, whereas Name and Sport represent sliced columns.

4. **(LO 1)** Which of the following statements about star schemas is *incorrect*?
   - **a.** Fact tables represent business transactions.
   - **b.** Compared to fact tables, dimension tables typically have fewer instances and more columns.
   - **c.** To make slicing data easier, most measures are defined as part of the dimension tables.
   - **d.** Dimension tables are used for what, when, and who analysis.
   - **e.** The grain represents the granularity level of a fact table.
5. **(LO 2)** Which of the following statements about data transformation are *correct*?
   1. Data transformation has three subprocesses: cleaning, restructuring, and integration.
   2. Cleaning aims to correct data anomalies.
   3. Restructuring does not change any data, just the way they are organized.
   4. Integration links the data together by defining relationships.

   - **a.** 1 and 2
   - **b.** 1 and 3
   - **c.** 1, 2, and 4
   - **d.** 1, 2, 3, and 4
6. **(LO 2)** Data cleaning can involve adding, modifying, and deleting data. In which of the following examples would data modifying be required?
   - **a.** A file of purchase orders is missing one month of purchase orders.
   - **b.** You are preparing an analysis of how many purchase orders are greater than $500. The file you are working with has a row at the beginning of each new month with the month name listed in the date column. There is no other data in the row with the month name.
   - **c.** While examining the purchase orders data, you note that the vendor's name, street address, city and state are listed in separate columns. In some cases, the state column displays the state name, in other cases the state abbreviation is shown.
   - **d.** During your analysis of purchase orders, you note that some of the orders include the vendor phone number.
7. **(LO 3)** Validation of data transfers is done by comparing the source data with the data in the ETL tool. Which of the following statements about data transfer is incorrect?
   - **a.** The number of rows helps validate the completeness of data transfers.
   - **b.** Sequence gap analysis helps validate the completeness of data transfers.
   - **c.** Comparing the averages of a numeric field helps validate the accuracy of data transfers.
   - **d.** Comparing column headers helps validate both completeness and accuracy.
   - **e.** Comparing the sum of numeric fields helps validate the accuracy of data transfers.
8. **(LO 3)** You are preparing an analysis of vendors and purchase orders and have transferred data from the ERP system for all purchase orders in the past year and the data for all vendors. You have been provided with the following information about the data:

1	Total number of purchase orders	15,786
2	Sum of all purchase orders	$1,567,679
3	Average purchase order amount	$99.31
4	Total number of vendors	672

   Which information is relevant to ensure the transfer of all data?
   - **a.** 1 and 2
   - **b.** 2 and 3
   - **c.** 1 and 4
   - **d.** 1, 2, 3, and 4
9. **(LO 4)** Lizelle is a financial analyst preparing an analytical database to analyze manufacturing data to determine the most efficient processes. She is currently identifying irrelevant and unreliable data in columns within a table. Which of the following is the most appropriate response when she identifies irrelevant or unreliable data?
   - **a.** Remove the column with irrelevant or unreliable data.
   - **b.** Abandon the project because the data are not valid and cannot be analyzed.
   - **c.** Unpivot the column so that the data are separated into distinct fields.
   - **d.** Put the relevant, irrelevant, reliable, and unreliable data in separate columns.

10. **(LO 4)** Hofflak is a large construction company in the Northwest U.S. Their ERP system keeps track of all their assets. The data dictionary for the asset table looks as follows:

Asset

Name	Description
ID	An asset's unique ID.
Description	An asset's description.
AssetCategory	An asset's category.
Price	The price paid for the asset.
Salvam	The estimated book value of the asset at the end of its useful life.
EstimatedLifetime	An asset's estimated lifespan.
DprMethod	The depreciation method that applies to the asset.

What column names would you change for data analysis purposes?

a. ID, Salvam, DprMethod
b. ID, Price, EstimatedLifetime, DprMethod
c. Price, Salvam, EstimatedLifetime, DprMethod
d. AssetCategory, Price
e. Salvam, EstimatedLifetime, DprMethod
f. EstimatedLifetime, DprMethod

11. **(LO 4)** To restructure a composite column you should

a. remove the column data that is not needed.
b. split the column data into separate columns.
c. create a separate table.
d. move the column to another table.

12. **(LO 4)** Courtier is a high-end luxury goods store that designs and manufactures watches and jewelry and sells them worldwide. The table provides a small sample of the customer information they track. They have a very simple loyalty program. Customers earn "diamond" status if they have bought more than $10,000,000 worth of products, "gold" status if they have bought for more than $1,000,000, and "no" status if they have bought less than $1,000,000.

Customer

Code	LoyaltyStatus	Country
1	Diamond	USA
2	Gold	Italy
3	N/A	US
4	Gold	

Based on the information given to you, how would you describe this data set's issues?

a. Incorrect, inconsistent, incomplete, invalid
b. Inconsistent, incomplete, invalid
c. Inconsistent, invalid
d. Incomplete, invalid

13. **(LO 5)** Which of the following statements is true about establishing a primary key in a table?

a. Primary keys must use the same column title in all related tables.
b. Primary keys must have a unique value for each row and have no null values.
c. Primary keys and foreign keys are identical information repeated in the same table.
d. If a table does not have a primary key, it cannot be included in an analytical database.

14. **(LO 5)** You are examining the columns in a data set that you are analyzing. The following columns are in the data set.

HotelPropertyID	Contains a unique identification number for each hotel.
HotelAge	How many years the hotel has been operating since opening.
OpeningDate	The date the hotel opened.
RoomsAvailable	The number of rooms in the hotel.
RoomsRented	The number of rooms rented during the year.
Revenue	The amount of money collected for rooms rented during the year.

After reviewing the column headings and description, which of the following would you note?

a. There is no overlap between the columns.
b. There is an overlap between HotelAge and OpeningDate.
c. There is redundancy between RoomsAvailable and RoomsRented.
d. There is dependency between HotelAge and OpeningDate.

15. **(LO 5)** Consider a scenario where the following validation rule was applied to the Employee table shown.

```
VALID =
IF EMPLOYEE.AGE < 24 AND EMPLOYEE.DEGREE = "COLLEGE",
THEN "YES",
ELSE "NO"
```

Employee

Code	Age	Degree
1	23	High School
2	23	College
3	25	College
4	21	

Which of the following statements is incorrect?

a. The value for the Valid column for the employee with code "1" is "No."
b. The value for the Valid column for the employee with code "2" is "No."
c. The value for the Valid column for the employee with code "3" is "No."
d. The value for the Valid column for the employee with code "4" is "No."

16. **(LO 6)** Mithali, a financial analyst, is preparing data to engage in a data analysis project to predict revenue for the next period based on environmental considerations. She is currently reviewing the tables to validate inter-table rules for referential integrity. Which of the following is true about referential integrity?

a. The primary and foreign key fields must have the same name.
b. The primary key must be single-valued, but the foreign key can be multi-valued.
c. The primary and foreign key fields can have different data types.
d. All values in a foreign key should also exist as values in the corresponding primary key.

17. **(LO 6)** Skyline is a national plumbing and heating company. Skyline's CFO is preparing the employee performance data for further analysis. Samples of four of the files are shown. How would you suggest combining these four files?

JanuaryPurchases

PurchaseID	Date	Amount	Employee
14399	1/1/2025	1,432.24	7
14400	1/1/2025	799.99	14
14401	1/1/2025	320	12
14402	1/1/2025	822.21	22

EmployeeDemographics

ID	Name	DateOfBirth
14399	Ben Jespers	3/3/1975
14400	Bei Shi	7/17/1985
14401	Michael Rodman	5/20/1984
14402	Benita Alvarez	11/11/1979

EmployeeWages

ID	HourlyWage
14399	25
14400	38
14401	36
14402	45

FebruaryPurchases

PurchaseID	Date	Amount	Employee
15021	2/1/2025	123.89	4
15022	2/1/2025	540.99	1
15023	2/1/2025	183.12	14
15024	2/1/2025	744.41	21

a. Merge JanuaryPurchases and EmployeeDemographics and merge FebruaryPurchases and EmployeeWages.

b. Union JanuaryPurchases and FebruaryPurchases and union EmployeeDemographics and EmployeeWages.

c. Union JanuaryPurchases and FebruaryPurchases and merge EmployeeDemographics and EmployeeWages.

d. Union JanuaryPurchases and EmployeeDemographics and merge FebruaryPurchases and EmployeeWages.

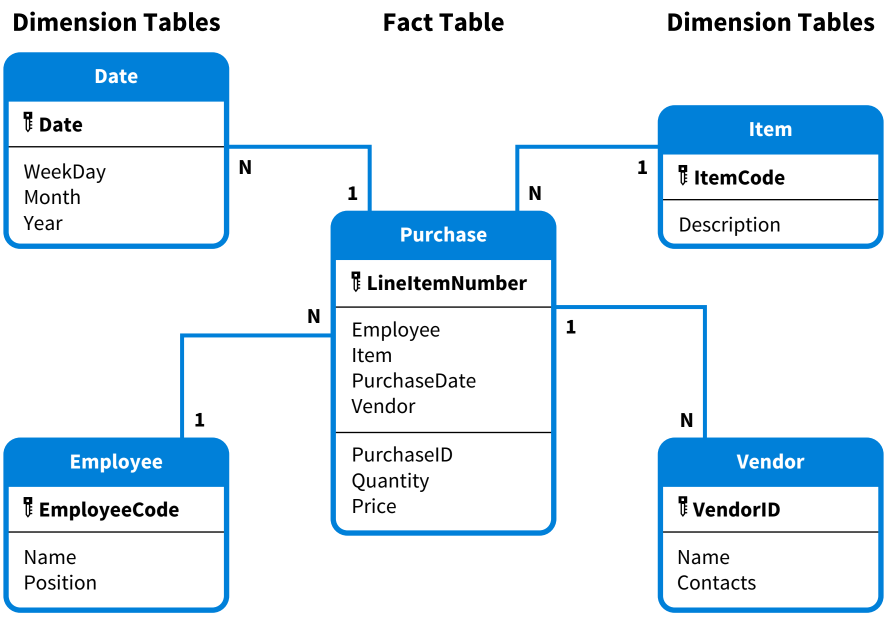
18. **(LO 6)** Which of the following would result in transforming this star schema into a snowflake schema?

a. EmployeeName is a combined field that contains both the employee's first name and last name.

b. For some employees, their position is not known.

c. It is possible to have multiple contacts for a vendor.

d. Weekday, Month, and Year are extensively used for analysis purposes.



19. **(LO 7)** Hamza is transferring data into an analytical database for analysis. Which steps should he take to identify whether all the data transferred?

   a. Review the columns for appropriate naming conventions.

   b. Establish data validation rules and identify which data do not comply with the data validation rules.

   c. Inspect distinct values and analyze frequencies.

   d. Compare the row count for the analytical database with the row count of the data set in the ETL tool.

20. **(LO 7)** You are preparing an analysis for a food delivery company called Dine At Home. Dine At Home picks up food orders from restaurants and delivers them to the customer. Customers can order many times, but an order is for one customer only. Restaurants may have many orders, but an order is only for one restaurant. Which of the following cardinality specifications is incorrect?

   a. 

   b. 

   c. 

   d. 

Review Questions

1. **(LO 1)** Describe data preparation. Why is data preparation necessary prior to analyzing data?
2. **(LO 1)** Assume you are preparing an analytic database. You have downloaded raw data from the information system and are now examining the data for consistency. Discuss two profiling techniques you might use to identify inconsistencies in the data.
3. **(LO 2)** Define and describe the three subprocesses of data transformation.
4. **(LO 2)** Compare and contrast the two forms of data integration: linking and combining.
5. **(LO 2)** What is data matching and why is it a challenge for data integration?
6. **(LO 3)** Describe what a data dictionary is and why it is an important part of data analysis.
7. **(LO 3)** Discuss why a pattern-based approach to data preparation is appropriate for data analysis projects.
8. **(LO 3)** Discuss the tools and features available in Microsoft Excel to ensure the transfer of all data when extracting data.
9. **(LO 4)** Compare and contrast the data types of whole number and text. Include how variables that are noted as whole numbers and variables that are coded as text are used in data analysis.
10. **(LO 4)** Discuss the difference between a composite column and a multi-valued column and give an example of each.
11. **(LO 4)** Describe the profiling techniques for detecting inconsistent values, and explain how to modify inconsistent values. Give an example of an inconsistent value.
12. **(LO 5)** Discuss the importance of minimizing redundant data among columns in an analytical database.
13. **(LO 5)** Discuss why it is important to verify the presence of primary keys. Explain how to identify a missing primary key and how to create one.
14. **(LO 6)** Describe dimensional modeling and discuss the importance of complying with dimensional modeling principles when building an analytical database.
15. **(LO 6)** What is the difference between a union and a merge? Give an example of when a union cannot be used.
16. **(LO 6)** Discuss the benefits of single-valued columns and flat tables for slicing data.

17. **(LO 6)** Explain how you would detect and correct the following issues.

Issues	Detection	Correction
A table in your analytical database does not have a primary key.		
A column in the fixed assets table has a column for date acquired (DateAcquired) and the asset age (AssetAge). Age represents how long the asset has been held since the date acquired.		
A table with vendor information is named VINFO.		
Invalid data might have been entered into one of the tables in the database.		

18. **(LO 6)** Discuss transforming models and how it differs from transforming columns.

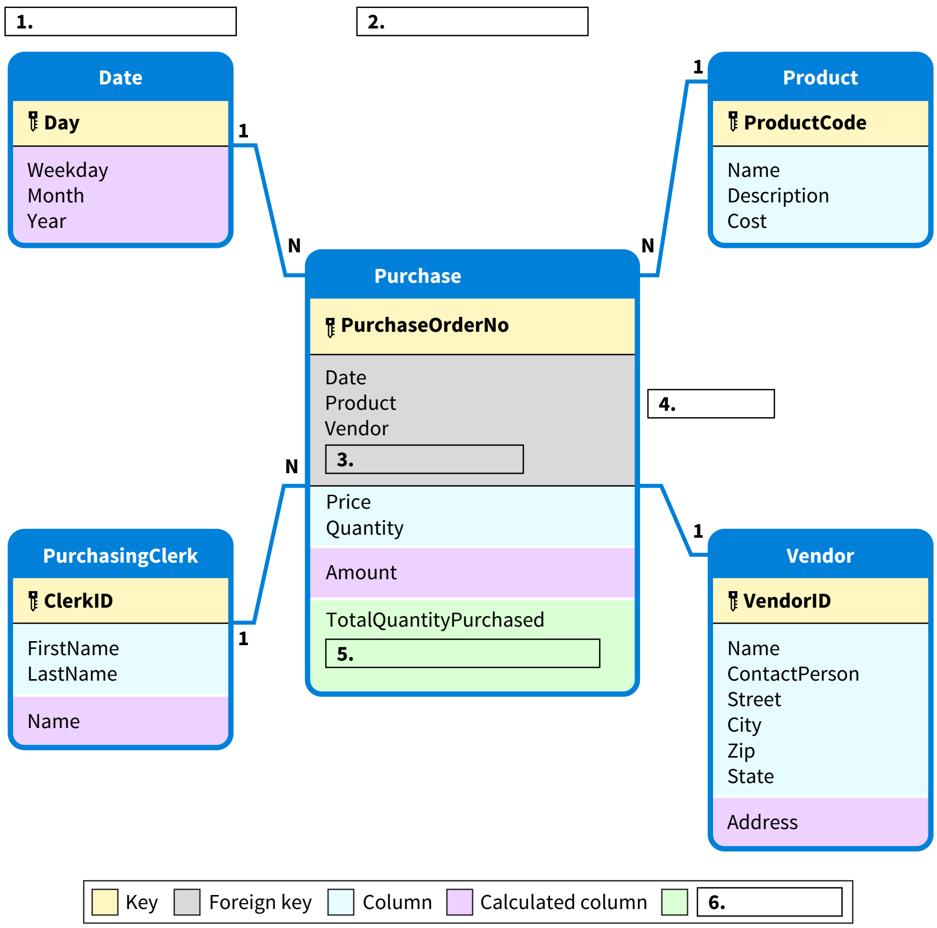
19. **(LO 7)** Discuss the importance of validating relationships when transferring data from the ETL tool to the analytical database.

20. **(LO 7)** Discuss why it is important to ensure correct transfer when loading data. Give an example of how to determine if the data are correct.

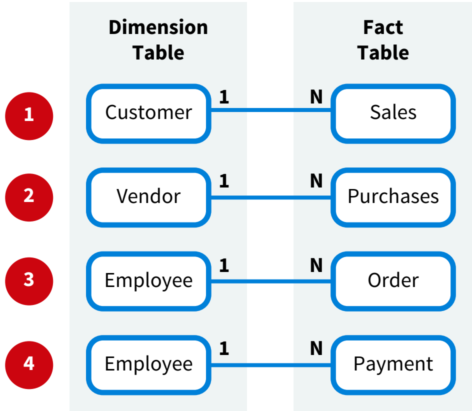
Brief Exercises

BE 5.1 **(LO 1)** **Auditing** You are an audit staff assigned to the Coleman Cable, Inc. audit. You have been asked to develop a star schema to illustrate the structure of the analytical data model for your sales analysis. The senior has provided you with a first draft of the star schema. Match each of the following terms to its appropriate location in the star schema.

a. Dimension tables
b. Fact table
c. N
d. Measure
e. TotalAmountPurchased
f. PurchasingClerk



BE 5.2 **(LO 1)** You are a financial analyst asked to debrief your manager on the relationships between tables in your analytical database. For each of the following four relationship specifications, identify the most appropriate sentence describing one of its cardinalities.



a. Only one vendor can be specified for a purchase.

b. For every one customer there can be many sales.

c. The same employee can place multiple orders.

d. There can be many customers for the same sale–for example, two people buying a house.

e. Only one employee is accountable for a payment.

BE 5.3 **(LO 1)** You have been provided with a data table that contains customer data collected for a retail company's customer loyalty program. An excerpt of that data table follows. Identify at least three data quality issues.

CustomerInfo	Gender	Date	Email
Steve Millier, 121 South Beach St	M	10/21/1978	steve.miller
Sara Stevens, 435 Parker St	Female	9/21/2008	sara.stevens@mail.com
Libby Ralston, 812 Foster Ave	Female	6/28/2956	libby123@mail.com
Able Meyers, 902	Male	1/4/2009	smileyone@mail.com
Spring Stevens, 639 Bulldog Ct	Undisclosed	6/14/1982	sparkie.steven
Kevin Hogan, 3098 N. Tier St. Apt 392	M	7/22/1955	doglover@mail.com
Libby Ralston, 812 Foster Ave	Female		
Sara Stevens, 435 Parker St	Female	9/21/2008	sara.stevens@mail.com
Cooper Mazzu, 2342 Lincoln Way	Male	4/17/1955	coop8623
Molly White	F	10/21/1977	molly.white.123@mail.com
Janet Jones, 909 Hwy 78	F		

BE 5.4 **(LO 2)** The following two tables must be integrated into one combined table. The combined table must include the following fields for each record: Description, QOH, Location, Cost, Price, and Sale. Review each table and identify three data matching challenges.

Table A

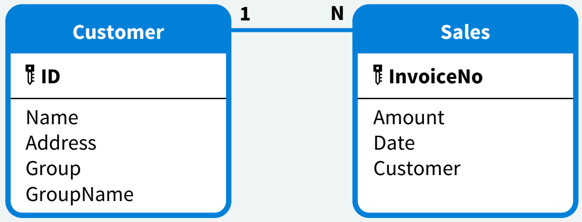
Description	QOH	Location
Blue Dog Collar	5	A10
Red Doc Collar	3	A11
7 Inch Rope Sturdy Lead	10	B10
10 Inch Rope Sturdy Lead - Green	3	B10
10 Inch Rope Sturdy Lead - Black	5	B10

Table B

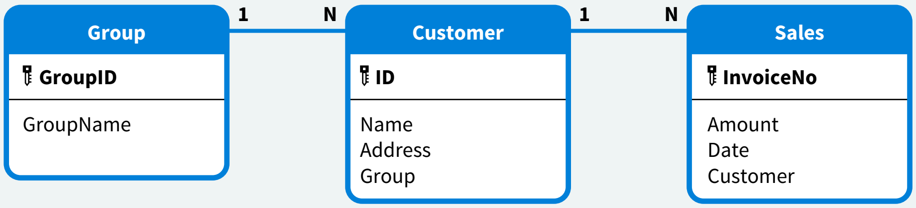
ProductDescription	Cost	Price	Sale
Dog Collar - Blue	12.54	16.93	Y
7 Inch Rope Sturdy Lead	15.44	20.85	N
Red Dog Collar	15.54	20.98	N
10 Inch Rope Sturdy Lead - Bk	20.38	27.52	N
Green 10 Inch Rope Sturdy Lead	28.34	38.26	Y

BE 5.5 **(LO 2)** Panels (A) and (B) represent alternative data models for the same data set. Compare and contrast the two models.

(A)



(B)



BE 5.6 **(LO 3)** **Tax Accounting** You have been asked to develop an analytical database to analyze local and federal sales tax compared to value-added tax. An excerpt of your data table is presented.

SaleID	Date	SaleLocation	SaleAmount	LocalTax	SalesTax	VAT
1001	5/25/2024	USA	$ 197	0.02	0.0685	
1002	5/26/2024	USA	$ 113	0.02	0.0685	
1003	5/27/2024	MEX	$ 155			0.16
1004	5/28/2024	MEX	$ 423			0.16
1005	5/29/2024	CAN	$ 88			0.05
1006	5/30/2024	AUS	$ 250			0.1

You have started the process and are now ready to create the data dictionary. For each of the blank spaces designated with a number in the following table, identify the appropriate sentence or term to create the data dictionary. Each term or sentence will be used only once.

a. SaleID

b. SaleAmount

c. Vat

d. LocalTax

e. The percentage of federal sales taxes collected at the time of the sale.

f. The location where the sale was made.

g. The date the sale was made.

Name	Description
1.	Unique identifier for each sale made.
Date	2.
SaleLocation	3.
4.	The total amount of the sale before taxes.
5.	The percentage of local taxes collected at the time of the sale.
SalesTax	6.
7.	The value-added-tax percentage to be collected related to the sale.

BE 5.7 **(LO 3, 4)** You face five different data preparation scenarios. Match a data quality issue to each scenario. Each data quality issue can be used only once.

a. The column names are incorrect or ambiguous.
b. All columns are not single-valued.
c. All the data have not been transferred correctly.
d. The columns do not have the correct data type.
e. All the data have not been transferred.

#	Scenario	Data Quality Issue
1.	Grace received a data set that had a column titled ClientInfo. Each record in that field contained a client's mailing address.	
2.	The data in the Sales column are the dollar amounts of the sale transactions. However, Quinn noted that the data type was classified as alphanumeric characters or text data.	
3.	Upon inspection of data transferred from the payroll system to a data warehouse, Molly noticed that employee payroll records only included employees with last names beginning A through S.	
4.	Barnes received a table of data from the IT group. One of the columns contains both customer name data and payment term data.	
5.	Devon extracted data from the company's shipping department log. In addition to shipment data, the log also contains summary statistics such as the total quantity shipped. Upon inspection of the extracted data, Devon notes that the total quantity shipped for the extracted data does not match the total quantity shipped number in the log.	

BE 5.8 **(LO 3)** **Data** You receive a text file with December purchase order information. Before you start your analysis, use Power Query (Excel or Power BI) to determine whether the data have been completely and correctly transferred. A total of 34 purchases with an average total cost per product (orderline) of $610.98 were made during the month of December.

1. Name the preparation pattern you used to verify all transactions were transferred and explain how you verified this.
2. Name the preparation pattern you used to verify that all data were transferred correctly and explain how you verified this.

BE 5.9 **(LO 4)** Simply Salon Beautiful collects customer information and assigns a unique customer ID to track the customer's rewards, purchases, and appointments.

CustomerID	CustomerName	Gender	DateOfBirth	Rewards	YTDSales	NumberOf AppointmentScheduled
A283	Emily Boyd	Female	10/21/1978	1	$ 490.12	1
A493	Artie Burns	Male	9/21/2008	3	$ 48.01	0
A393	Mario Edwards Jr.	Male	6/28/2906	6	$ 287.89	1
B382	Javon Wims	Male	1/4/2009	2	$ 476.45	1
C494	Tierna Davidson	Female	6/14/1982	4	$ 393.99	0
C948	Sarah Gorden	Female	7/22/1955	6	$ 593.07	1

For each column, identify the most appropriate data type. Each data type can be used once, more than once, or not at all.

a. Date
b. Date/Time
c. Decimal number
d. Percentage
e. Text
f. Time
g. Whole number

Column	Data Type
CustomerId	1.
CustomerName	2.
Gender	3.
DateofBirth	4.
Rewards	5.
YTDSales	6.
NumberofAppointmentsScheduled	7.

BE 5.10 (LO 4) Data With the provided data, use an ETL tool to perform the following.

1. Modify the data type for the CustomerId column from text to whole number. How many records report an error after this change?
2. Determine the highest number of rewards a customer earned.
3. Identify any incorrect data in the AppointmentDuration column.

BE 5.11 (LO 5) Data Your boss gives you a text file that contains sales information including amounts, discounts, and coupons. An excerpt is shown here.

InvoiceNo	SaleDate	Amount	Discount	Coupon	Customer
1203	1/6/2025	$ 621	0	25	23
1204	1/6/2025	$ 682	2	0	144
Cash	1/7/2025	$ 477	0	5	
1206	1/7/2025	$ 779	5	25	233
1207	1/8/2025	$ 742	5	0	17
Cash	1/8/2025	$ 452	2	0	
1208	1/8/2025	$ 233	0	10	101
1209	1/9/2025	$ 539	0	5	224
Cash	1/10/2025	$ 985	0	2	
Cash	1/10/2025	$ 320	0	0	
1210	1/10/2025	$ 398	5	0	301
1211	1/10/2025	$ 354	10	0	24
Cash	1/11/2025	$ 863	0	5	
1212	1/11/2025	$ 944	2	15	144

Based on a discussion with your boss, you put together the following data dictionary.

Name	Description
InvoiceNo	A unique code given to Credit sales for which an invoice is prepared. The label Cash is used for cash sales.
SaleDate	The date the sale occurred.
Amount	The dollar amount the customer owes before discounts and coupons.
Discount	The discount percentage given to the customer.
Coupon	A coupon with a dollar provided by the customer. It should be noted that a coupon can only be considered when no discount applies.
Customer	A customer's ID. For cash sales, this field is left blank.

List the table transformation patterns (patterns 11–14) that are relevant for this data set and explain how you would apply each.

BE 5.12 (LO 5) Your supervisor has asked you to understand sales and the discounts and coupons applied to them. They provided you with a spreadsheet for your analysis. Here is an excerpt of that spreadsheet:

SaleDate	Amount	Discount	Coupon	Approved
10/21/2025	$ 197	0	0	1
10/22/2025	$ 113	2	32	2
10/23/2025	$ 155	0	1	1
10/24/2025	$ 423	12	0	
10/25/2025	$ 88	10	0	2
10/26/2026	$ 250	3	45	
10/27/2025	$ 4,680	2	0	3
10/28/2025	$ 584	0	0	
10/29/2025	$ 997	1	1	1
10/30/2025	$ 258	2	10	1

1. Apply data preparation Pattern 11. The table is titled Amounts & Discounts. Propose a table name that is less ambiguous.
2. Apply data preparation pattern 12. Does the table have a primary key? If not, describe how you would create a primary key for this table.

BE 5.13 **(LO 6)** **Data** You are given two files. One file contains employee information and the other contains employee performance information. These are the data dictionaries for both tables.

Employee (Generic employee information)	
Name	**Description**
LastName	An employee's last name.
FirstName	An employee's first name.
Location	The city where the employee is located.
Gender	An employee's gender.

EmployeePerformance (Information about employee performance)	
Name	**Description**
LastName	An employee's last name.
FirstName	An employee's first name.
Level	An employee's level with 1 being the lowest and 7 being the highest.
2024Hours	The number of hours an employee worked in 2024.
RatePerHour	An employee's hourly rate.

Using data preparation pattern 15, combine both files to achieve the result shown here.

	ABC LastName	ABC FirstName	ABC Location	ABC Gender	123 Level	123 2024Hours	123 RateperHour
1	James	Kayne	Houston	Male	1	3,521	27
2	Lenk	Ivan	Texarcane	Male	3	3,811	35
3	Richards	Mick	Chicago	Male	5	2,901	39
4	Lozano	Juan	Boston	Male	4	3,477	37
5	David	Gail	Chicago	Female	7	3,420	40
6	Petroni	Ann	Houston	Female	1	1,709	27
7	Diamond	Shiela	Boston	Female	3	2,331	35
8	Sutherland	Yvonne	Chicago	Female	4	3,871	37
9	Clay	Marcus	Boston	Male	5	1,942	39
10	McCarthy	Molly	Chicago	Female	1	3,665	27

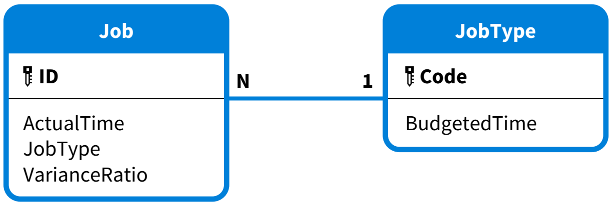
BE 5.14 **(LO 6)** **Data** HoneyBees is a high-end restaurant in Northern California. They just installed tableside point-of-sale (POS) systems on each of their 20 tables. They provide you the data for the first two days of using the new system. The data dictionary describes the data captured by the POS systems on both days.

HoneyBees Day 1 Data Dictionary	
Name	**Description**
Date	The day the sales transaction occurred.
Time	The time the sales transaction occurred.
Amount	The dollar amount of the sales transaction.
Tip	The tip received as a percentage.
Customer	Name of the customer–name on the credit card used.
Server	The name of the server (employee).
POS	The POS number–between 1 and 20–that was used to record the sales transaction.

HoneyBees Day 2 Data Dictionary	
Name	**Description**
Date	The day sales transaction occurred.
Time	The time the sales transaction occurred.
Amount	The dollar amount of the sales transaction.
Tip	The tip received as a percentage.
Server	The name of the server (employee).
Customer	Name of the customer–name on the credit card used.
POS	The POS number–between 1 and 20–that was used to record the sales transaction.

Combine the data in the two files for analysis purposes. The new file should have 46 transactions.

BE 5.15 **(LO 6)** You are evaluating a data set to determine if the data are valid. The data model for the data set is shown.



VarianceRatio is a measure, and this is its formula:

VarianceRatio = (ActualTime – BudgetedTime) / BudgetedTime

You note that the minimum budgeted time for all jobs is two hours. Your team designs a number of validation rules, including the following inter-table validation rule:

IF VARIANCERATIO > 1
THEN RED
OTHERWISE GREEN

1. Specify the validation rule in terms that are easy to understand for a business person.
2. Discuss what might cause the formula to generate the value red.

BE 5.16 **(LO 7)** Tin Fos, a midsize information service company, received a data set from one of its clients. They are trying to design analytics reports, but the results are either empty or wrong. Evaluate the following data dictionary and data model to identify the two issues that are causing the problems.

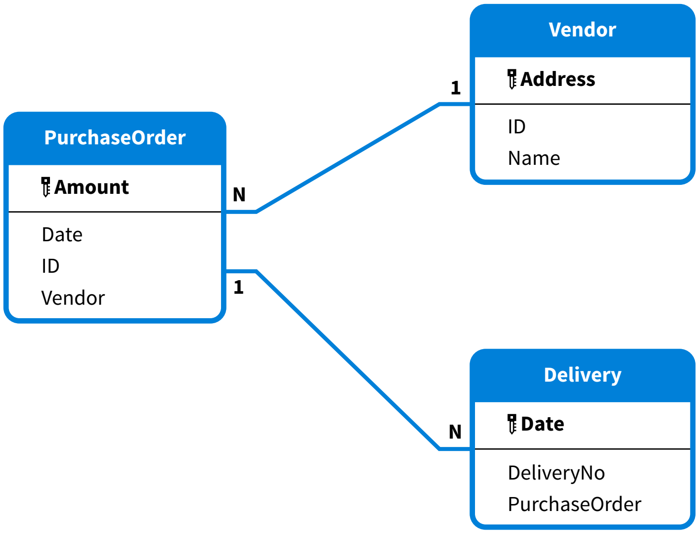
Data Dictionary

PurchaseOrder (Information about orders placed)	
Name	**Description**
Amount	The dollar amount to be paid to the vendor.
Date	The date on which the order was placed.
ID	The purchase order's unique ID.
Vendor	The ID of the vendor from whom the goods were purchased.

Delivery (Information about goods delivered)	
Name	**Description**
Date	The date on which the goods were delivered.
DeliveryNo	The delivery's unique ID.
PurchaseOrder	The purchase order number associated with the delivery.

Vendor (Information about vendors)	
Name	**Description**
Address	The vendor's address.
ID	The vendor's unique ID.
Name	The vendor's name.

Data Model:



Relationships:

Manage Relationships

Active	From: Table (Column)	To: Table (Column)
☑	Delivery (DeliveryNo)	PurchaseOrder (ID)
☑	PurchaseOrder (Vendor)	Vendor (Name)

Exercises

EX 5.1 (LO 1, 2, 4, 5) Data Basic Data Transformations You are running your own accounting practice. Your clients are small to midsize companies that need assistance with transaction processing, bookkeeping, and financial reporting. Your newest client, Healthy Pets, is a pet supply retailer with ten locations across the region. The company owner has provided you with customer loyalty data. Each customer can join the customer loyalty program to earn points for discounted purchases.

1. Remove duplicate customer entries (Hint: There are 11 records in the original table; the clean table should only include 9 records).
2. Replace values in the gender column so that records labeled "M" are replaced with Male, and records with "F" are replaced with Female.
3. Rename the Date column as DOB, (date of birth of the customer).
4. Transform the CustInfo column into two single-valued columns. Label one column Name and the second column Street.

EX 5.2 (LO 1, 2, 4, 6) Data Column and Model Transformations You are an accountant for Bargain, which is a franchise that buys items all over North America, fixes and cleans them, and then sells them with a substantial markup. You have been asked to create an analytical database to understand purchases in anticipation of a business cycle review and efficiency project. Your supervisor provided you with three CSV files and a PDF file with the data dictionaries for each. Apply the appropriate data preparation patterns to the three files to create an analytical database that is consistent with the one shown here.

BargainVendors

	ABC ID	ABC Name	ABC Email
1	US1	Cohen, Sandra	scohen@vstores.com
2	US2	Gray, Sev	sev-gray@Ibox.com
3	US3	Leroy, Shanice	sleroy@google.com
4	US4	Morales, Benita	bmor@hf.com
5	US5	Wang, CeCe	cece@cch.com
6	CAN1	Zhu, Jim	jzhu@outlook.com
7	CAN2	Coyle, Irene	Irene@coyle.com
8	CAN3	Dabrowski, Elzbieta	ElzDab@gmail.com
9	CAN4	King, Laura	LauraKing@fmh.com
10	CAN5	Jackson, Kiara	kiaraj@ncn.com

BargainPurchases

	123 LineItemID	123 OrderNo	Date	ABC Type	ABC Category	123 Price	123 Quantity	ABC Vendor
1	1	8992	1/2/2025	Rice Cooker	Kitchen	$ 69	2	US1
2	2	8993	1/2/2025	Phone	Electronics	$ 113	1	CAN3
3	3	8994	1/2/2025	Fryer	Kitchen	$ 181	1	CAN5
4	4	8894	1/2/2025	Stuffed Animal	Toys	$ 14	5	US2
5	5	8896	1/2/2025	TV	Electro	$ 79	1	CAN2
6	6	8897	1/3/2025	Stuffed Animal	Toys	$ 67	1	CAN1
7	7	8897	1/3/2025	Microwave	Kitchen	$ 205	1	US1
8	8	8898	1/3/2025	Headphone	Electronics	$ 45	1	US4
9	9	8898	1/3/2025	Jewelry	Toys	$ 13	2	CAN1
10	10	8899	1/3/2025	Printer	Electro	$ 199	2	CAN3

EX 5.3 **(LO 1, 2, 3, 4, 5, 7)** **Data** **Financial Accounting** **Extraction, Column and Table Transformations** As the accountant for El Azteco, a restaurant in Knoxville, TN, you are asked to prepare data for analysis of the restaurant's sales. The data dictionary details the data recorded for each sales transaction.

Name	Description
ID	A number that uniquely identifies the sale.
SaleDate	The date the sale occurred.
Amount	The dollar amount the customer owes before discounts and coupons.
Discount	A discount percentage given to the customer.
LoyaltyDiscount	A loyalty discount percentage given to the customer.
LolyaltyNumber	A unique number given to loyalty customers. Only customers enrolled in El Azteco's loyalty program can receive a loyalty discount.

All data recorded thus far is given to you in a CSV file. The data are also shown here. Notice that currently, all data are stored in one column. Further, Fernando, El Azteco's owner, informs you that the average sales amount thus far is $212.1.

A
SalesInformation
1,1/6/2025,97,0,0
2,1/6/2025,113,2,3,49
3,1/6/2025,55,0,1,23
4,1/7/2025,423,2,0
5,1/7/2025,88,0,0
6,1/7/2025,250,3,5,53
7,1/8/2025,680,2,0
8,1/8/2025,58,0,0
9,1/8/2025,99,1,1
10,1/8/2025,258,2,0

Examples of questions you need to answer using this data include:

- What is the actual amount paid for each sale?
- What is the average discount per sale?
- What is the total discount given to loyal customers?

Apply the necessary data preparation patterns to prepare an analytical database that makes it easy to answer these questions. Make sure to consider the patterns that help with detecting and correcting inconsistent and invalid data.

EX 5.4 **(LO 4, 5, 7)** **Managerial Accounting** **Apply Principles of Dimensional Modeling** MEQ is a medical equipment manufacturer that sells their products in the U.S., Canada, and several European countries. All sales are conducted by salespeople. As MEQ's accountant, you are asked to perform an analysis of your entity's sales by salesperson. You will calculate key performance indicators (KPIs) for each salesperson as part of their annual performance review. Currently, all information is stored in a single table (named Sales) in a relational database. The data dictionary is given here.

Name	Description
Address	A customer's address.
Category	An item's category.
CCode	A code that uniquely identifies a customer.
CName	A customer's name.
Date	The day a sale occurred.
Description	An item's name.
DOH	The day a salesperson was hired.

(Continued)

Name	Description
InvoiceId	An ID that uniquely identifies a sale.
ItemCode	A code that uniquely identifies an item or product.
LiId	A unique ID given to each line item. A sale can have multiple line items. A line item specifies an item's ID and description, the quantity sold, and the price asked per unit.
PaymentTerms	The payment terms that apply to a customer.
Price	The price paid for a line item as part of a sale.
Quantity	The quantity sold of a line item as part of a sale.
QOH	Current quantity on hand for a product.
Region	The different regions in which a salesperson operates. Each region is assigned to exactly one salesperson.
SpCode	A code that uniquely identifies a salesperson.
SpName	A salesperson's name.

1. Apply the principles of dimensional modeling to the Sales table by creating a star or snowflake schema.
2. List the patterns you would use to detect and correct data issues, and explain why you would use each pattern.

EX 5.5 (LO 6) Data Auditing Combine Tables and Identify Invalid Data Assume you are a financial analyst working in the controller's group at a public company. You are asked to prepare an analytical database that shows the consolidated fixed assets for your entity. This analytical database will be used by the auditor to verify depreciation expenses and by your controller when they are preparing the financial statement footnotes. Your company has locations in the United States and Mexico. Each location's technology group provided you with a data file that contains data about fixed assets and depreciation. Apply the appropriate data preparation patterns to the data files to create an analytical database that helps verify the 2025 depreciation expenses and the accumulated depreciation expenses. You can assume that all data were transferred correctly.

EX 5.6 (LO 4) Data Use Patterns to Detect and Correct Data Issues Dunn Motors is a Honda dealership in Tallahassee, Fl. They keep a list of all vehicles available for the salespeople in their used car department. A sample of the list and the table's data dictionary follows.

ID	ProductCode	Price
3044	Accord-Black-2017-*13,14,15*	12350
3045	CRV-White-2019-*11,22,27*	16990
3046	Accord-Green-2012-*4,7*	9766
3047	Civic-White-2015-*4,5,6*	10889
3048	Pilot-Red-2018-*1,7,11,18*	22998
3049	Civic-Gray-2017-*2,6,9,20*	11150
3050	Odyssey-White-2020-*4,7,11,12,17,*	24551
3051	Ridgeline-Black-2012-*1,13*	12990
3052	Pilot-White-2016-*7*	17000
3053	Accord-Black-2018-*12,14,17*	13500

Name	Description
ID	A car's unique ID.
ProductCode	A car's code that captures the following information: model, color, year, and list of extra features (between **). Each feature has an internal code between 1 and 30.
Price	A car's price (before negotiation).

1. Describe the table's data problems.
2. Address these issues by transforming the table.

EX 5.7 **(LO 2, 5, 6)** **Data** **Financial Accounting** **Prepare Data for Financial Analysis** Hikko is a technology firm headquartered in California with sales in three regions: U.S., Europe, and Asia. It publishes the following report regarding its quarterly revenues.

Region	2022:Q1	2022:Q2	2022:Q3	2022:Q4	2023:Q1	2023:Q2	2023:Q3	2023:Q4	2024:Q1	2024:Q2	2024:Q3	2024:Q4
U.S	77,265,889	57,176,758	57,748,525	78,811,207	97,265,889	71,976,758	80,613,969	94,347,912	107,467,822	78,451,510	89,434,721	102,094,431
EUROPE	53,761,998	47,848,178	51,197,551	53,224,378	65,534,311	58,325,537	67,074,367	67,500,340	73,666,129	54,512,935	54,512,935	72,929,468
ASIA	32,188,799	27,038,591	31,094,380	32,832,575	31,777,112	23,197,292	23,661,238	32,094,883	29,778,112	22,035,803	24,239,383	28,289,206

You would like to compare Hikko's revenues across regions, quarters, and years. How would you reorganize the data to make such analysis easy?

EX 5.8 **(LO 1, 4, 5, 6)** **Data** **Managerial Accounting** **Prepare Data for Cost Analysis** Wilkinson, a luxury custom home builder in southern Nevada, uses job time cards to track labor costs for its different properties. They create weekly reports. Samples of the 2025 week 1 and week 2 reports are shown here.

WilkinsonWeek1

ID	Employee	JobNo	WeekNo	Thursday: 1/2/2025	Friday: 1/3/2025	Saturday: 1/4/2025
3405	1	ERF007	1	4	8	4
3406	1	ERF008	1	4		4
3407	13	ERF007	1	8	8	8
3408	25	IR68	1	4	8	0

WilkinsonWeek2

ID	Employee	JobNo	WeekNo	Monday: 1/6/2025	Tuesday: 1/7/2025	Wednesday: 1/8/2025	Thursday: 1/9/2025	Friday: 1/10/2025	Saturday: 1/11/2025
4608	25	ERF008	2	8	8	8	8	8	4
4609	13	ERF008	2	8	8	4			4
4610	13	IR68	2			4	8	8	
4611	1	IR101	2	6	8	10	8	8	4

How would you prepare Wilkinson's data for labor cost analysis purposes?

EX 5.9 **(LO 4)** **Data** **Detect Data Issues** HomePrinter is a distributor of laser and inkjet printers for the home office. You receive a data set that describes the different models they buy (and sell) and their January purchases. Profile the data set and identify any data issues.

EX 5.10 **(LO 5)** **Data** **Detect Data Issues** The Creighton Group is a small accounting firm with twenty employees. They keep track of their employees' salaries, bonuses and other information in a data file. Profile this data set and identify any data issues.

EX 5.11 **(LO 6)** **Data** **Create and Test Validation Rules** Vroomba manufactures and sells state-of-the art robot vacuums. All units sell for $279. Salespeople receive a large bonus if they sell more than 1,000 units in a single week. Salespeople can sell to a pre-approved list of customers, which are distributors, only. Every customer (distributor) has a unique name. They provide you with a sample of sales transactions for the week of January 6, 2025. What validation rule would you test? Are there any transactions that require further investigation?

EX 5.12 **(LO 5)** **Data** **Identify Inconsistent, Incomplete, or Invalid Data** Libel is a new ride-sharing company. They believe their employess are the keys to their success. They keep track of a wide range of employee information in a custom cloud application. On January 1, 2025, they provide you with a sample of employee information. The data dictionary for the Driver table follows.

Name	Description
DriverNumber	A Libel driver's unique number.
DateOfBirth	A driver's birthday.
State	The state in which the driver lives.
DateDriverLicense	The day a driver license was obtained by the driver.
AverageRating	The driver's average rating, with 1 being the worst possible rating and 5 being the best possible.
Customer	A six-digit unique code used to text a driver pickup information.

Profile this data set and identify any inconsistent, incomplete, or invalid data problems.

EX 5.13 (LO 1, 4) Data Prepare Data for Analysis The Muffin House is an industrial bakery in Ogunquit, Maine. They produce baked goods in large batches. For each batch they record a range of information. A sample of this information is provided in a data file. The data dictionary for the Batch table follows.

BATCH

Name	Description
ID	Uniquely identifies a batch.
Product	The product, a baked good, produced by the batch.
EstThrough	The estimated throughput time for the batch. How many minutes it should take to complete the batch.
ActThrough	The actual throughput time for the batch. How many minutes actually it takes to complete the batch.
EstQuantity	The estimated number of baked goods units that will be produced by the batch.
ActQuantity	The actual number of baked goods units produced by the batch.
Facility	Records which of the two facilities, A or B, produced the batch.
Supervisor	The employee who supervised the batch.

What changes would you make to the file before it is used for analysis?

Professional Application Case: Fluffy

Fluffy is a fast-growing company in North Dakota that creates and ships customized stuffed animals to all fifty U.S. states. There are several options for customization, from customers submitting their own designs to sending the company a rough design idea that Fluffy then develops and creates. While business is going well, the company's information system is a disaster and they are not able to do any analytics. The priority is to integrate their data into an analytical database. They hire your firm to help them.

PAC 5.1 Auditing: Identify Data Issues

Auditing You are given access to Fluffy's vendors, purchases, and employees files. Samples of the three files appear here.

Vendors

Code	Name	Address
1	Discount store	4326 Lochmere Lane, Groton, CT, 06340
2	Lebron Jordan	1177 Gore Street, Houston, TX, 77027
3	Duffy Warehouse	4457 Lonely Oak Drive, Mobile, AL, 36603
4	Bob Dylan	3764 Courtright Street, Fargo, ND, 58102
5	Segher's Fabrics	2170 Metz Lane, Camden, NJ, 08102

Purchases

Number	Date	Amount	Employee
1	1/6/2025	$ 4,176	1
2	1/7/2025	$ 660	3
3	1/9/2025	$ 1,680	5
4	1/14/2025	$ 4,703	4
6	1/20/2025	$ 3,108	2
7	1/21/2025	$ 2,263	4
8	1/23/2025	$ 1,597	6
8	1/23/2025	$ 1,597	6
9	1/30/2025	$ 3,433	2
10	1/31/2025	$ 1,833	4

Employees

Code	Name	Address
1	Joan Waddington	Phillps Circle 12, Fargo, ND, 58102
2	Matt Anthony	1012 Catherine Drive, St. Thomas, ND, 58276
3	Elizabeth Petroni	2490 Findley Avenue, Hope, ND, 58046
4	Andrea Dylan	3764 Courtright Street, Fargo, ND, 58102
5	Jimmy John	4000 Findley Avenue, Minot, ND, 58701

Using the data preparation patterns, review the sample data and identify at least three issues. Where applicable, specify which pattern you applied to detect and correct an issue.

PAC 5.2 Financial Accounting: Determine and Analyze Accounts Receivable

Data **Financial Accounting** Fluffy allows their customers to pay in installments. In the current system, sales orders and payments are recorded in two separate files. The SalesOrders file records the amount the customer owes Fluffy and the payments that have been received. The CashReceipts file records all payments received. Samples of both files are shown here.

SalesOrders

OrderNo	Date	Amount	Payments
1	1/6/2025	$ 838.46	1
2	1/7/2025	$ 100.03	2
3	1/9/2025	$ 245.67	3,10
4	1/13/2025	$ 3,632.16	
5	1/15/2025	$ 386.90	5
6	1/20/2025	$ 753.15	4
7	1/20/2025	$ 611.00	7,9,11
8	1/23/2025	$ 63.00	
9	1/25/2025	$ 1,496.72	
10	1/28/2025	$ 194.80	11

CashReceipts

CashReceiptNo	Date	Amount	Type
1	1/6/2025	$ 838.46	Cash
2	1/7/2025	$ 100.3	Credit Card
3	1/9/2025	$ 150	Check
4	1/20/2025	$ 350	Credit Card
5	1/24/2025	$ 386.9	Cash
6	1/27/2025	$ 200	Check
7	1/27/2025	$ 500	Check
8	1/30/2025	$ 403.15	Check
9	1/30/2025	$ 200	Check
10	1/31/2025	$ 95.67	Check
11	1/31/2025	$ 194.8	Check

1. Identify and explain the primary issue in the data set.
2. Describe how you would reorganize the data to enable the calculation of accounts receivable.

PAC 5.3 Managerial Accounting: Analyze Costs

Data **Managerial Accounting** Cost analysis is extremely important for Fluffy. In addition to producing the stuffed animals and providing standard shipping, they also offer a range of other services, such as express shipping, packing, and design. In fact, any service requested by a customer will be considered by Fluffy. When the company outsources a service, such as express shipping, the customer simply pays the amount charged to Fluffy. Fluffy has a standard markup for everything produced in-house, the stuffed animals, designs, and packing.

No	Price:100	StandardShipping:0	ExpressShipping:0	Packing:25	Design:50
1010	700	38	0	50	0
1011	280	0	154	125	225
1012	1,590	50	0	0	0
1013	906	57	0	25	0
1014	142	49	0	250	150
1015	1,728	44	0	125	0
1016	664	0	252	35	0
1017	208	66	0	35	0
1018	778	60	0	0	0
1019	296	62	0	10	0
1020	800	75	0	0	0

The sample data file shows how they currently record their cost and markup information. For example, for Order # 1010:

- The cost to make the products sold is $350. The price they charge is $700. As indicated by the column header, the markup on stuffed animals is 100%.
- There is no markup for shipping or express shipping.
- The markup for packing is 25%. The packing cost for Order # 1010 is $40. There is a 25% markup ($10), and the customer pays $50 ($40 + $10.)
- The overall cost for Order 1010 is $350 (stuffed animals) + $38 (standard shipping) + $40 (packing) = $428. The markup for the same order is $350 (stuffed animals) + $10 (packing) = $360. The total price to be paid by the customer is $788.

Fluffy would like to use their data to answer questions such as:

- What is the total revenue generated?
- What is the overall cost structure of orders based on the services offered?
- How do additional services affect the overall profit margin?

How would you restructure the data to make it easier to answer these questions through analysis?

PAC 5.4 Tax Accounting: Validate Sales Tax Amounts

Tax Accounting Currently, Fluffy's salespeople are responsible for creating and sending orders to customers. All salespeople have access to the Excel file that contains the sales tax rates for each state. They use that information to manually determine the sales tax of an order and enter it in the accounting system.

SalesTaxRate

State	SalesTaxRate
Connecticut	6.35
Delaware	N/A
Hawaii	4.17
Kentucky	6
Maryland	6
Michigan	6
Rhode Island	7

At the end of each week, Fluffy's accounting system generates a list of all orders called Sales Orders. As Fluffy's accountant, you must collect sales taxes and remit them to applicable states. You must check whether the sales tax amounts are accurate. A sample with ten orders from the sales orders file is shown next.

SalesOrder

OrderNo	Amount	SalesTax	State
1	$ 791	$ 47.46	KY
2	$ 88	$ 12.03	RI
3	$ 231	$ 14.67	CT
4	$ 3,488	$ 144.16	RI
5	$ 365	$ 21.90	MD
6	$ 723	$ 30.15	HI
7	$ 611	$ 0.00	DE
8	$ 55	$ 8.00	MD
9	$ 1,412	$ 84.72	MI
10	$ 187	$ 7.80	HI

It would help if you could combine the information in both tables to automatically calculate the sales tax per order and compare the tax information by state. Are there any data issues that would prevent this? List the patterns you used to identify the issues.

Le Grind Continuing Case: Use Data Preparation Patterns to Clean and Structure Data for Analysis Purposes

Data Visit Wiley's online learning platform for the case background, additional questions, data, and more details about the continuing case.